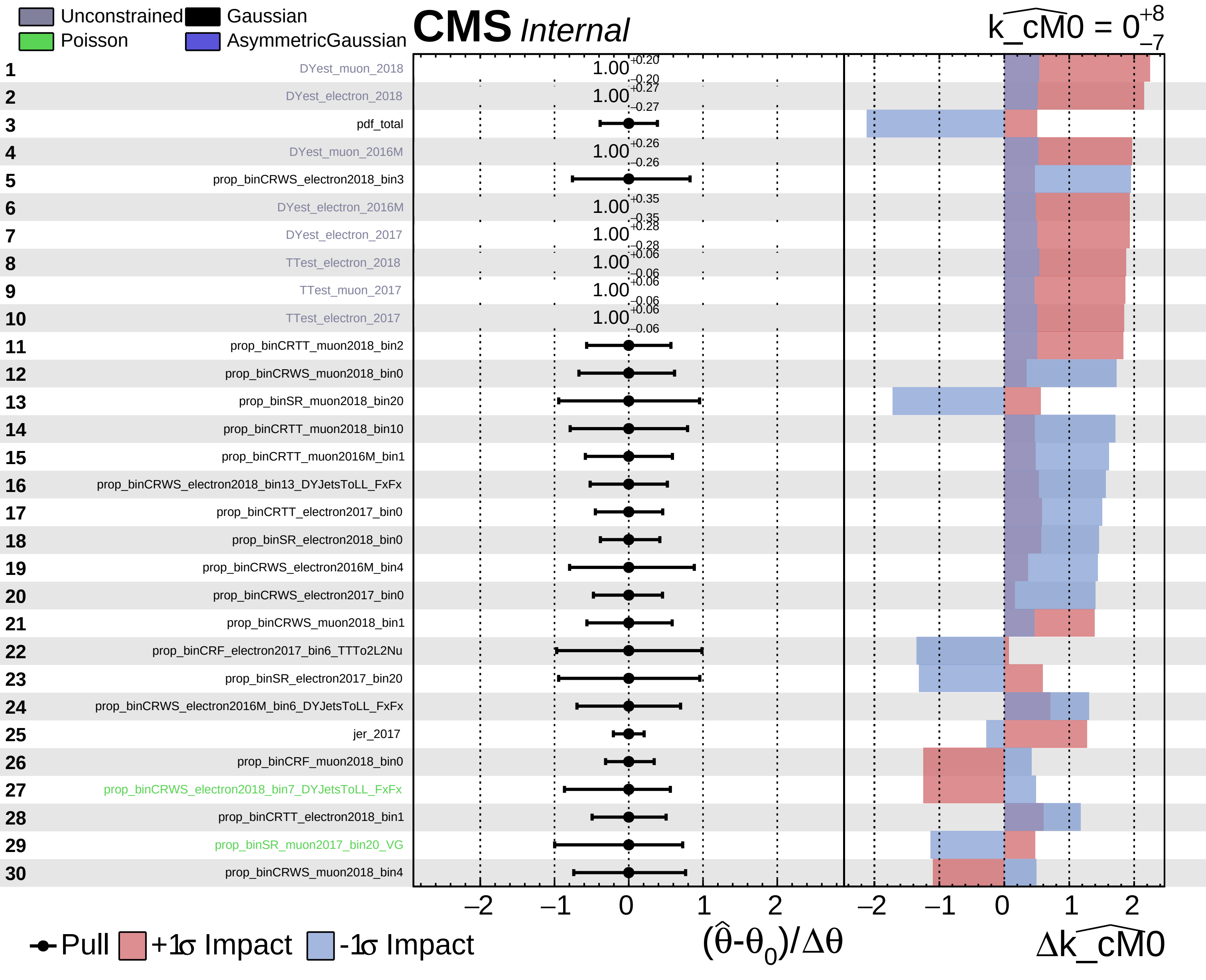
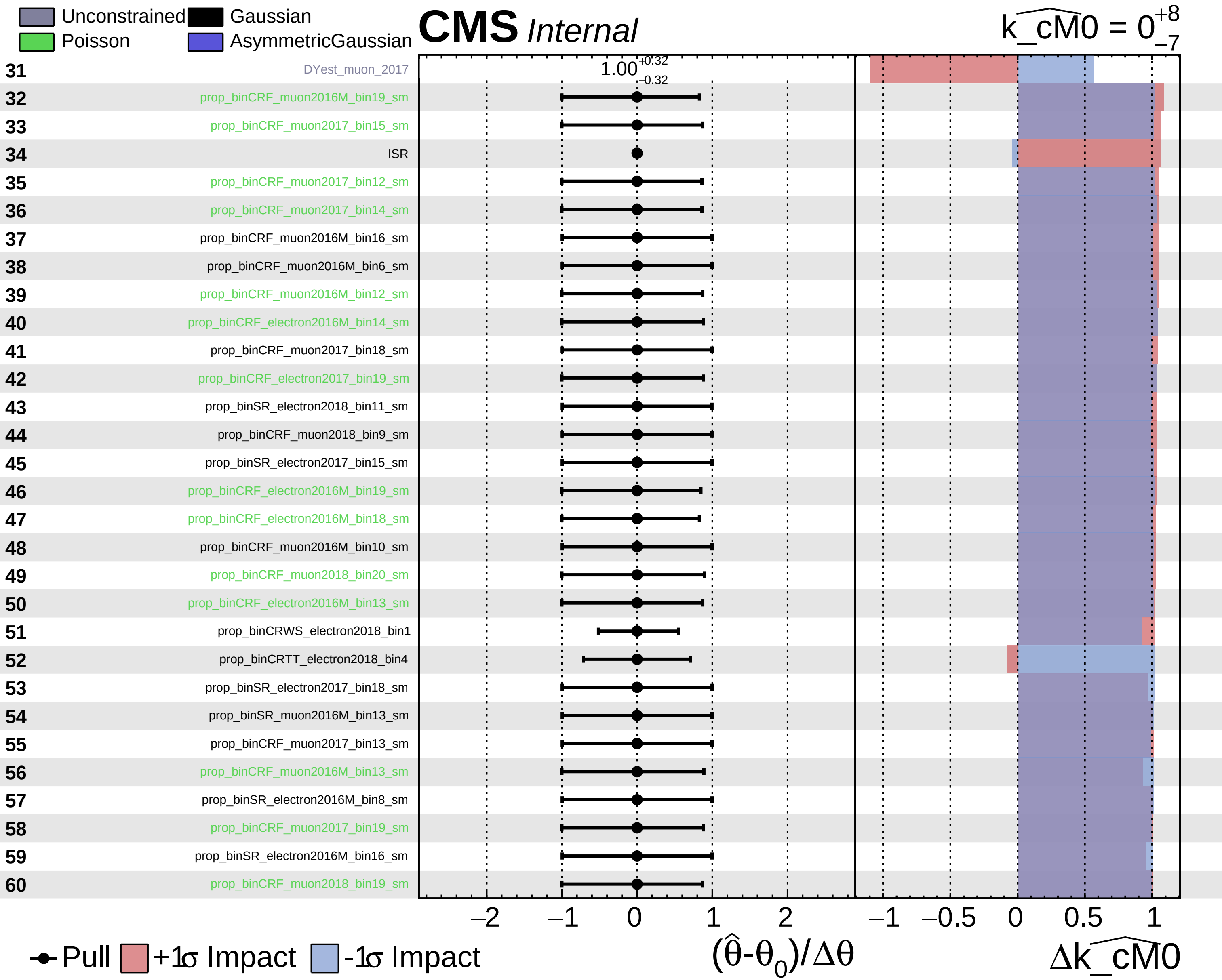
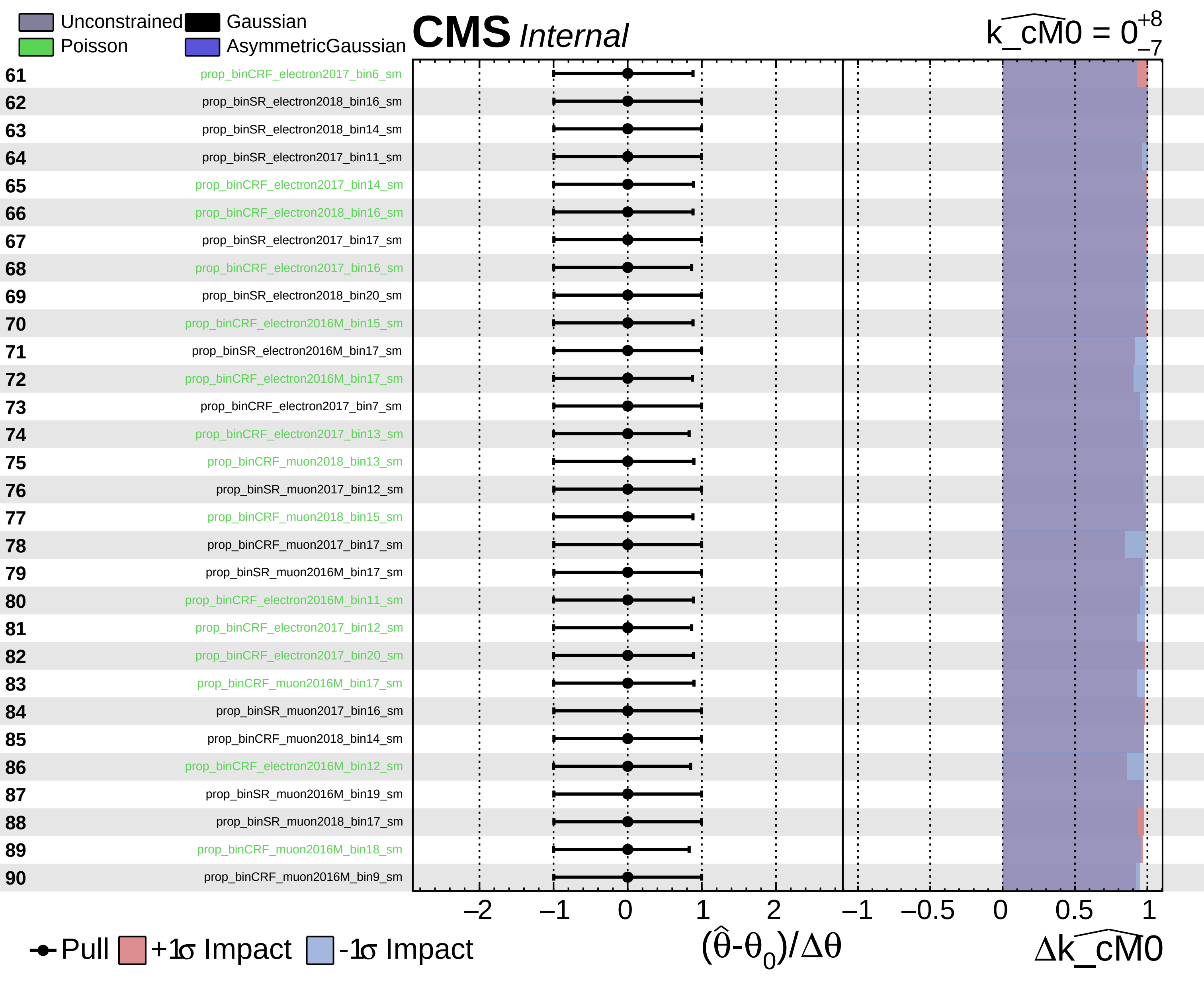


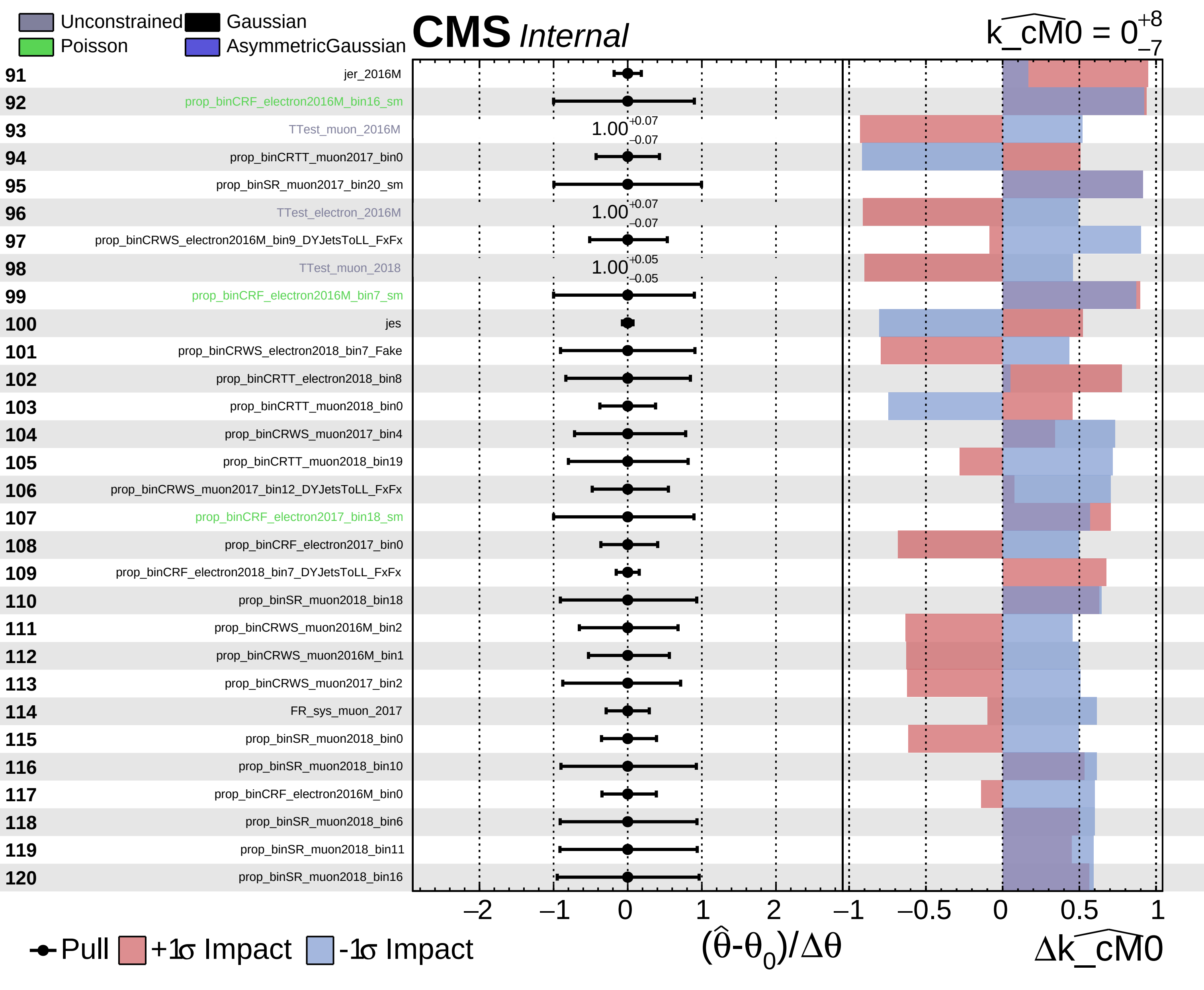
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
1 DYest_muon_2018 $1.00^{+0.20}_{-0.20}$
2 DYest_electron_2018 $1.00^{+0.27}_{-0.27}$
3 pdf_total
4 DYest_muon_2016M $1.00^{+0.26}_{-0.26}$
5 prop_binCRWS_electron2018_bin3
6 DYest_electron_2016M $1.00^{+0.35}_{-0.35}$
7 DYest_electron_2017 $1.00^{+0.28}_{-0.28}$
8 TTest_electron_2018 $1.00^{+0.06}_{-0.06}$
9 TTest_muon_2017 $1.00^{+0.06}_{-0.06}$
10 TTest_electron_2017 $1.00^{+0.06}_{-0.06}$
11 prop_binCRTT_muon2018_bin2
12 prop_binCRWS_muon2018_bin0
13 prop_binSR_muon2018_bin20
14 prop_binCRTT_muon2018_bin10
15 prop_binCRTT_muon2016M_bin1
16 prop_binCRWS_electron2018_bin13_DYJetsToLL_FxFx
17 prop_binCRTT_electron2017_bin0
18 prop_binSR_electron2018_bin0
19 prop_binCRWS_electron2016M_bin4
20 prop_binCRWS_electron2017_bin0
21 prop_binCRWS_muon2018_bin1
22 prop_binCRF_electron2017_bin6_TTTo2L2Nu
23 prop_binSR_electron2017_bin20
24 prop_binCRWS_electron2016M_bin6_DYJetsToLL_FxFx
25 jer_2017
26 prop_binCRF_muon2018_bin0
27 prop_binCRWS_electron2018_bin7_DYJetsToLL_FxFx
28 prop_binCRTT_electron2018_bin1
29 prop_binSR_muon2017_bin20_VG
30 prop_binCRWS_muon2018_bin4
−2 −1 0 1 2
−2 −1 0 1 2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



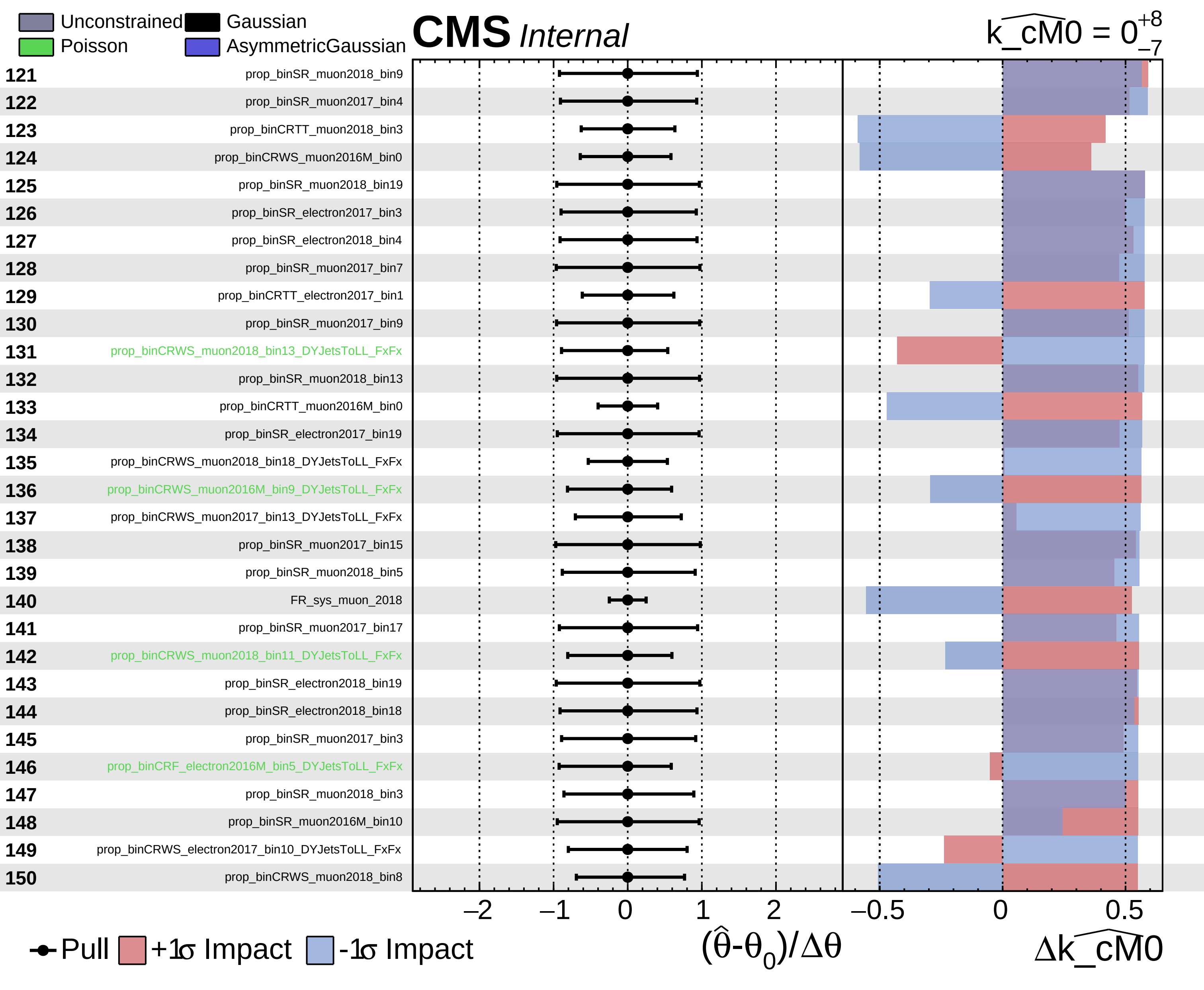
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
31 DYest_muon_2017
$1.00^{+0.32}_{-0.32}$
32 prop_binCRF_muon2016M_bin19_sm
33 prop_binCRF_muon2017_bin15_sm
34 ISR
35 prop_binCRF_muon2017_bin12_sm
36 prop_binCRF_muon2017_bin14_sm
37 prop_binCRF_muon2016M_bin16_sm
38 prop_binCRF_muon2016M_bin6_sm
39 prop_binCRF_muon2016M_bin12_sm
40 prop_binCRF_electron2016M_bin14_sm
41 prop_binCRF_muon2017_bin18_sm
42 prop_binCRF_electron2017_bin19_sm
43 prop_binSR_electron2018_bin11_sm
44 prop_binCRF_muon2018_bin9_sm
45 prop_binSR_electron2017_bin15_sm
46 prop_binCRF_electron2016M_bin19_sm
47 prop_binCRF_electron2016M_bin18_sm
48 prop_binCRF_muon2016M_bin10_sm
49 prop_binCRF_muon2018_bin20_sm
50 prop_binCRF_electron2016M_bin13_sm
51 prop_binCRWS_electron2018_bin1
52 prop_binCRTT_electron2018_bin4
53 prop_binSR_electron2017_bin18_sm
54 prop_binSR_muon2016M_bin13_sm
55 prop_binCRF_muon2017_bin13_sm
56 prop_binCRF_muon2016M_bin13_sm
57 prop_binSR_electron2016M_bin8_sm
58 prop_binCRF_muon2017_bin19_sm
59 prop_binSR_electron2016M_bin16_sm
60 prop_binCRF_muon2018_bin19_sm
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



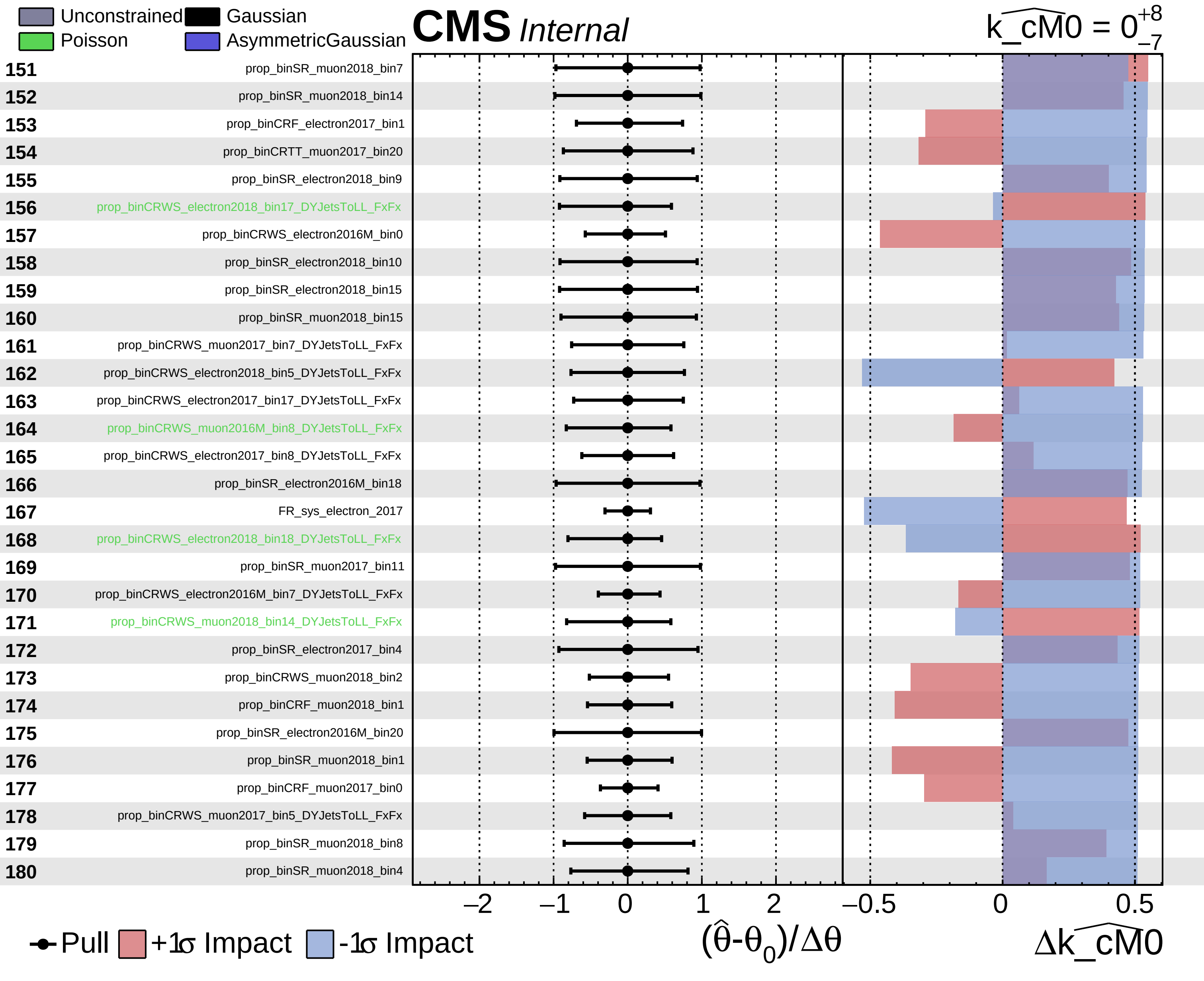
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0+8 −7
61 prop_binCRF_electron2017_bin6_sm
62 prop_binSR_electron2018_bin16_sm
63 prop_binSR_electron2018_bin14_sm
64 prop_binSR_electron2017_bin11_sm
65 prop_binCRF_electron2017_bin14_sm
66 prop_binCRF_electron2018_bin16_sm
67 prop_binSR_electron2017_bin17_sm
68 prop_binCRF_electron2017_bin16_sm
69 prop_binSR_electron2018_bin20_sm
70 prop_binCRF_electron2016M_bin15_sm
71 prop_binSR_electron2016M_bin17_sm
72 prop_binCRF_electron2016M_bin17_sm
73 prop_binCRF_electron2017_bin7_sm
74 prop_binCRF_electron2017_bin13_sm
75 prop_binCRF_muon2018_bin13_sm
76 prop_binSR_muon2017_bin12_sm
77 prop_binCRF_muon2018_bin15_sm
78 prop_binCRF_muon2017_bin17_sm
79 prop_binSR_muon2016M_bin17_sm
80 prop_binCRF_electron2016M_bin11_sm
81 prop_binCRF_electron2017_bin12_sm
82 prop_binCRF_electron2017_bin20_sm
83 prop_binCRF_muon2016M_bin17_sm
84 prop_binSR_muon2017_bin16_sm
85 prop_binCRF_muon2018_bin14_sm
86 prop_binCRF_electron2016M_bin12_sm
87 prop_binSR_muon2016M_bin19_sm
88 prop_binSR_muon2018_bin17_sm
89 prop_binCRF_muon2016M_bin18_sm
90 prop_binCRF_muon2016M_bin9_sm
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0



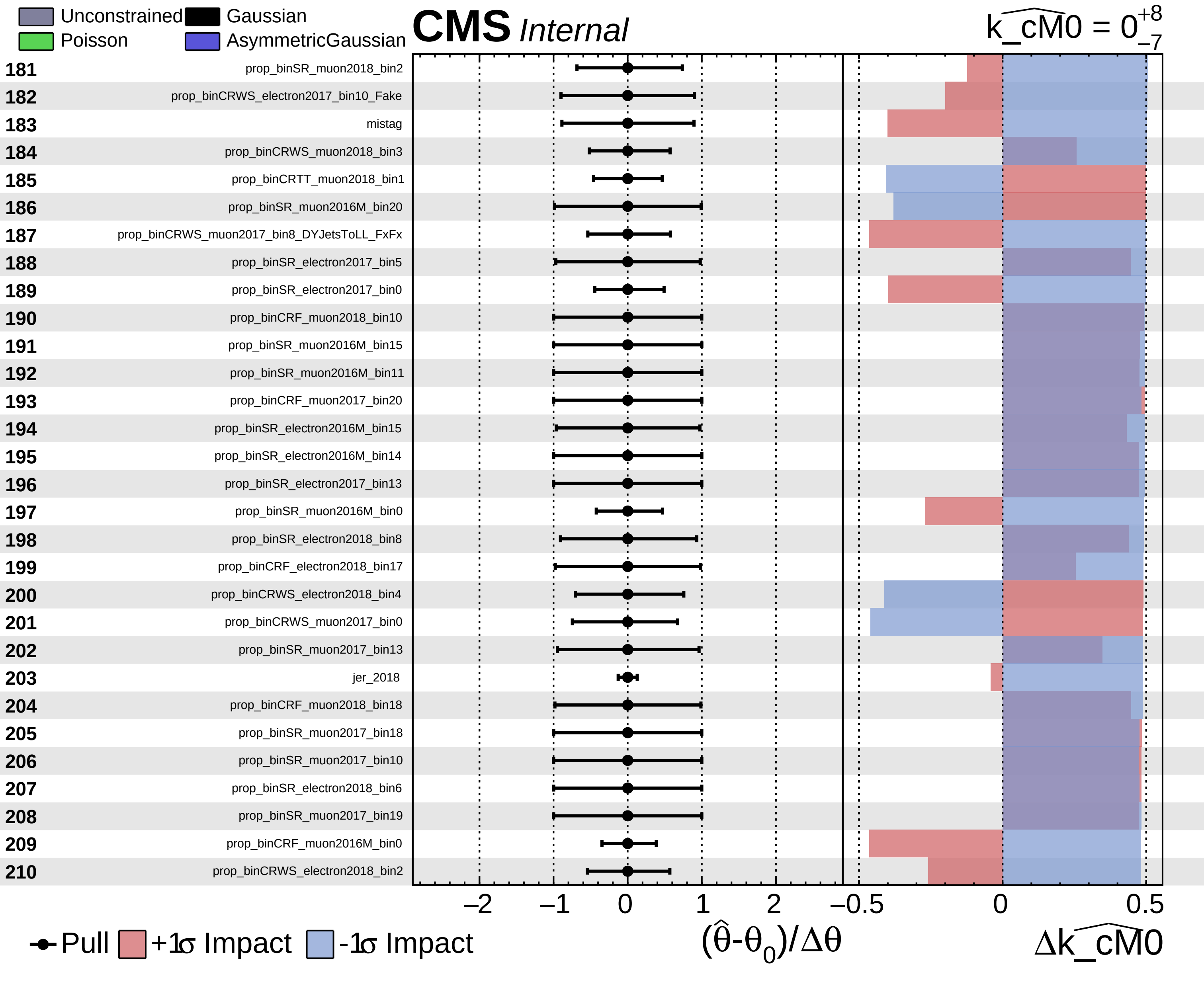
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
91 jer_2016M
92 prop_binCRF_electron2016M_bin16_sm
93 TTest_muon_2016M
$1.00^{+0.07}_{-0.07}$
94 prop_binCRTT_muon2017_bin0
95 prop_binSR_muon2017_bin20_sm
96 TTest_electron_2016M
$1.00^{+0.07}_{-0.07}$
97 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
98 TTest_muon_2018
$1.00^{+0.05}_{-0.05}$
99 prop_binCRF_electron2016M_bin7_sm
100 jes
101 prop_binCRWS_electron2018_bin7_Fake
102 prop_binCRTT_electron2018_bin8
103 prop_binCRTT_muon2018_bin0
104 prop_binCRWS_muon2017_bin4
105 prop_binCRTT_muon2018_bin19
106 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
107 prop_binCRF_electron2017_bin18_sm
108 prop_binCRF_electron2017_bin0
109 prop_binCRF_electron2018_bin7_DYJetsToLL_FxFx
110 prop_binSR_muon2018_bin18
111 prop_binCRWS_muon2016M_bin2
112 prop_binCRWS_muon2016M_bin1
113 prop_binCRWS_muon2017_bin2
114 FR_sys_muon_2017
115 prop_binSR_muon2018_bin0
116 prop_binSR_muon2018_bin10
117 prop_binCRF_electron2016M_bin0
118 prop_binSR_muon2018_bin6
119 prop_binSR_muon2018_bin11
120 prop_binSR_muon2018_bin16
−2
−1
0
1
2
−1
−0.5
0
0.5
1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



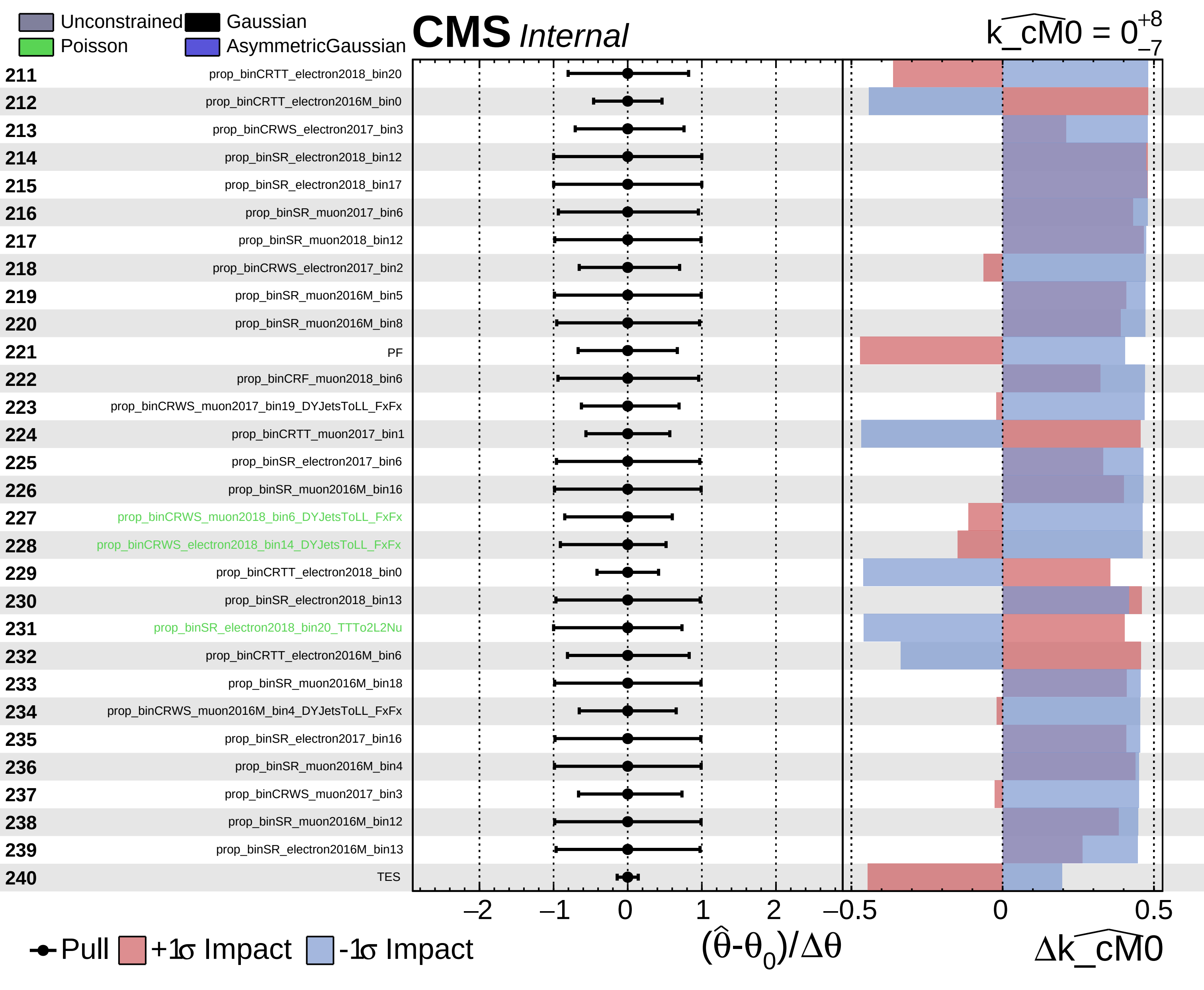
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
121 prop_binSR_muon2018_bin9
122 prop_binSR_muon2017_bin4
123 prop_binCRTT_muon2018_bin3
124 prop_binCRWS_muon2016M_bin0
125 prop_binSR_muon2018_bin19
126 prop_binSR_electron2017_bin3
127 prop_binSR_electron2018_bin4
128 prop_binSR_muon2017_bin7
129 prop_binCRTT_electron2017_bin1
130 prop_binSR_muon2017_bin9
131 prop_binCRWS_muon2018_bin13_DYJetsToLL_FxFx
132 prop_binSR_muon2018_bin13
133 prop_binCRTT_muon2016M_bin0
134 prop_binSR_electron2017_bin19
135 prop_binCRWS_muon2018_bin18_DYJetsToLL_FxFx
136 prop_binCRWS_muon2016M_bin9_DYJetsToLL_FxFx
137 prop_binCRWS_muon2017_bin13_DYJetsToLL_FxFx
138 prop_binSR_muon2017_bin15
139 prop_binSR_muon2018_bin5
140 FR_sys_muon_2018
141 prop_binSR_muon2017_bin17
142 prop_binCRWS_muon2018_bin11_DYJetsToLL_FxFx
143 prop_binSR_electron2018_bin19
144 prop_binSR_electron2018_bin18
145 prop_binSR_muon2017_bin3
146 prop_binCRF_electron2016M_bin5_DYJetsToLL_FxFx
147 prop_binSR_muon2018_bin3
148 prop_binSR_muon2016M_bin10
149 prop_binCRWS_electron2017_bin10_DYJetsToLL_FxFx
150 prop_binCRWS_muon2018_bin8
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$

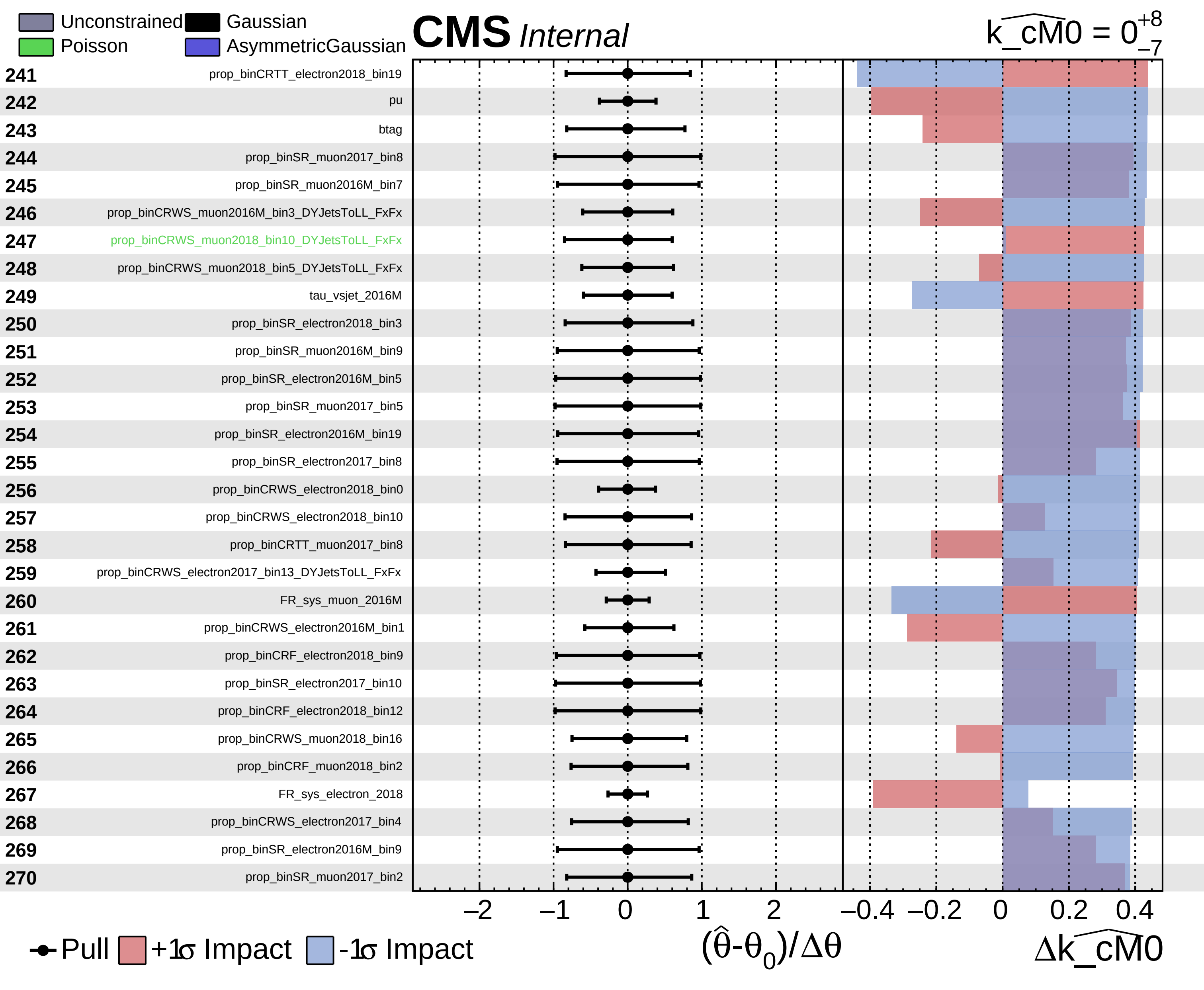


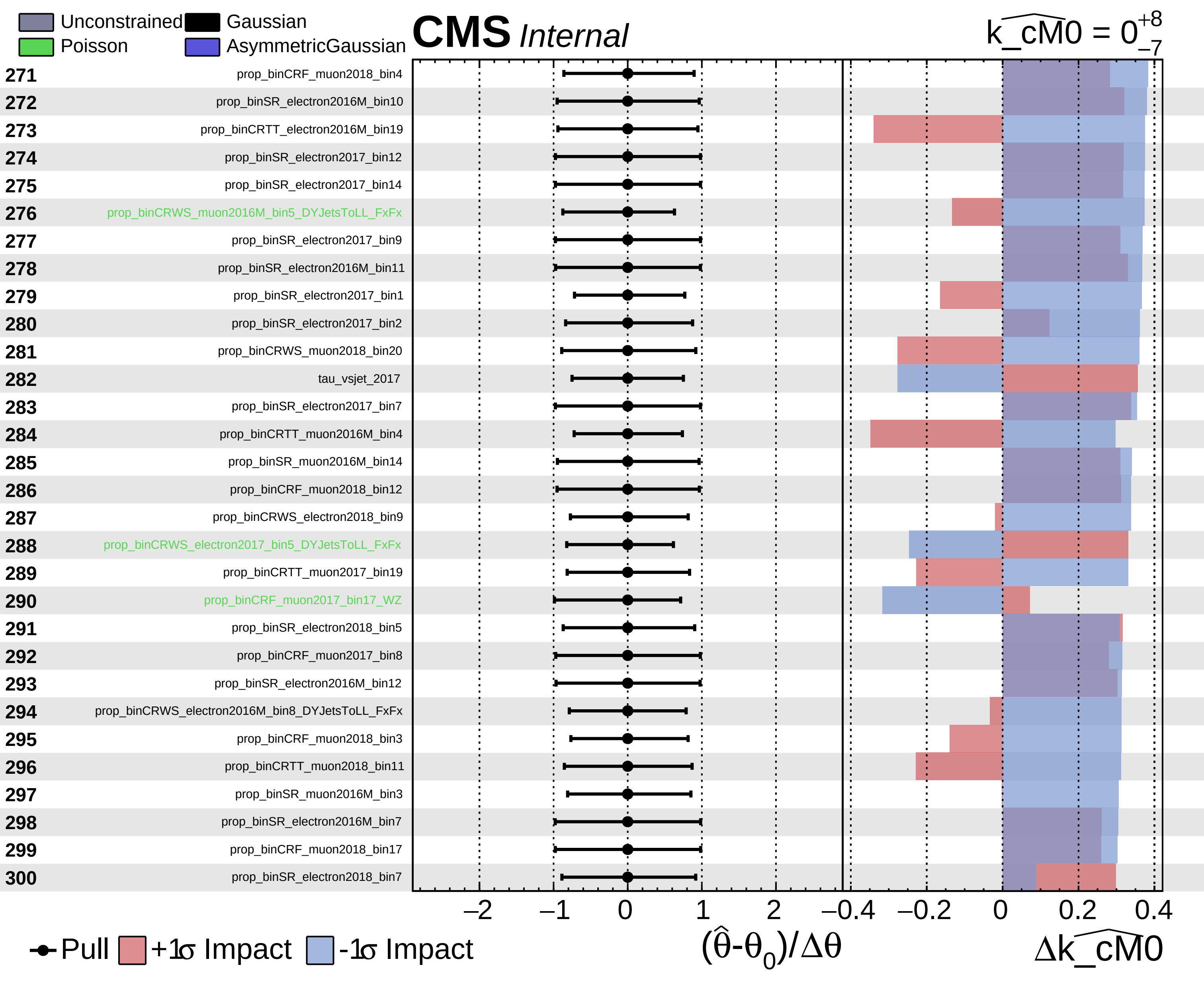
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0+8−7
151 prop_binSR_muon2018_bin7
152 prop_binSR_muon2018_bin14
153 prop_binCRF_electron2017_bin1
154 prop_binCRTT_muon2017_bin20
155 prop_binSR_electron2018_bin9
156 prop_binCRWS_electron2018_bin17_DYJetsToLL_FxFx
157 prop_binCRWS_electron2016M_bin0
158 prop_binSR_electron2018_bin10
159 prop_binSR_electron2018_bin15
160 prop_binSR_muon2018_bin15
161 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
162 prop_binCRWS_electron2018_bin5_DYJetsToLL_FxFx
163 prop_binCRWS_electron2017_bin17_DYJetsToLL_FxFx
164 prop_binCRWS_muon2016M_bin8_DYJetsToLL_FxFx
165 prop_binCRWS_electron2017_bin8_DYJetsToLL_FxFx
166 prop_binSR_electron2016M_bin18
167 FR_sys_electron_2017
168 prop_binCRWS_electron2018_bin18_DYJetsToLL_FxFx
169 prop_binSR_muon2017_bin11
170 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
171 prop_binCRWS_muon2018_bin14_DYJetsToLL_FxFx
172 prop_binSR_electron2017_bin4
173 prop_binCRWS_muon2018_bin2
174 prop_binCRF_muon2018_bin1
175 prop_binSR_electron2016M_bin20
176 prop_binSR_muon2018_bin1
177 prop_binCRF_muon2017_bin0
178 prop_binCRWS_muon2017_bin5_DYJetsToLL_FxFx
179 prop_binSR_muon2018_bin8
180 prop_binSR_muon2018_bin4
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

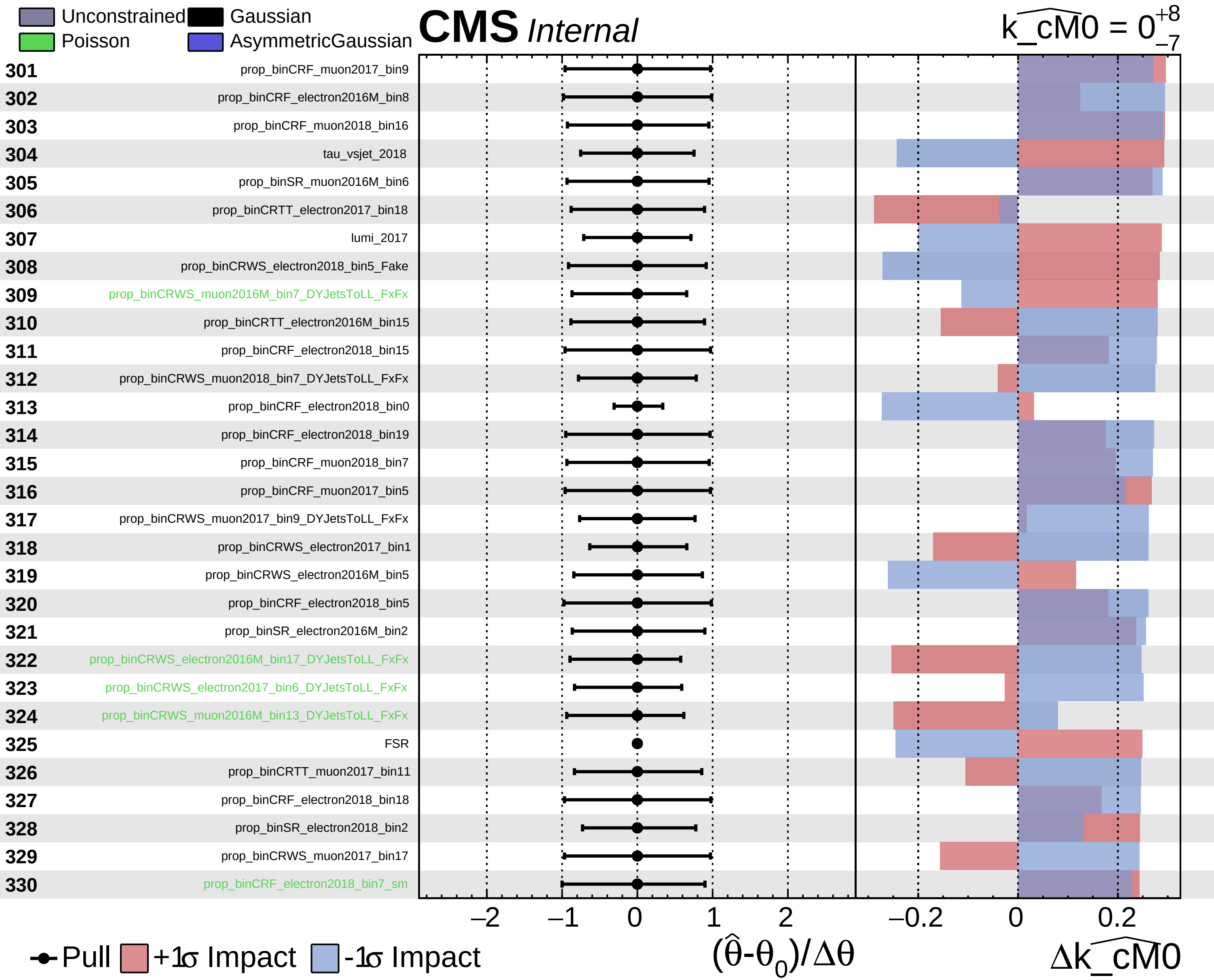


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0+8 −7
181 prop_binSR_muon2018_bin2
182 prop_binCRWS_electron2017_bin10_Fake
183 mistag
184 prop_binCRWS_muon2018_bin3
185 prop_binCRTT_muon2018_bin1
186 prop_binSR_muon2016M_bin20
187 prop_binCRWS_muon2017_bin8_DYJetsToLL_FxFx
188 prop_binSR_electron2017_bin5
189 prop_binSR_electron2017_bin0
190 prop_binCRF_muon2018_bin10
191 prop_binSR_muon2016M_bin15
192 prop_binSR_muon2016M_bin11
193 prop_binCRF_muon2017_bin20
194 prop_binSR_electron2016M_bin15
195 prop_binSR_electron2016M_bin14
196 prop_binSR_electron2017_bin13
197 prop_binSR_muon2016M_bin0
198 prop_binSR_electron2018_bin8
199 prop_binCRF_electron2018_bin17
200 prop_binCRWS_electron2018_bin4
201 prop_binCRWS_muon2017_bin0
202 prop_binSR_muon2017_bin13
203 jer_2018
204 prop_binCRF_muon2018_bin18
205 prop_binSR_muon2017_bin18
206 prop_binSR_muon2017_bin10
207 prop_binSR_electron2018_bin6
208 prop_binSR_muon2017_bin19
209 prop_binCRF_muon2016M_bin0
210 prop_binCRWS_electron2018_bin2
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

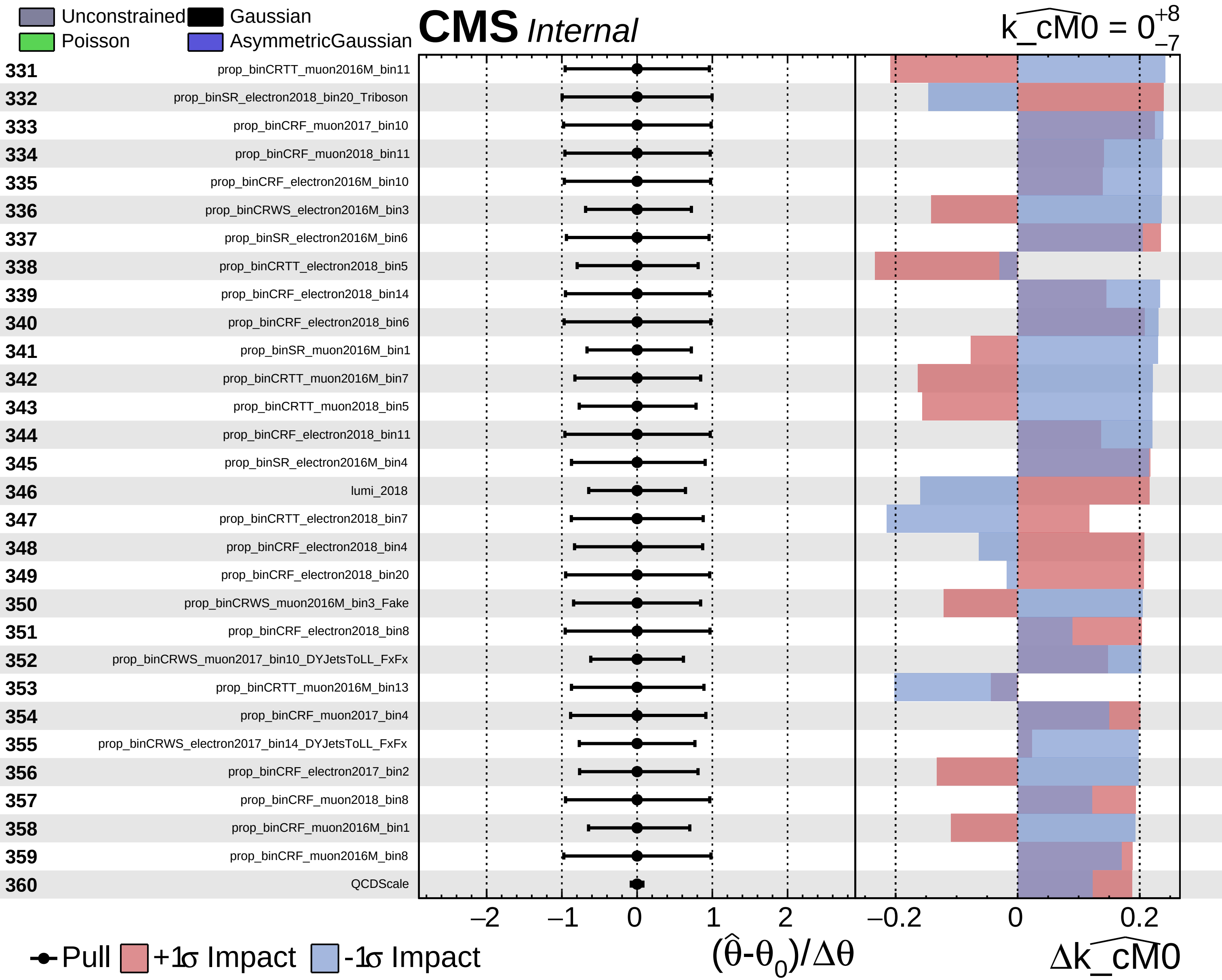

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$k\_\widehat{cM0} = 0^{+8}_{-7}$
211 prop_binCRTT_electron2018_bin20
212 prop_binCRTT_electron2016M_bin0
213 prop_binCRWS_electron2017_bin3
214 prop_binSR_electron2018_bin12
215 prop_binSR_electron2018_bin17
216 prop_binSR_muon2017_bin6
217 prop_binSR_muon2018_bin12
218 prop_binCRWS_electron2017_bin2
219 prop_binSR_muon2016M_bin5
220 prop_binSR_muon2016M_bin8
221 PF
222 prop_binCRF_muon2018_bin6
223 prop_binCRWS_muon2017_bin19_DYJetsToLL_FxFx
224 prop_binCRTT_muon2017_bin1
225 prop_binSR_electron2017_bin6
226 prop_binSR_muon2016M_bin16
227 prop_binCRWS_muon2018_bin6_DYJetsToLL_FxFx
228 prop_binCRWS_electron2018_bin14_DYJetsToLL_FxFx
229 prop_binCRTT_electron2018_bin0
230 prop_binSR_electron2018_bin13
231 prop_binSR_electron2018_bin20_TTTo2L2Nu
232 prop_binCRTT_electron2016M_bin6
233 prop_binSR_muon2016M_bin18
234 prop_binCRWS_muon2016M_bin4_DYJetsToLL_FxFx
235 prop_binSR_electron2017_bin16
236 prop_binSR_muon2016M_bin4
237 prop_binCRWS_muon2017_bin3
238 prop_binSR_muon2016M_bin12
239 prop_binSR_electron2016M_bin13
240 TES
−2 −1 0 1 2
−0.5 0 0.5
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta k\_\widehat{cM0}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = $0^{+8}_{-7}$
241 prop_binCRTT_electron2018_bin19
242 pu
243 btag
244 prop_binSR_muon2017_bin8
245 prop_binSR_muon2016M_bin7
246 prop_binCRWS_muon2016M_bin3_DYJetsToLL_FxFx
247 prop_binCRWS_muon2018_bin10_DYJetsToLL_FxFx
248 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
249 tau_vsjet_2016M
250 prop_binSR_electron2018_bin3
251 prop_binSR_muon2016M_bin9
252 prop_binSR_electron2016M_bin5
253 prop_binSR_muon2017_bin5
254 prop_binSR_electron2016M_bin19
255 prop_binSR_electron2017_bin8
256 prop_binCRWS_electron2018_bin0
257 prop_binCRWS_electron2018_bin10
258 prop_binCRTT_muon2017_bin8
259 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
260 FR_sys_muon_2016M
261 prop_binCRWS_electron2016M_bin1
262 prop_binCRF_electron2018_bin9
263 prop_binSR_electron2017_bin10
264 prop_binCRF_electron2018_bin12
265 prop_binCRWS_muon2018_bin16
266 prop_binCRF_muon2018_bin2
267 FR_sys_electron_2018
268 prop_binCRWS_electron2017_bin4
269 prop_binSR_electron2016M_bin9
270 prop_binSR_muon2017_bin2
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM0

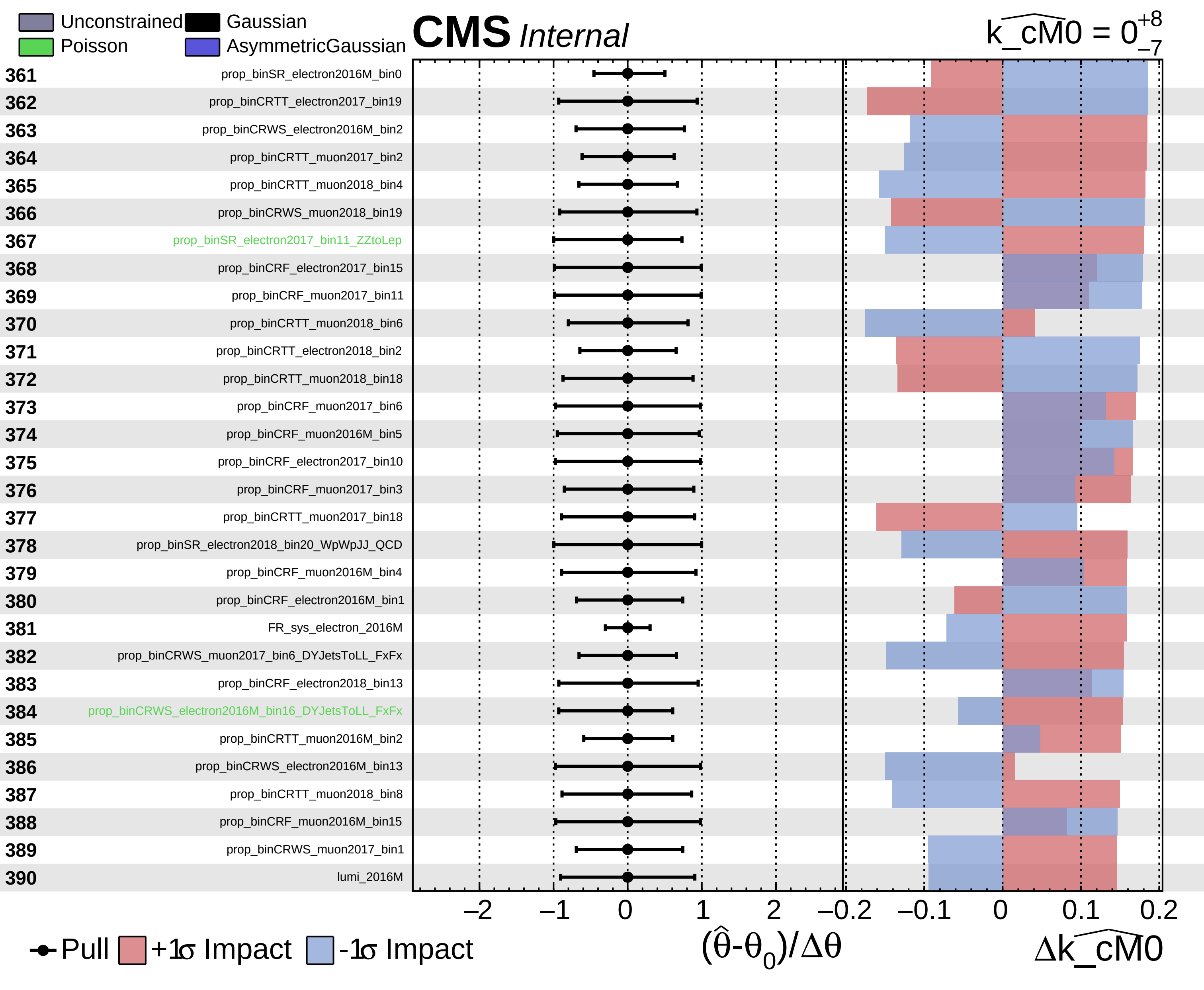

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0+8−7
271 prop_binCRF_muon2018_bin4
272 prop_binSR_electron2016M_bin10
273 prop_binCRTT_electron2016M_bin19
274 prop_binSR_electron2017_bin12
275 prop_binSR_electron2017_bin14
276 prop_binCRWS_muon2016M_bin5_DYJetsToLL_FxFx
277 prop_binSR_electron2017_bin9
278 prop_binSR_electron2016M_bin11
279 prop_binSR_electron2017_bin1
280 prop_binSR_electron2017_bin2
281 prop_binCRWS_muon2018_bin20
282 tau_vsjet_2017
283 prop_binSR_electron2017_bin7
284 prop_binCRTT_muon2016M_bin4
285 prop_binSR_muon2016M_bin14
286 prop_binCRF_muon2018_bin12
287 prop_binCRWS_electron2018_bin9
288 prop_binCRWS_electron2017_bin5_DYJetsToLL_FxFx
289 prop_binCRTT_muon2017_bin19
290 prop_binCRF_muon2017_bin17_WZ
291 prop_binSR_electron2018_bin5
292 prop_binCRF_muon2017_bin8
293 prop_binSR_electron2016M_bin12
294 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
295 prop_binCRF_muon2018_bin3
296 prop_binCRTT_muon2018_bin11
297 prop_binSR_muon2016M_bin3
298 prop_binSR_electron2016M_bin7
299 prop_binCRF_muon2018_bin17
300 prop_binSR_electron2018_bin7
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0



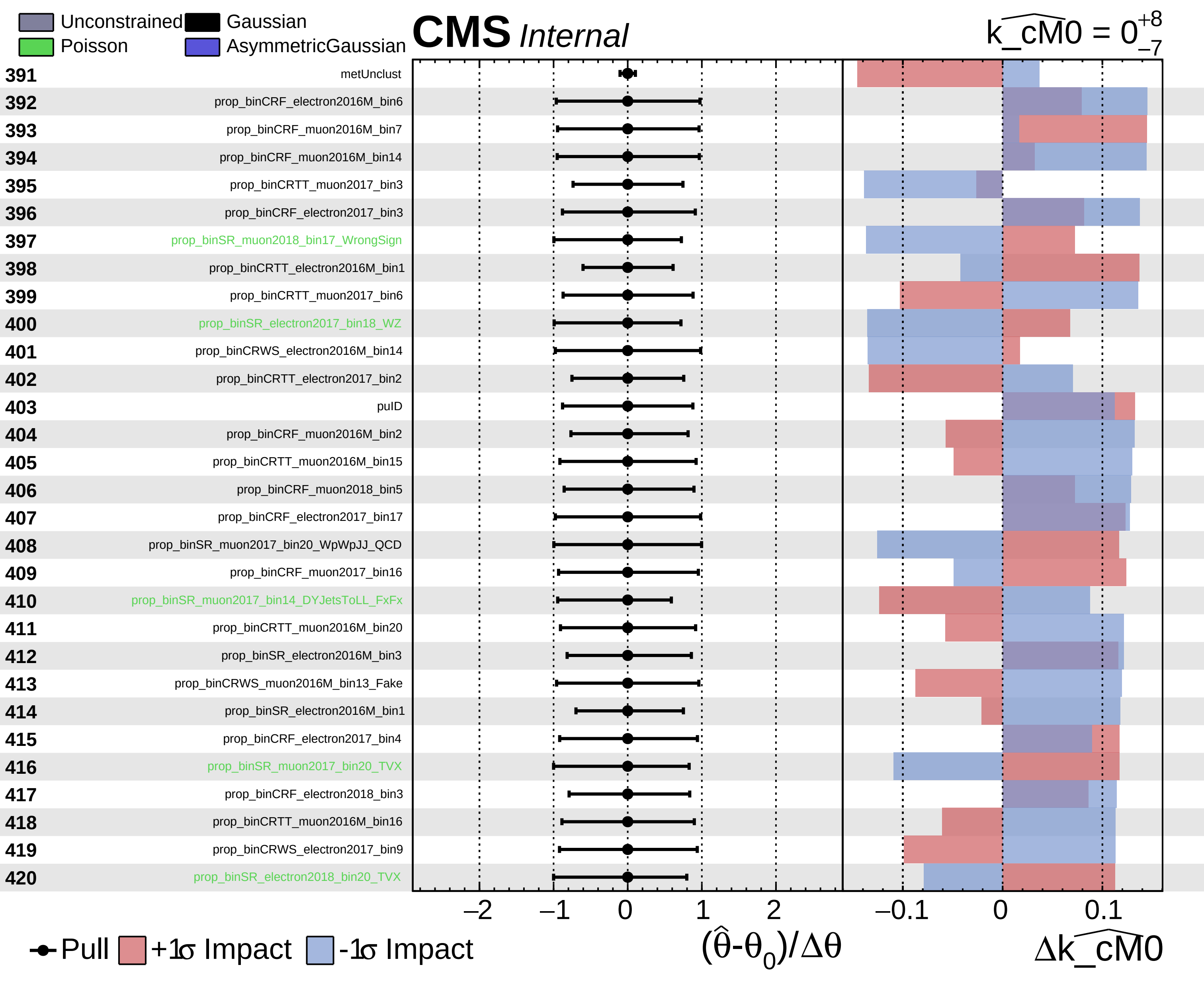
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0 +8 −7
301 prop_binCRF_muon2017_bin9
302 prop_binCRF_electron2016M_bin8
303 prop_binCRF_muon2018_bin16
304 tau_vsjet_2018
305 prop_binSR_muon2016M_bin6
306 prop_binCRTT_electron2017_bin18
307 lumi_2017
308 prop_binCRWS_electron2018_bin5_Fake
309 prop_binCRWS_muon2016M_bin7_DYJetsToLL_FxFx
310 prop_binCRTT_electron2016M_bin15
311 prop_binCRF_electron2018_bin15
312 prop_binCRWS_muon2018_bin7_DYJetsToLL_FxFx
313 prop_binCRF_electron2018_bin0
314 prop_binCRF_electron2018_bin19
315 prop_binCRF_muon2018_bin7
316 prop_binCRF_muon2017_bin5
317 prop_binCRWS_muon2017_bin9_DYJetsToLL_FxFx
318 prop_binCRWS_electron2017_bin1
319 prop_binCRWS_electron2016M_bin5
320 prop_binCRF_electron2018_bin5
321 prop_binSR_electron2016M_bin2
322 prop_binCRWS_electron2016M_bin17_DYJetsToLL_FxFx
323 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
324 prop_binCRWS_muon2016M_bin13_DYJetsToLL_FxFx
325 FSR
326 prop_binCRTT_muon2017_bin11
327 prop_binCRF_electron2018_bin18
328 prop_binSR_electron2018_bin2
329 prop_binCRWS_muon2017_bin17
330 prop_binCRF_electron2018_bin7_sm
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0



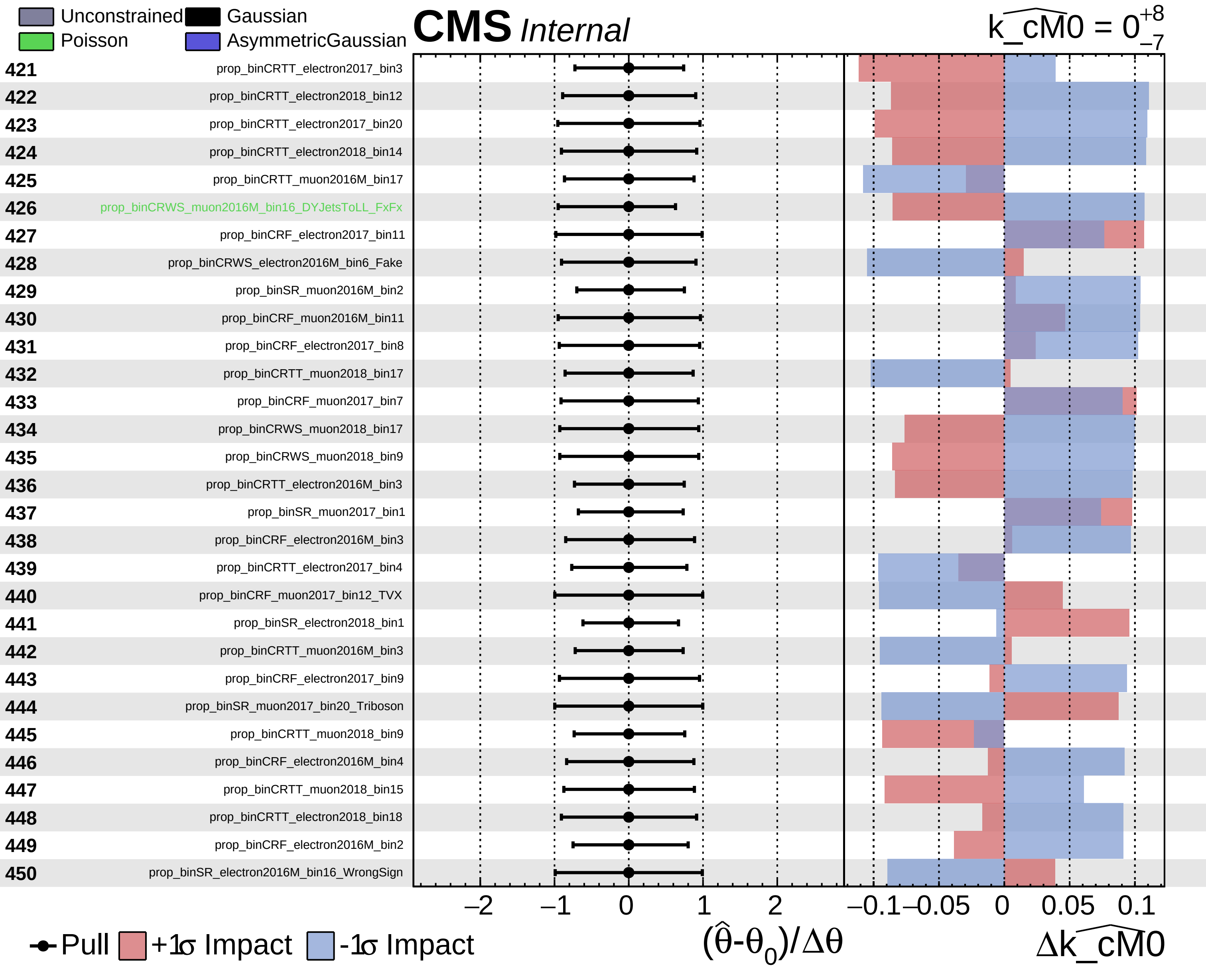
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0 +8 −7
331 prop_binCRTT_muon2016M_bin11
332 prop_binSR_electron2018_bin20_Triboson
333 prop_binCRF_muon2017_bin10
334 prop_binCRF_muon2018_bin11
335 prop_binCRF_electron2016M_bin10
336 prop_binCRWS_electron2016M_bin3
337 prop_binSR_electron2016M_bin6
338 prop_binCRTT_electron2018_bin5
339 prop_binCRF_electron2018_bin14
340 prop_binCRF_electron2018_bin6
341 prop_binSR_muon2016M_bin1
342 prop_binCRTT_muon2016M_bin7
343 prop_binCRTT_muon2018_bin5
344 prop_binCRF_electron2018_bin11
345 prop_binSR_electron2016M_bin4
346 lumi_2018
347 prop_binCRTT_electron2018_bin7
348 prop_binCRF_electron2018_bin4
349 prop_binCRF_electron2018_bin20
350 prop_binCRWS_muon2016M_bin3_Fake
351 prop_binCRF_electron2018_bin8
352 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
353 prop_binCRTT_muon2016M_bin13
354 prop_binCRF_muon2017_bin4
355 prop_binCRWS_electron2017_bin14_DYJetsToLL_FxFx
356 prop_binCRF_electron2017_bin2
357 prop_binCRF_muon2018_bin8
358 prop_binCRF_muon2016M_bin1
359 prop_binCRF_muon2016M_bin8
360 QCDScale
−2
−1
0
1
2
−0.2
0
0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0



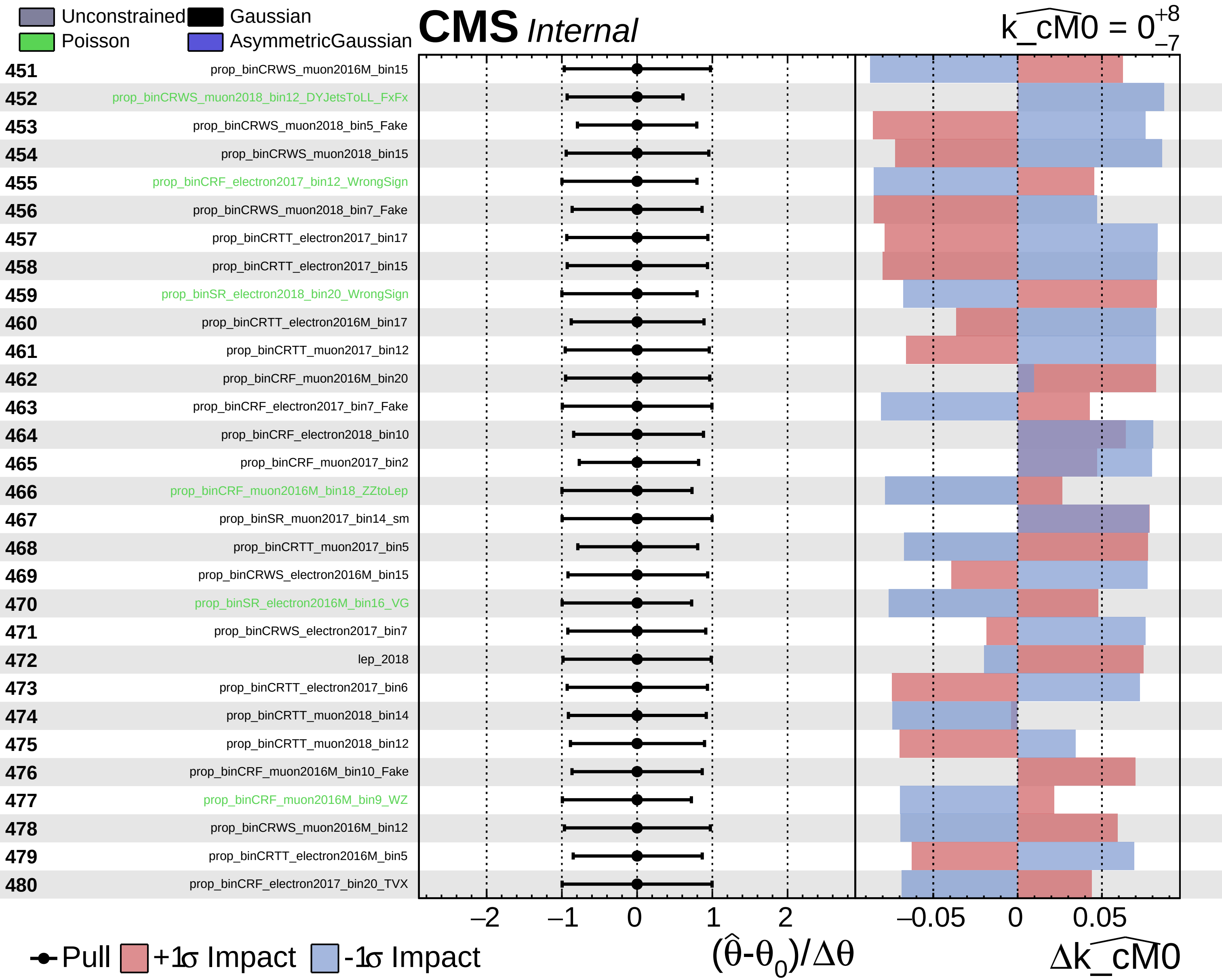
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
361 prop_binSR_electron2016M_bin0
362 prop_binCRTT_electron2017_bin19
363 prop_binCRWS_electron2016M_bin2
364 prop_binCRTT_muon2017_bin2
365 prop_binCRTT_muon2018_bin4
366 prop_binCRWS_muon2018_bin19
367 prop_binSR_electron2017_bin11_ZZtoLep
368 prop_binCRF_electron2017_bin15
369 prop_binCRF_muon2017_bin11
370 prop_binCRTT_muon2018_bin6
371 prop_binCRTT_electron2018_bin2
372 prop_binCRTT_muon2018_bin18
373 prop_binCRF_muon2017_bin6
374 prop_binCRF_muon2016M_bin5
375 prop_binCRF_electron2017_bin10
376 prop_binCRF_muon2017_bin3
377 prop_binCRTT_muon2017_bin18
378 prop_binSR_electron2018_bin20_WpWpJJ_QCD
379 prop_binCRF_muon2016M_bin4
380 prop_binCRF_electron2016M_bin1
381 FR_sys_electron_2016M
382 prop_binCRWS_muon2017_bin6_DYJetsToLL_FxFx
383 prop_binCRF_electron2018_bin13
384 prop_binCRWS_electron2016M_bin16_DYJetsToLL_FxFx
385 prop_binCRTT_muon2016M_bin2
386 prop_binCRWS_electron2016M_bin13
387 prop_binCRTT_muon2018_bin8
388 prop_binCRF_muon2016M_bin15
389 prop_binCRWS_muon2017_bin1
390 lumi_2016M
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$

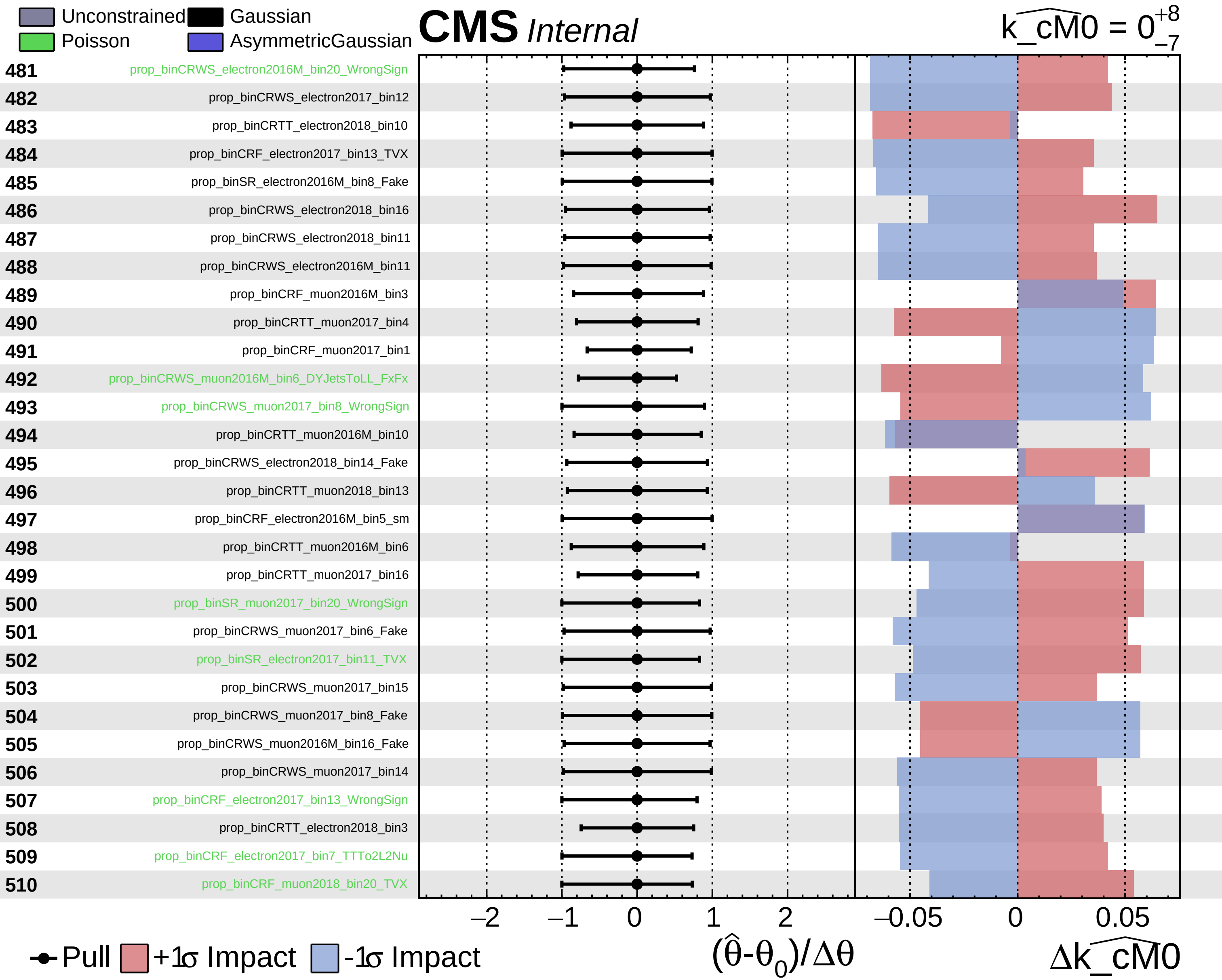


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0 +8 −7
391 metUnclust
392 prop_binCRF_electron2016M_bin6
393 prop_binCRF_muon2016M_bin7
394 prop_binCRF_muon2016M_bin14
395 prop_binCRTT_muon2017_bin3
396 prop_binCRF_electron2017_bin3
397 prop_binSR_muon2018_bin17_WrongSign
398 prop_binCRTT_electron2016M_bin1
399 prop_binCRTT_muon2017_bin6
400 prop_binSR_electron2017_bin18_WZ
401 prop_binCRWS_electron2016M_bin14
402 prop_binCRTT_electron2017_bin2
403 puID
404 prop_binCRF_muon2016M_bin2
405 prop_binCRTT_muon2016M_bin15
406 prop_binCRF_muon2018_bin5
407 prop_binCRF_electron2017_bin17
408 prop_binSR_muon2017_bin20_WpWpJJ_QCD
409 prop_binCRF_muon2017_bin16
410 prop_binSR_muon2017_bin14_DYJetsToLL_FxFx
411 prop_binCRTT_muon2016M_bin20
412 prop_binSR_electron2016M_bin3
413 prop_binCRWS_muon2016M_bin13_Fake
414 prop_binSR_electron2016M_bin1
415 prop_binCRF_electron2017_bin4
416 prop_binSR_muon2017_bin20_TVX
417 prop_binCRF_electron2018_bin3
418 prop_binCRTT_muon2016M_bin16
419 prop_binCRWS_electron2017_bin9
420 prop_binSR_electron2018_bin20_TVX
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

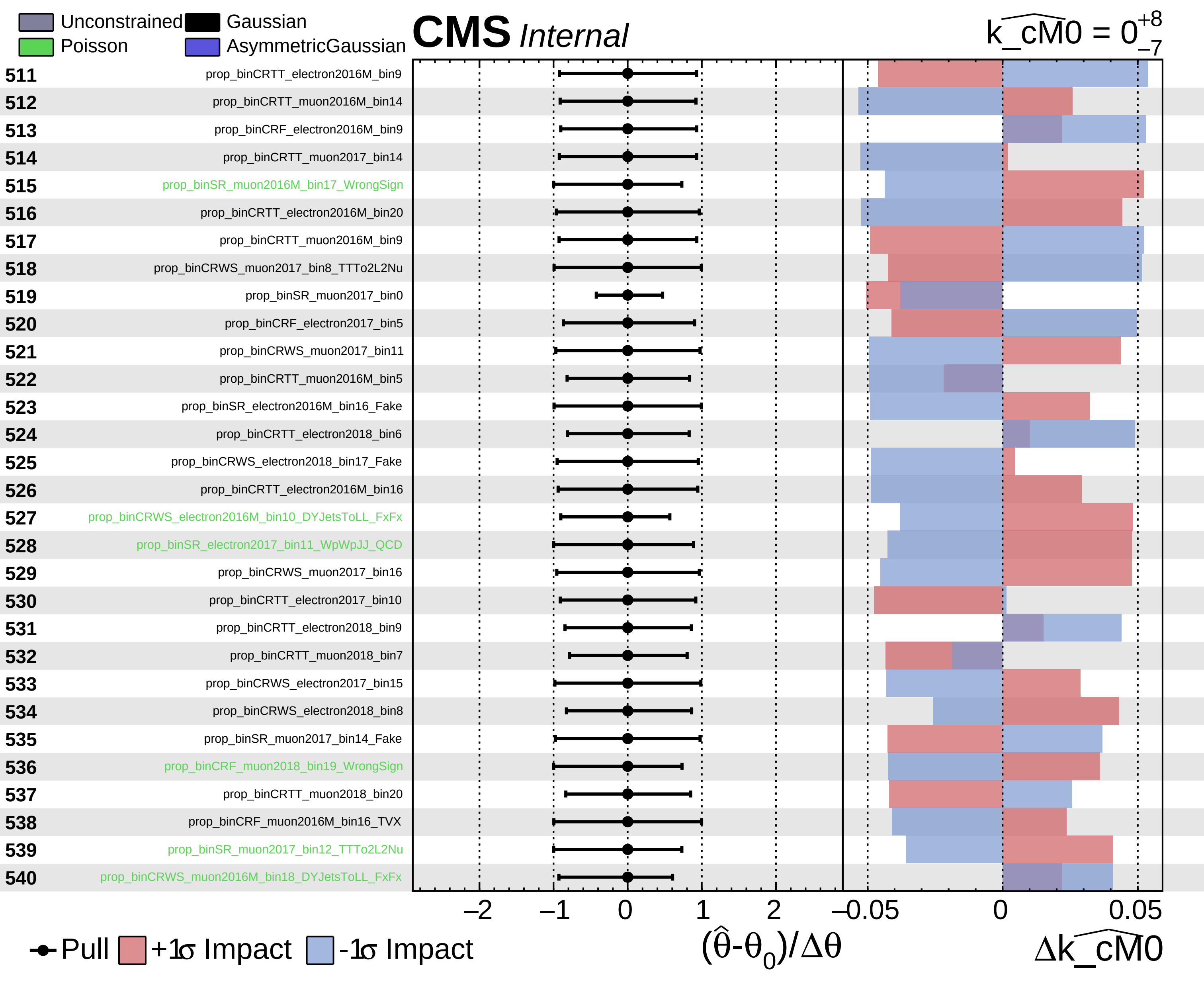


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0+8−7
421 prop_binCRTT_electron2017_bin3
422 prop_binCRTT_electron2018_bin12
423 prop_binCRTT_electron2017_bin20
424 prop_binCRTT_electron2018_bin14
425 prop_binCRTT_muon2016M_bin17
426 prop_binCRWS_muon2016M_bin16_DYJetsToLL_FxFx
427 prop_binCRF_electron2017_bin11
428 prop_binCRWS_electron2016M_bin6_Fake
429 prop_binSR_muon2016M_bin2
430 prop_binCRF_muon2016M_bin11
431 prop_binCRF_electron2017_bin8
432 prop_binCRTT_muon2018_bin17
433 prop_binCRF_muon2017_bin7
434 prop_binCRWS_muon2018_bin17
435 prop_binCRWS_muon2018_bin9
436 prop_binCRTT_electron2016M_bin3
437 prop_binSR_muon2017_bin1
438 prop_binCRF_electron2016M_bin3
439 prop_binCRTT_electron2017_bin4
440 prop_binCRF_muon2017_bin12_TVX
441 prop_binSR_electron2018_bin1
442 prop_binCRTT_muon2016M_bin3
443 prop_binCRF_electron2017_bin9
444 prop_binSR_muon2017_bin20_Triboson
445 prop_binCRTT_muon2018_bin9
446 prop_binCRF_electron2016M_bin4
447 prop_binCRTT_muon2018_bin15
448 prop_binCRTT_electron2018_bin18
449 prop_binCRF_electron2016M_bin2
450 prop_binSR_electron2016M_bin16_WrongSign
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

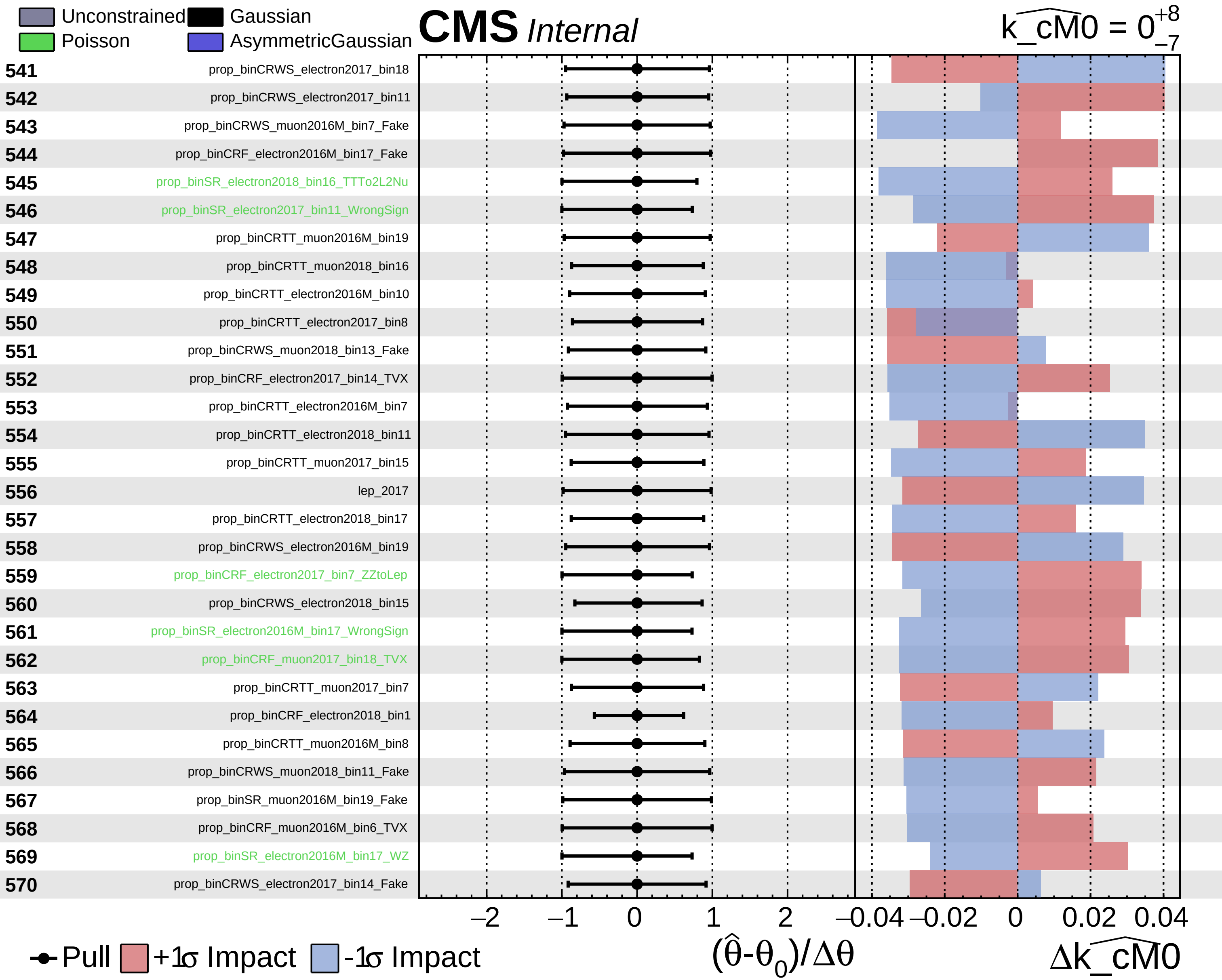

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
451 prop_binCRWS_muon2016M_bin15
452 prop_binCRWS_muon2018_bin12_DYJetsToLL_FxFx
453 prop_binCRWS_muon2018_bin5_Fake
454 prop_binCRWS_muon2018_bin15
455 prop_binCRF_electron2017_bin12_WrongSign
456 prop_binCRWS_muon2018_bin7_Fake
457 prop_binCRTT_electron2017_bin17
458 prop_binCRTT_electron2017_bin15
459 prop_binSR_electron2018_bin20_WrongSign
460 prop_binCRTT_electron2016M_bin17
461 prop_binCRTT_muon2017_bin12
462 prop_binCRF_muon2016M_bin20
463 prop_binCRF_electron2017_bin7_Fake
464 prop_binCRF_electron2018_bin10
465 prop_binCRF_muon2017_bin2
466 prop_binCRF_muon2016M_bin18_ZZtoLep
467 prop_binSR_muon2017_bin14_sm
468 prop_binCRTT_muon2017_bin5
469 prop_binCRWS_electron2016M_bin15
470 prop_binSR_electron2016M_bin16_VG
471 prop_binCRWS_electron2017_bin7
472 lep_2018
473 prop_binCRTT_electron2017_bin6
474 prop_binCRTT_muon2018_bin14
475 prop_binCRTT_muon2018_bin12
476 prop_binCRF_muon2016M_bin10_Fake
477 prop_binCRF_muon2016M_bin9_WZ
478 prop_binCRWS_muon2016M_bin12
479 prop_binCRTT_electron2016M_bin5
480 prop_binCRF_electron2017_bin20_TVX
−2 −1 0 1 2
−0.05 0 0.05
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



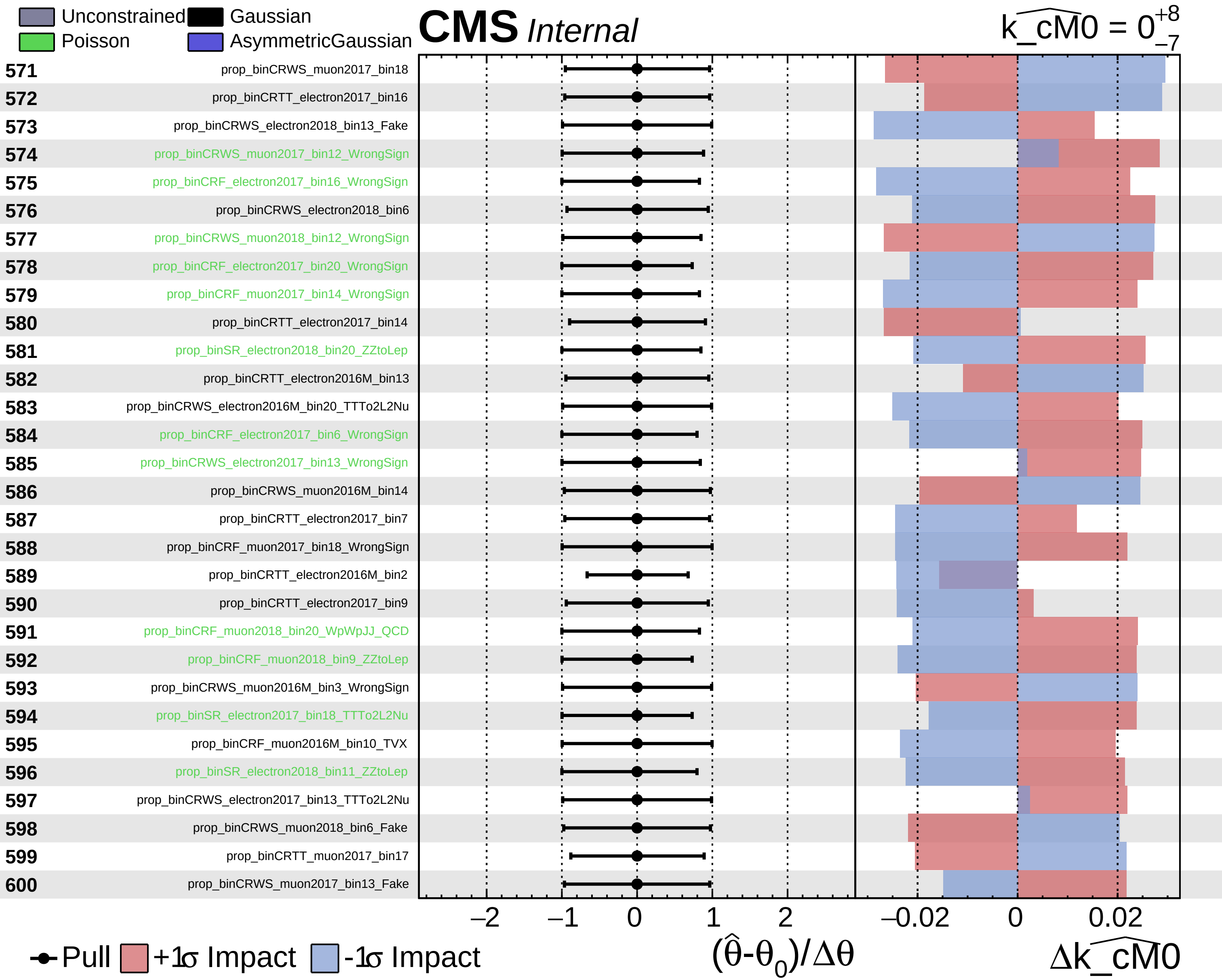
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0^{+8}_{−7}
481 prop_binCRWS_electron2016M_bin20_WrongSign
482 prop_binCRWS_electron2017_bin12
483 prop_binCRTT_electron2018_bin10
484 prop_binCRF_electron2017_bin13_TVX
485 prop_binSR_electron2016M_bin8_Fake
486 prop_binCRWS_electron2018_bin16
487 prop_binCRWS_electron2018_bin11
488 prop_binCRWS_electron2016M_bin11
489 prop_binCRF_muon2016M_bin3
490 prop_binCRTT_muon2017_bin4
491 prop_binCRF_muon2017_bin1
492 prop_binCRWS_muon2016M_bin6_DYJetsToLL_FxFx
493 prop_binCRWS_muon2017_bin8_WrongSign
494 prop_binCRTT_muon2016M_bin10
495 prop_binCRWS_electron2018_bin14_Fake
496 prop_binCRTT_muon2018_bin13
497 prop_binCRF_electron2016M_bin5_sm
498 prop_binCRTT_muon2016M_bin6
499 prop_binCRTT_muon2017_bin16
500 prop_binSR_muon2017_bin20_WrongSign
501 prop_binCRWS_muon2017_bin6_Fake
502 prop_binSR_electron2017_bin11_TVX
503 prop_binCRWS_muon2017_bin15
504 prop_binCRWS_muon2017_bin8_Fake
505 prop_binCRWS_muon2016M_bin16_Fake
506 prop_binCRWS_muon2017_bin14
507 prop_binCRF_electron2017_bin13_WrongSign
508 prop_binCRTT_electron2018_bin3
509 prop_binCRF_electron2017_bin7_TTTo2L2Nu
510 prop_binCRF_muon2018_bin20_TVX
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0



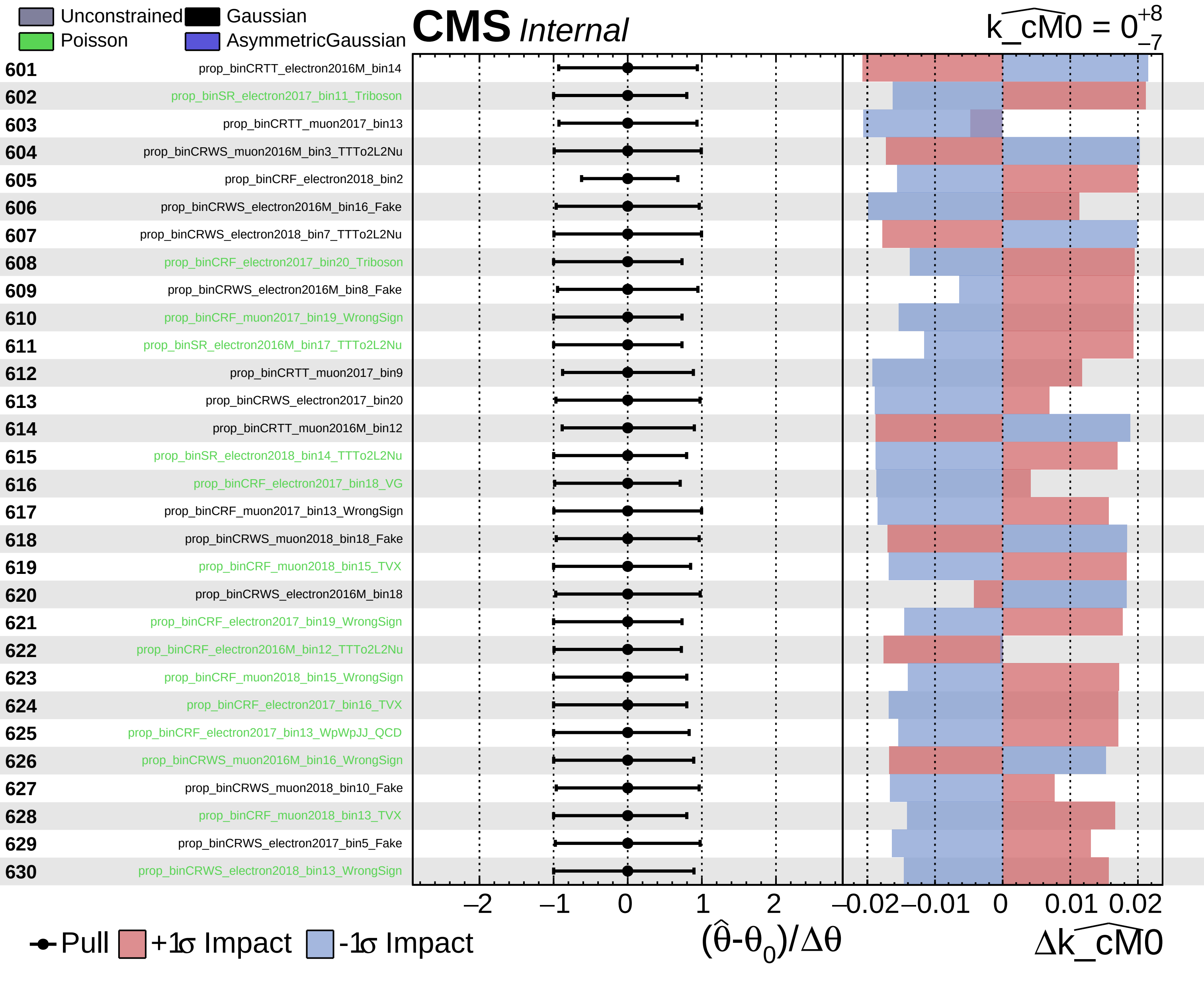
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0 +8 −7
511 prop_binCRTT_electron2016M_bin9
512 prop_binCRTT_muon2016M_bin14
513 prop_binCRF_electron2016M_bin9
514 prop_binCRTT_muon2017_bin14
515 prop_binSR_muon2016M_bin17_WrongSign
516 prop_binCRTT_electron2016M_bin20
517 prop_binCRTT_muon2016M_bin9
518 prop_binCRWS_muon2017_bin8_TTTo2L2Nu
519 prop_binSR_muon2017_bin0
520 prop_binCRF_electron2017_bin5
521 prop_binCRWS_muon2017_bin11
522 prop_binCRTT_muon2016M_bin5
523 prop_binSR_electron2016M_bin16_Fake
524 prop_binCRTT_electron2018_bin6
525 prop_binCRWS_electron2018_bin17_Fake
526 prop_binCRTT_electron2016M_bin16
527 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
528 prop_binSR_electron2017_bin11_WpWpJJ_QCD
529 prop_binCRWS_muon2017_bin16
530 prop_binCRTT_electron2017_bin10
531 prop_binCRTT_electron2018_bin9
532 prop_binCRTT_muon2018_bin7
533 prop_binCRWS_electron2017_bin15
534 prop_binCRWS_electron2018_bin8
535 prop_binSR_muon2017_bin14_Fake
536 prop_binCRF_muon2018_bin19_WrongSign
537 prop_binCRTT_muon2018_bin20
538 prop_binCRF_muon2016M_bin16_TVX
539 prop_binSR_muon2017_bin12_TTTo2L2Nu
540 prop_binCRWS_muon2016M_bin18_DYJetsToLL_FxFx
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

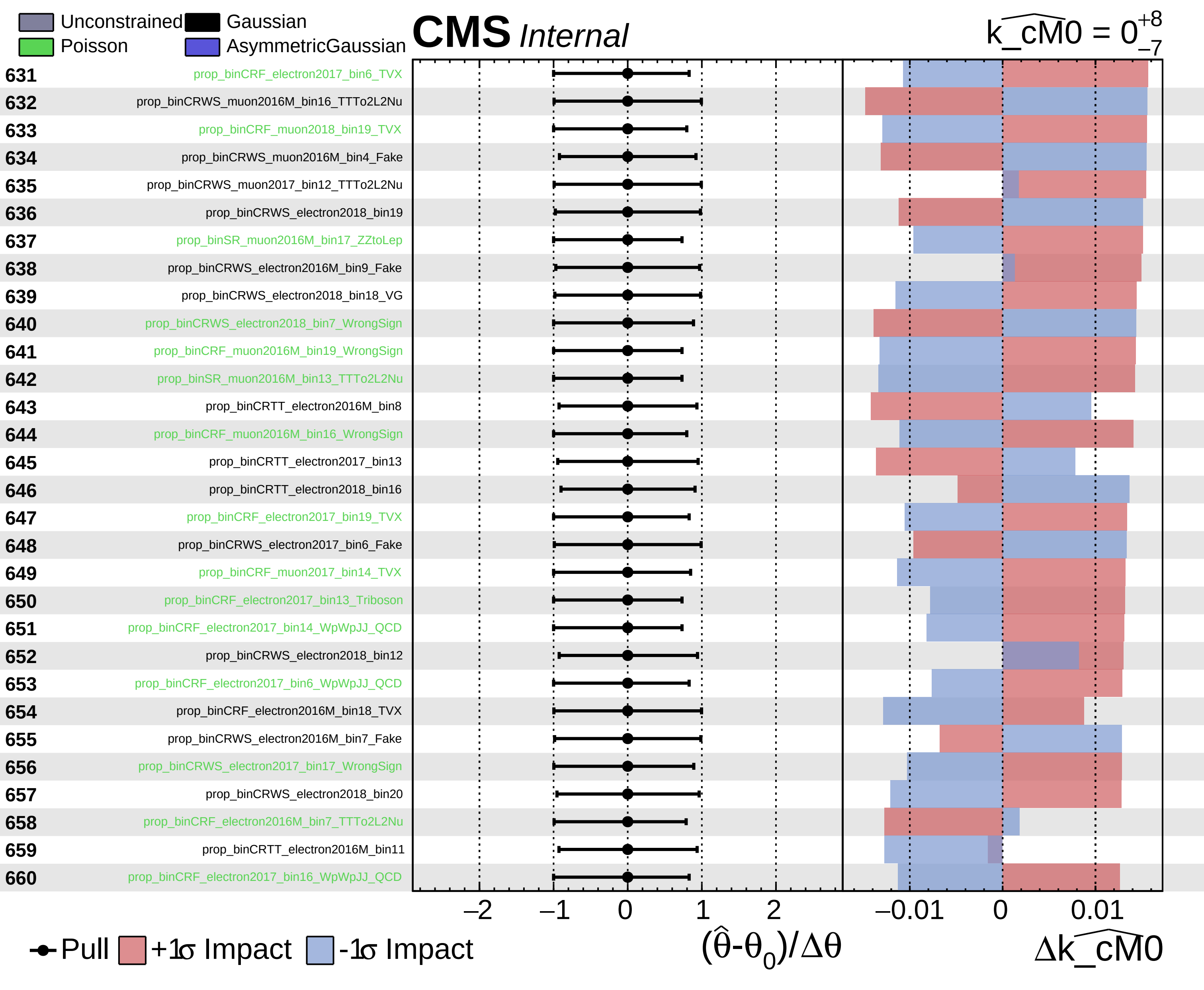


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
541 prop_binCRWS_electron2017_bin18
542 prop_binCRWS_electron2017_bin11
543 prop_binCRWS_muon2016M_bin7_Fake
544 prop_binCRF_electron2016M_bin17_Fake
545 prop_binSR_electron2018_bin16_TTTo2L2Nu
546 prop_binSR_electron2017_bin11_WrongSign
547 prop_binCRTT_muon2016M_bin19
548 prop_binCRTT_muon2018_bin16
549 prop_binCRTT_electron2016M_bin10
550 prop_binCRTT_electron2017_bin8
551 prop_binCRWS_muon2018_bin13_Fake
552 prop_binCRF_electron2017_bin14_TVX
553 prop_binCRTT_electron2016M_bin7
554 prop_binCRTT_electron2018_bin11
555 prop_binCRTT_muon2017_bin15
556 lep_2017
557 prop_binCRTT_electron2018_bin17
558 prop_binCRWS_electron2016M_bin19
559 prop_binCRF_electron2017_bin7_ZZtoLep
560 prop_binCRWS_electron2018_bin15
561 prop_binSR_electron2016M_bin17_WrongSign
562 prop_binCRF_muon2017_bin18_TVX
563 prop_binCRTT_muon2017_bin7
564 prop_binCRF_electron2018_bin1
565 prop_binCRTT_muon2016M_bin8
566 prop_binCRWS_muon2018_bin11_Fake
567 prop_binSR_muon2016M_bin19_Fake
568 prop_binCRF_muon2016M_bin6_TVX
569 prop_binSR_electron2016M_bin17_WZ
570 prop_binCRWS_electron2017_bin14_Fake
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$

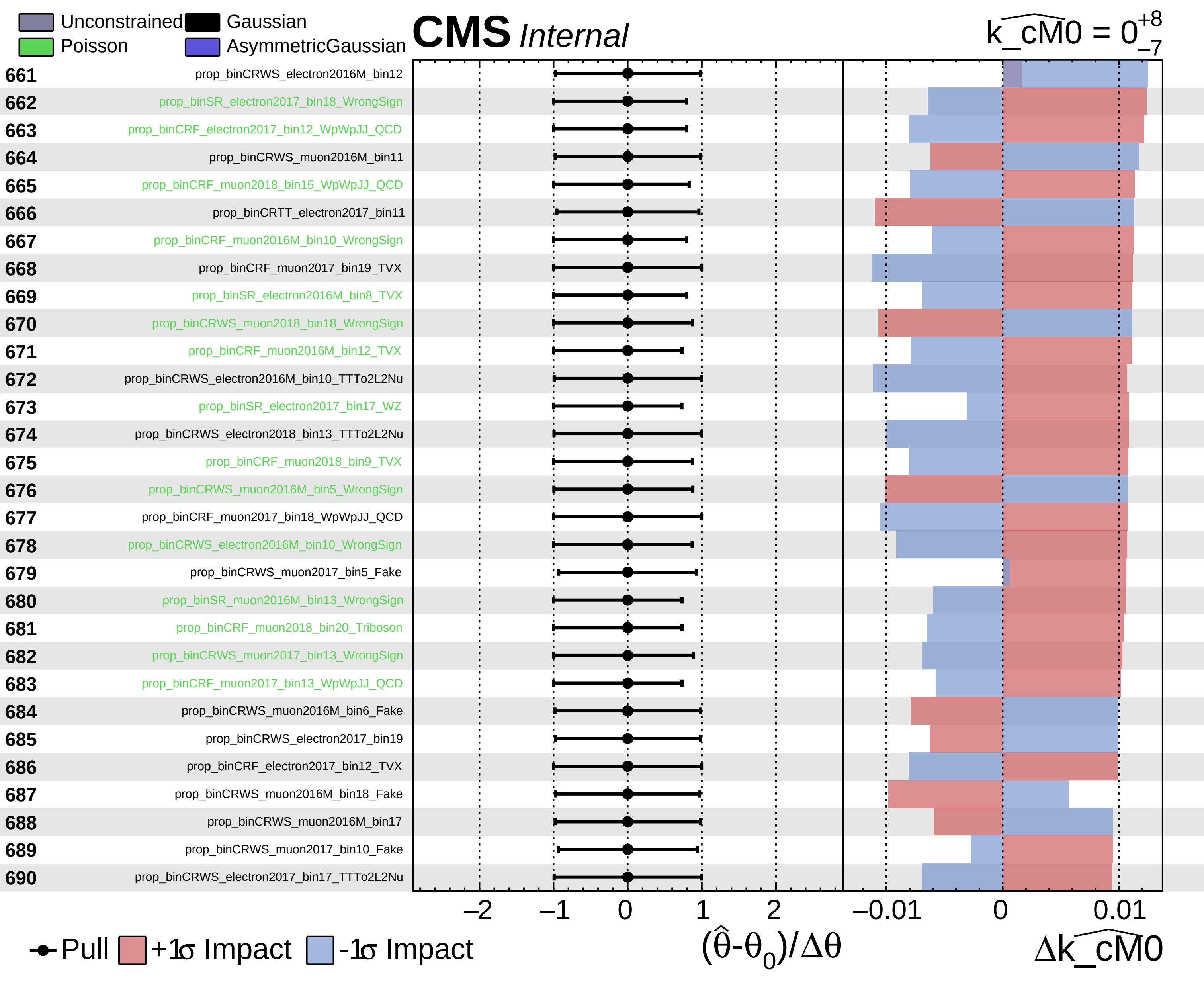


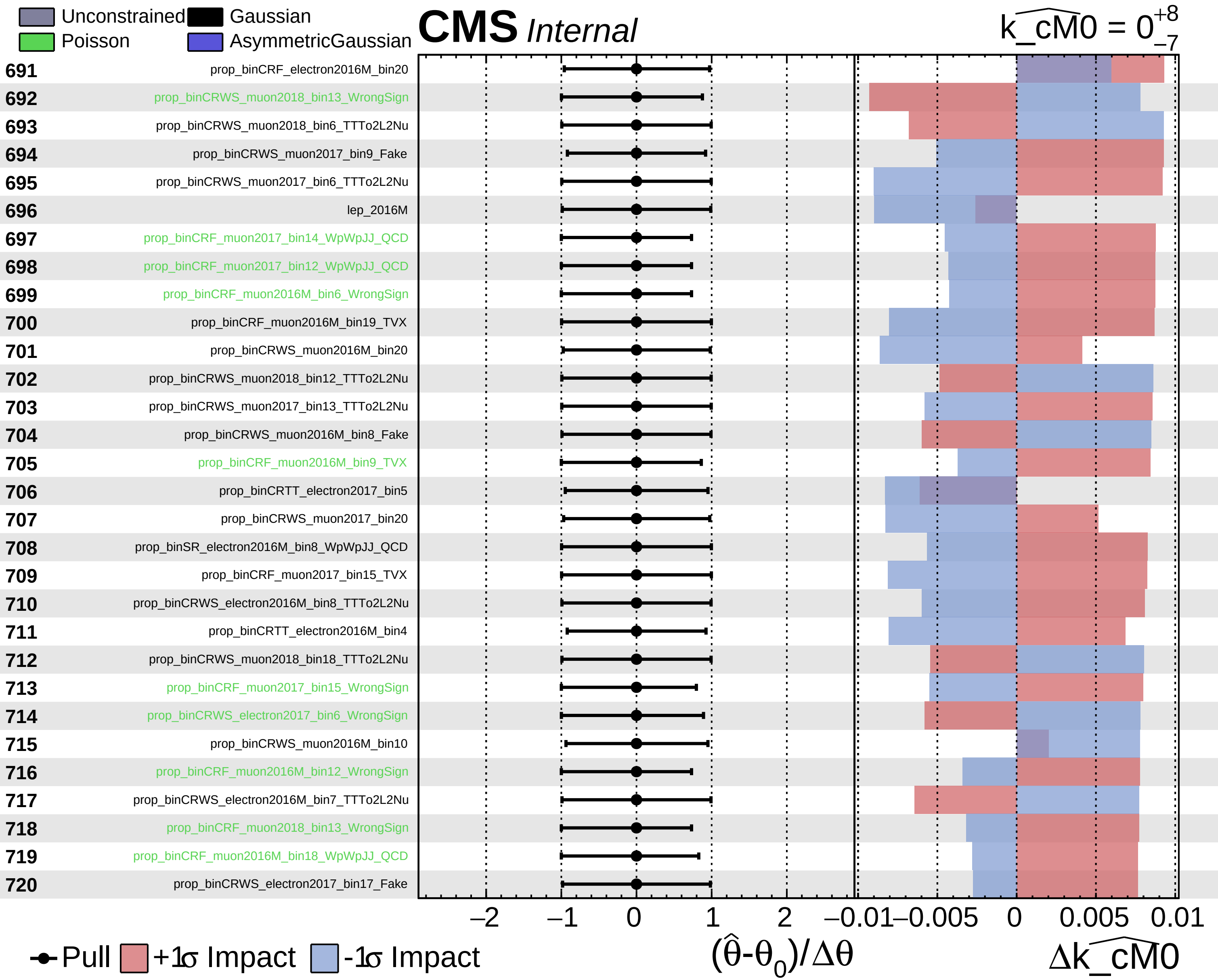
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
571 prop_binCRWS_muon2017_bin18
572 prop_binCRTT_electron2017_bin16
573 prop_binCRWS_electron2018_bin13_Fake
574 prop_binCRWS_muon2017_bin12_WrongSign
575 prop_binCRF_electron2017_bin16_WrongSign
576 prop_binCRWS_electron2018_bin6
577 prop_binCRWS_muon2018_bin12_WrongSign
578 prop_binCRF_electron2017_bin20_WrongSign
579 prop_binCRF_muon2017_bin14_WrongSign
580 prop_binCRTT_electron2017_bin14
581 prop_binSR_electron2018_bin20_ZZtoLep
582 prop_binCRTT_electron2016M_bin13
583 prop_binCRWS_electron2016M_bin20_TTTo2L2Nu
584 prop_binCRF_electron2017_bin6_WrongSign
585 prop_binCRWS_electron2017_bin13_WrongSign
586 prop_binCRWS_muon2016M_bin14
587 prop_binCRTT_electron2017_bin7
588 prop_binCRF_muon2017_bin18_WrongSign
589 prop_binCRTT_electron2016M_bin2
590 prop_binCRTT_electron2017_bin9
591 prop_binCRF_muon2018_bin20_WpWpJJ_QCD
592 prop_binCRF_muon2018_bin9_ZZtoLep
593 prop_binCRWS_muon2016M_bin3_WrongSign
594 prop_binSR_electron2017_bin18_TTTo2L2Nu
595 prop_binCRF_muon2016M_bin10_TVX
596 prop_binSR_electron2018_bin11_ZZtoLep
597 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
598 prop_binCRWS_muon2018_bin6_Fake
599 prop_binCRTT_muon2017_bin17
600 prop_binCRWS_muon2017_bin13_Fake
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$

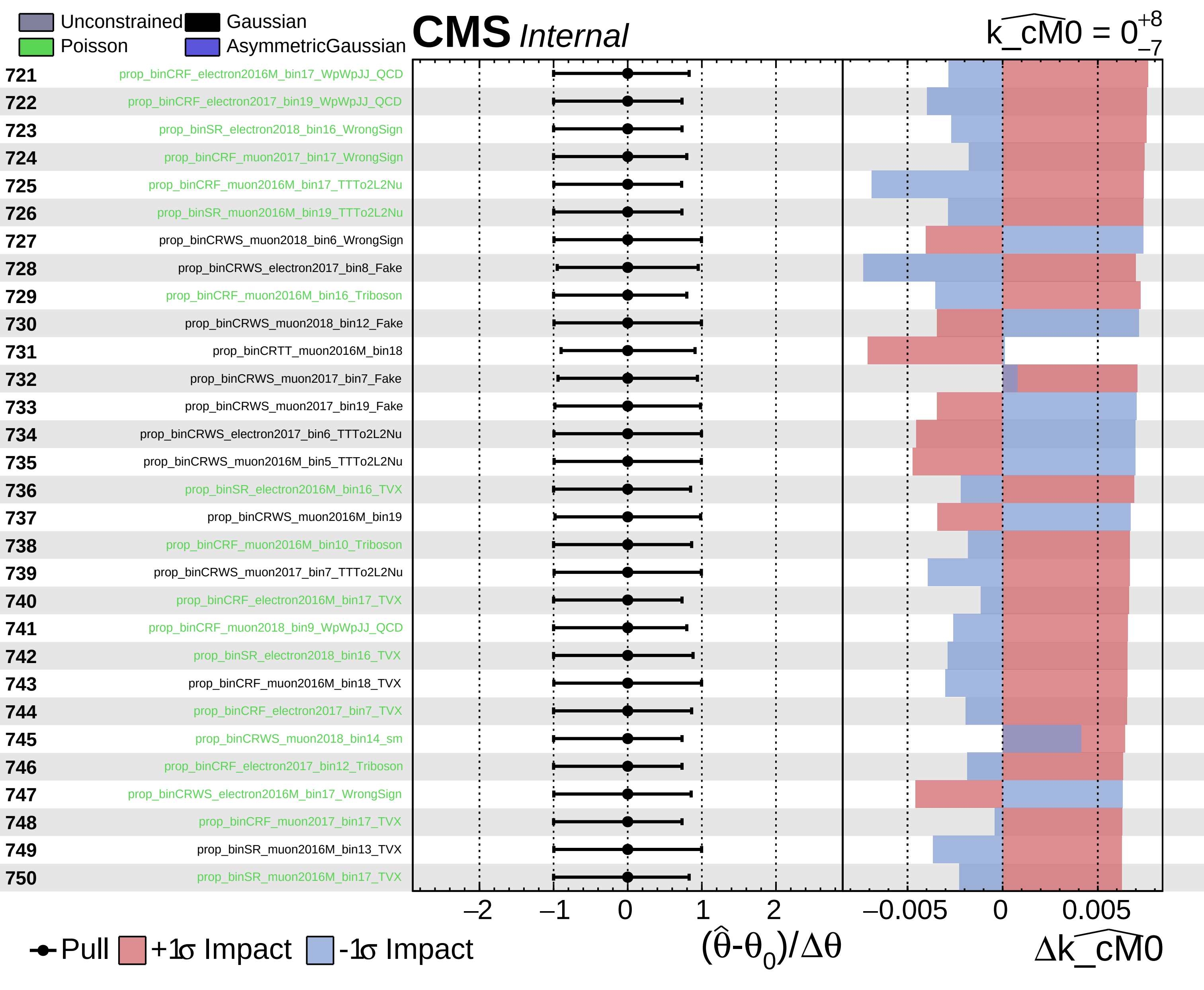

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = $0^{+8}_{-7}$
601 prop_binCRTT_electron2016M_bin14
602 prop_binSR_electron2017_bin11_Triboson
603 prop_binCRTT_muon2017_bin13
604 prop_binCRWS_muon2016M_bin3_TTTo2L2Nu
605 prop_binCRF_electron2018_bin2
606 prop_binCRWS_electron2016M_bin16_Fake
607 prop_binCRWS_electron2018_bin7_TTTo2L2Nu
608 prop_binCRF_electron2017_bin20_Triboson
609 prop_binCRWS_electron2016M_bin8_Fake
610 prop_binCRF_muon2017_bin19_WrongSign
611 prop_binSR_electron2016M_bin17_TTTo2L2Nu
612 prop_binCRTT_muon2017_bin9
613 prop_binCRWS_electron2017_bin20
614 prop_binCRTT_muon2016M_bin12
615 prop_binSR_electron2018_bin14_TTTo2L2Nu
616 prop_binCRF_electron2017_bin18_VG
617 prop_binCRF_muon2017_bin13_WrongSign
618 prop_binCRWS_muon2018_bin18_Fake
619 prop_binCRF_muon2018_bin15_TVX
620 prop_binCRWS_electron2016M_bin18
621 prop_binCRF_electron2017_bin19_WrongSign
622 prop_binCRF_electron2016M_bin12_TTTo2L2Nu
623 prop_binCRF_muon2018_bin15_WrongSign
624 prop_binCRF_electron2017_bin16_TVX
625 prop_binCRF_electron2017_bin13_WpWpJJ_QCD
626 prop_binCRWS_muon2016M_bin16_WrongSign
627 prop_binCRWS_muon2018_bin10_Fake
628 prop_binCRF_muon2018_bin13_TVX
629 prop_binCRWS_electron2017_bin5_Fake
630 prop_binCRWS_electron2018_bin13_WrongSign
−2
−1
0
1
2
−0.02
−0.01
0
0.01
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM0



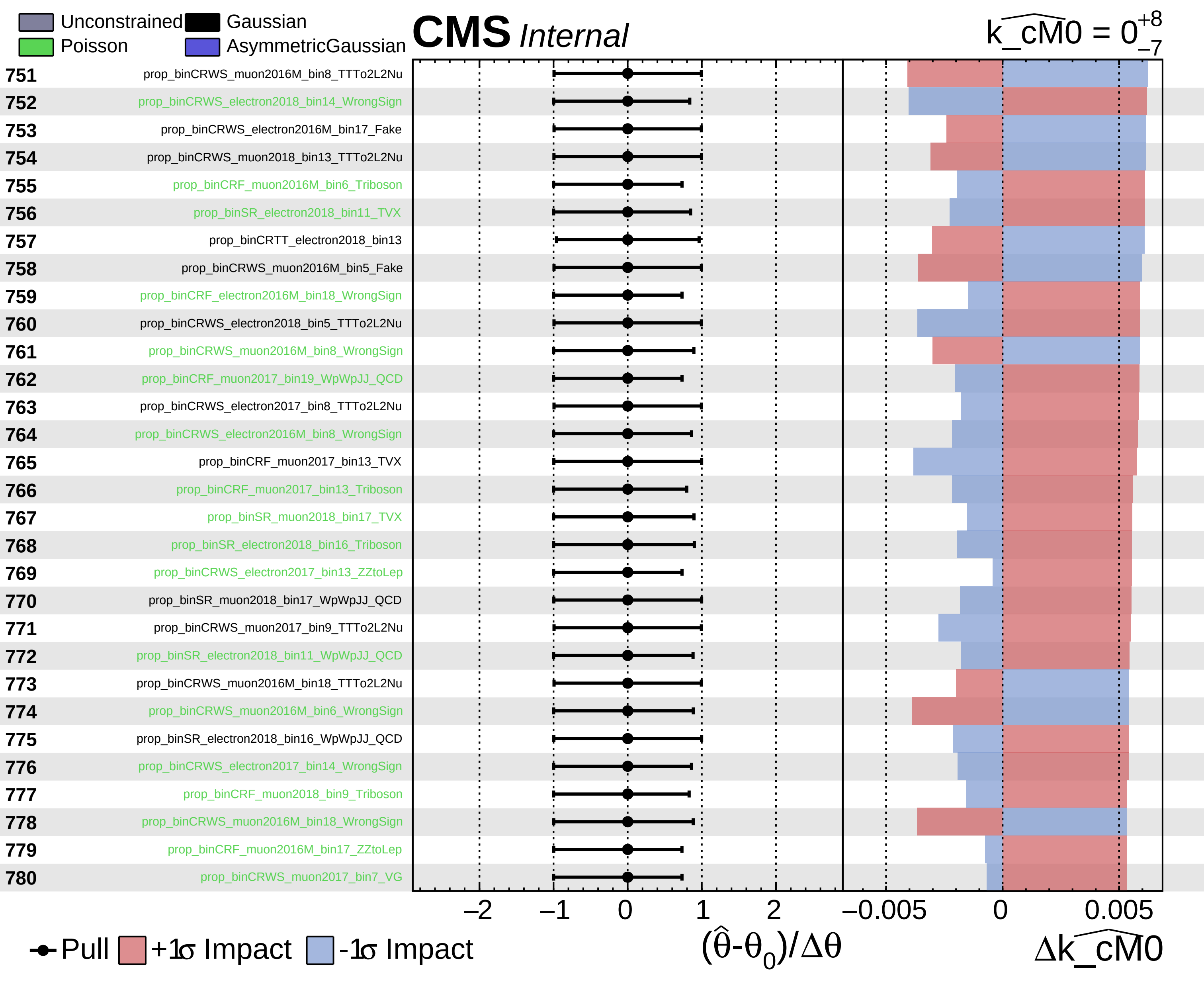
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0 +8 −7
631 prop_binCRF_electron2017_bin6_TVX
632 prop_binCRWS_muon2016M_bin16_TTTo2L2Nu
633 prop_binCRF_muon2018_bin19_TVX
634 prop_binCRWS_muon2016M_bin4_Fake
635 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
636 prop_binCRWS_electron2018_bin19
637 prop_binSR_muon2016M_bin17_ZZtoLep
638 prop_binCRWS_electron2016M_bin9_Fake
639 prop_binCRWS_electron2018_bin18_VG
640 prop_binCRWS_electron2018_bin7_WrongSign
641 prop_binCRF_muon2016M_bin19_WrongSign
642 prop_binSR_muon2016M_bin13_TTTo2L2Nu
643 prop_binCRTT_electron2016M_bin8
644 prop_binCRF_muon2016M_bin16_WrongSign
645 prop_binCRTT_electron2017_bin13
646 prop_binCRTT_electron2018_bin16
647 prop_binCRF_electron2017_bin19_TVX
648 prop_binCRWS_electron2017_bin6_Fake
649 prop_binCRF_muon2017_bin14_TVX
650 prop_binCRF_electron2017_bin13_Triboson
651 prop_binCRF_electron2017_bin14_WpWpJJ_QCD
652 prop_binCRWS_electron2018_bin12
653 prop_binCRF_electron2017_bin6_WpWpJJ_QCD
654 prop_binCRF_electron2016M_bin18_TVX
655 prop_binCRWS_electron2016M_bin7_Fake
656 prop_binCRWS_electron2017_bin17_WrongSign
657 prop_binCRWS_electron2018_bin20
658 prop_binCRF_electron2016M_bin7_TTTo2L2Nu
659 prop_binCRTT_electron2016M_bin11
660 prop_binCRF_electron2017_bin16_WpWpJJ_QCD
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM0

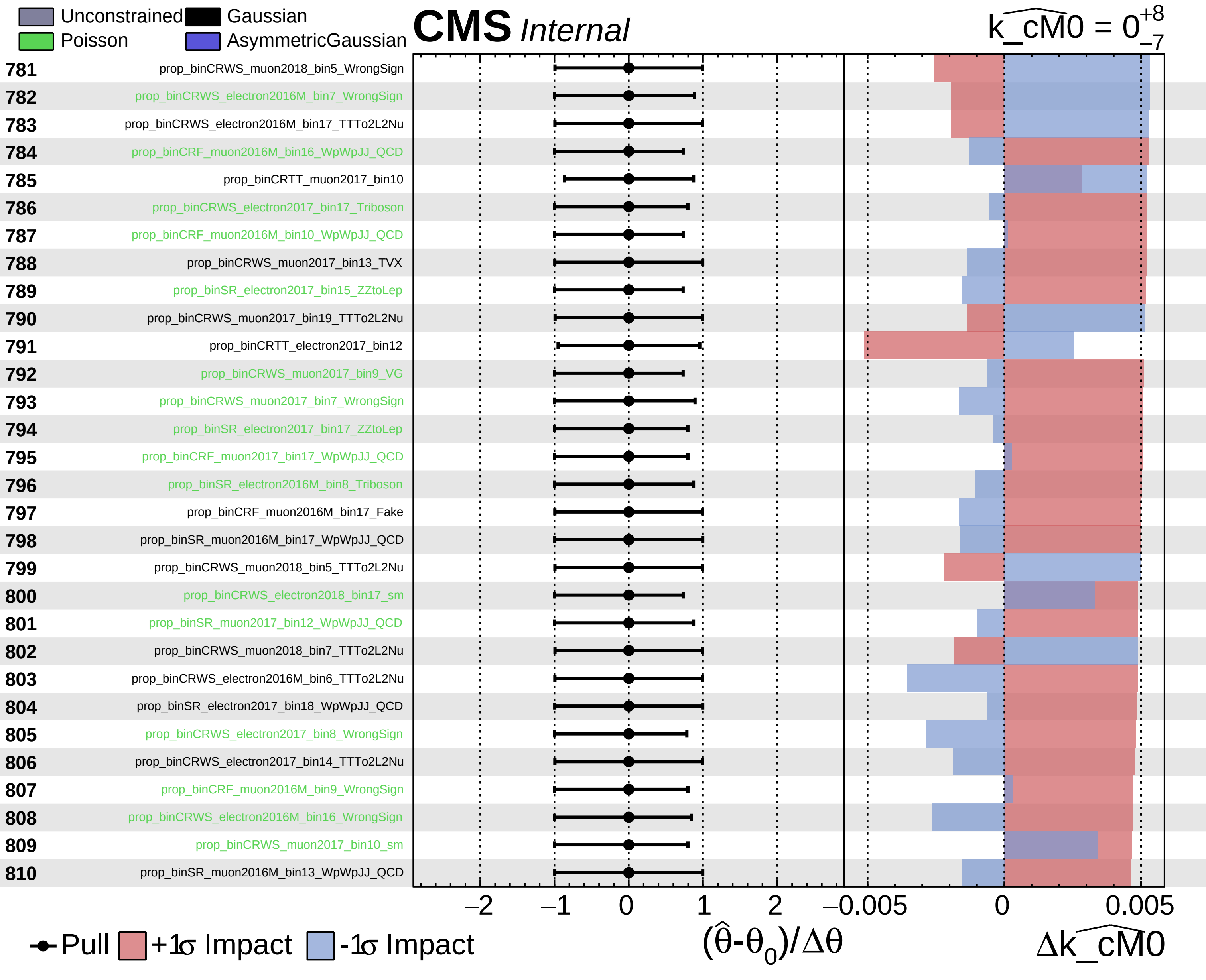

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0+8 −7
661 prop_binCRWS_electron2016M_bin12
662 prop_binSR_electron2017_bin18_WrongSign
663 prop_binCRF_electron2017_bin12_WpWpJJ_QCD
664 prop_binCRWS_muon2016M_bin11
665 prop_binCRF_muon2018_bin15_WpWpJJ_QCD
666 prop_binCRTT_electron2017_bin11
667 prop_binCRF_muon2016M_bin10_WrongSign
668 prop_binCRF_muon2017_bin19_TVX
669 prop_binSR_electron2016M_bin8_TVX
670 prop_binCRWS_muon2018_bin18_WrongSign
671 prop_binCRF_muon2016M_bin12_TVX
672 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
673 prop_binSR_electron2017_bin17_WZ
674 prop_binCRWS_electron2018_bin13_TTTo2L2Nu
675 prop_binCRF_muon2018_bin9_TVX
676 prop_binCRWS_muon2016M_bin5_WrongSign
677 prop_binCRF_muon2017_bin18_WpWpJJ_QCD
678 prop_binCRWS_electron2016M_bin10_WrongSign
679 prop_binCRWS_muon2017_bin5_Fake
680 prop_binSR_muon2016M_bin13_WrongSign
681 prop_binCRF_muon2018_bin20_Triboson
682 prop_binCRWS_muon2017_bin13_WrongSign
683 prop_binCRF_muon2017_bin13_WpWpJJ_QCD
684 prop_binCRWS_muon2016M_bin6_Fake
685 prop_binCRWS_electron2017_bin19
686 prop_binCRF_electron2017_bin12_TVX
687 prop_binCRWS_muon2016M_bin18_Fake
688 prop_binCRWS_muon2016M_bin17
689 prop_binCRWS_muon2017_bin10_Fake
690 prop_binCRWS_electron2017_bin17_TTTo2L2Nu
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0^{+8}_{−7}
691 prop_binCRF_electron2016M_bin20
692 prop_binCRWS_muon2018_bin13_WrongSign
693 prop_binCRWS_muon2018_bin6_TTTo2L2Nu
694 prop_binCRWS_muon2017_bin9_Fake
695 prop_binCRWS_muon2017_bin6_TTTo2L2Nu
696 lep_2016M
697 prop_binCRF_muon2017_bin14_WpWpJJ_QCD
698 prop_binCRF_muon2017_bin12_WpWpJJ_QCD
699 prop_binCRF_muon2016M_bin6_WrongSign
700 prop_binCRF_muon2016M_bin19_TVX
701 prop_binCRWS_muon2016M_bin20
702 prop_binCRWS_muon2018_bin12_TTTo2L2Nu
703 prop_binCRWS_muon2017_bin13_TTTo2L2Nu
704 prop_binCRWS_muon2016M_bin8_Fake
705 prop_binCRF_muon2016M_bin9_TVX
706 prop_binCRTT_electron2017_bin5
707 prop_binCRWS_muon2017_bin20
708 prop_binSR_electron2016M_bin8_WpWpJJ_QCD
709 prop_binCRF_muon2017_bin15_TVX
710 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
711 prop_binCRTT_electron2016M_bin4
712 prop_binCRWS_muon2018_bin18_TTTo2L2Nu
713 prop_binCRF_muon2017_bin15_WrongSign
714 prop_binCRWS_electron2017_bin6_WrongSign
715 prop_binCRWS_muon2016M_bin10
716 prop_binCRF_muon2016M_bin12_WrongSign
717 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
718 prop_binCRF_muon2018_bin13_WrongSign
719 prop_binCRF_muon2016M_bin18_WpWpJJ_QCD
720 prop_binCRWS_electron2017_bin17_Fake
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ_0)/Δθ
Δk_cM0

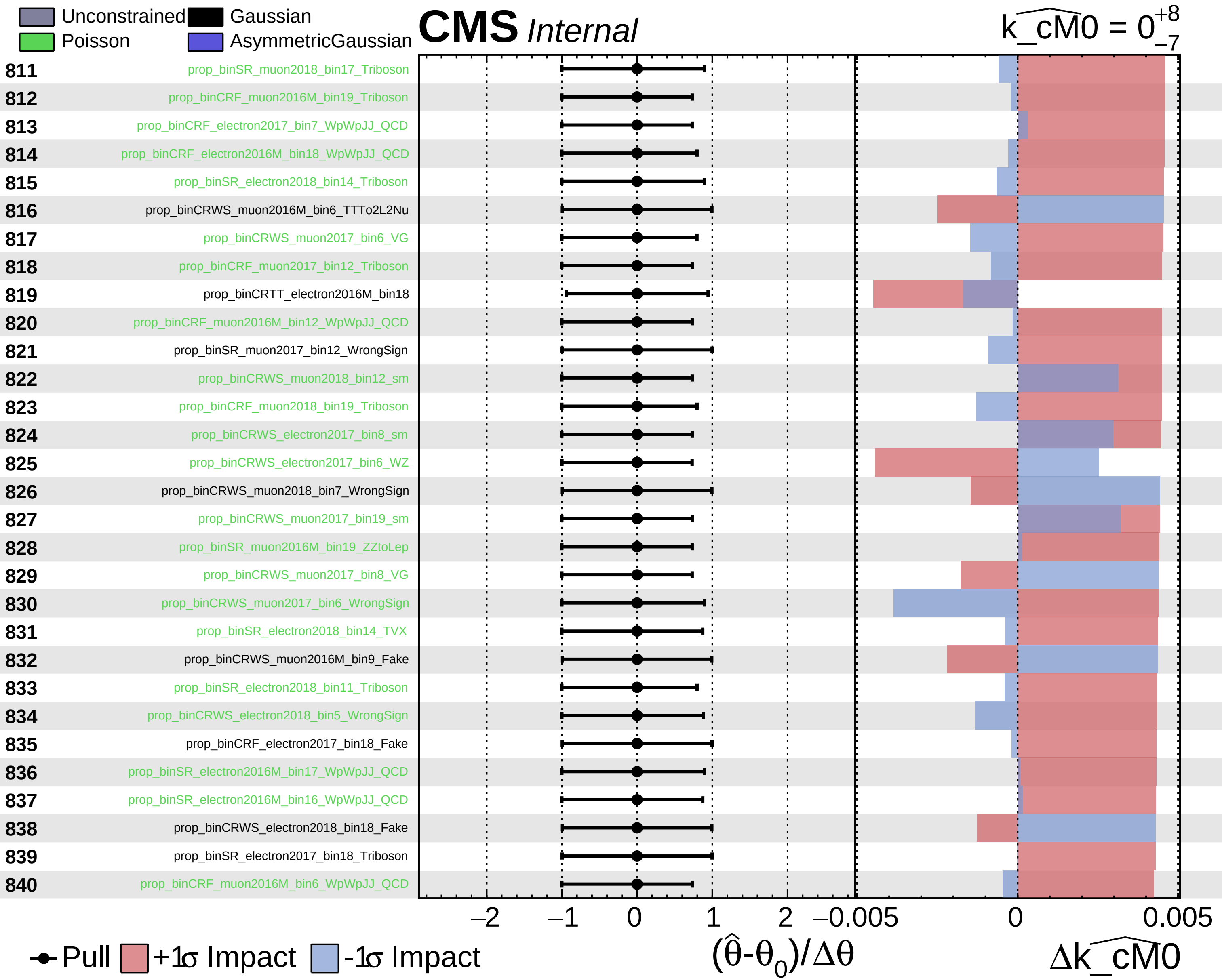


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0$^{+8}_{-7}$
721 prop_binCRF_electron2016M_bin17_WpWpJJ_QCD
722 prop_binCRF_electron2017_bin19_WpWpJJ_QCD
723 prop_binSR_electron2018_bin16_WrongSign
724 prop_binCRF_muon2017_bin17_WrongSign
725 prop_binCRF_muon2016M_bin17_TTTo2L2Nu
726 prop_binSR_muon2016M_bin19_TTTo2L2Nu
727 prop_binCRWS_muon2018_bin6_WrongSign
728 prop_binCRWS_electron2017_bin8_Fake
729 prop_binCRF_muon2016M_bin16_Triboson
730 prop_binCRWS_muon2018_bin12_Fake
731 prop_binCRTT_muon2016M_bin18
732 prop_binCRWS_muon2017_bin7_Fake
733 prop_binCRWS_muon2017_bin19_Fake
734 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
735 prop_binCRWS_muon2016M_bin5_TTTo2L2Nu
736 prop_binSR_electron2016M_bin16_TVX
737 prop_binCRWS_muon2016M_bin19
738 prop_binCRF_muon2016M_bin10_Triboson
739 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
740 prop_binCRF_electron2016M_bin17_TVX
741 prop_binCRF_muon2018_bin9_WpWpJJ_QCD
742 prop_binSR_electron2018_bin16_TVX
743 prop_binCRF_muon2016M_bin18_TVX
744 prop_binCRF_electron2017_bin7_TVX
745 prop_binCRWS_muon2018_bin14_sm
746 prop_binCRF_electron2017_bin12_Triboson
747 prop_binCRWS_electron2016M_bin17_WrongSign
748 prop_binCRF_muon2017_bin17_TVX
749 prop_binSR_muon2016M_bin13_TVX
750 prop_binSR_muon2016M_bin17_TVX
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM0

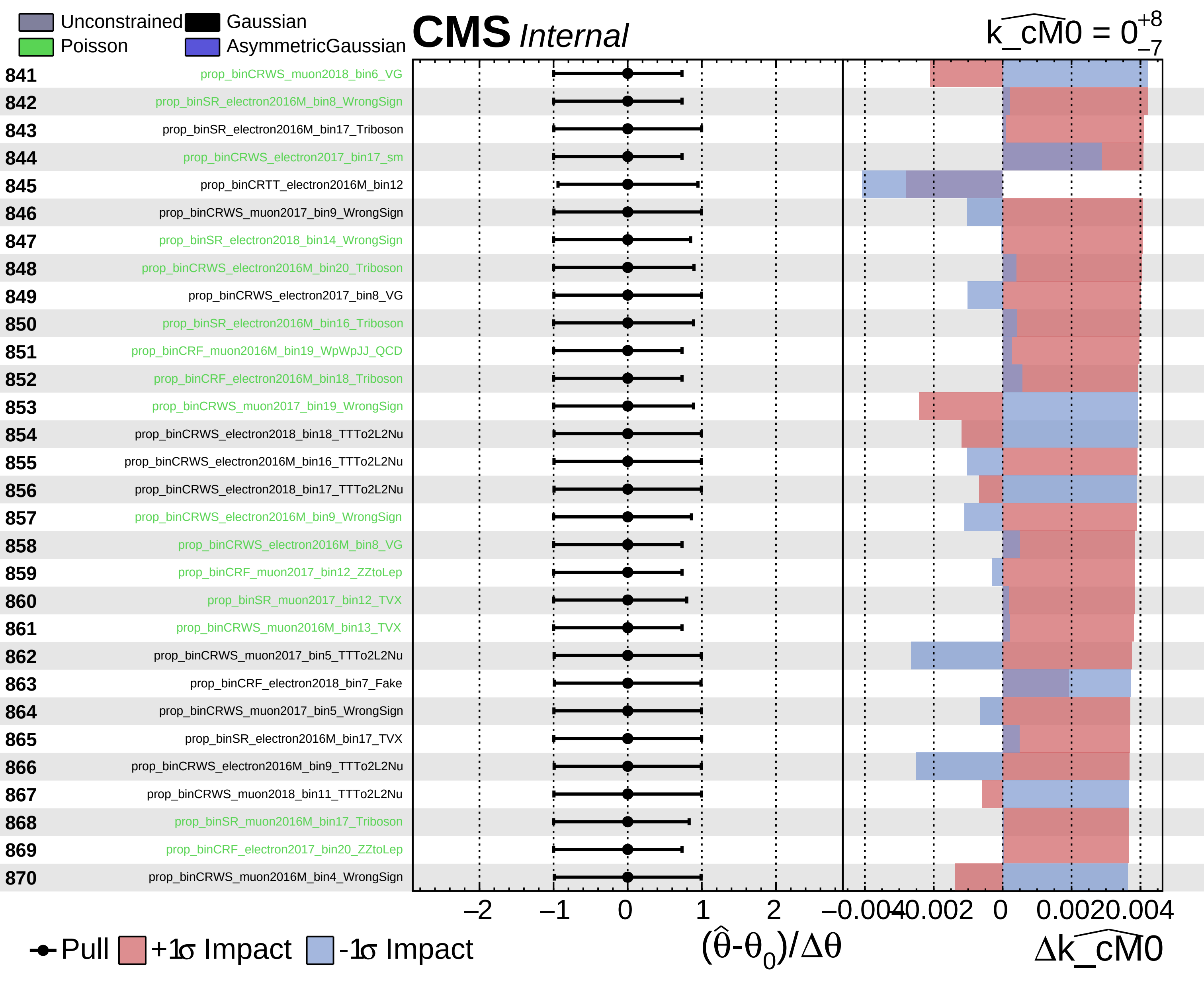

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
751 prop_binCRWS_muon2016M_bin8_TTTo2L2Nu
752 prop_binCRWS_electron2018_bin14_WrongSign
753 prop_binCRWS_electron2016M_bin17_Fake
754 prop_binCRWS_muon2018_bin13_TTTo2L2Nu
755 prop_binCRF_muon2016M_bin6_Triboson
756 prop_binSR_electron2018_bin11_TVX
757 prop_binCRTT_electron2018_bin13
758 prop_binCRWS_muon2016M_bin5_Fake
759 prop_binCRF_electron2016M_bin18_WrongSign
760 prop_binCRWS_electron2018_bin5_TTTo2L2Nu
761 prop_binCRWS_muon2016M_bin8_WrongSign
762 prop_binCRF_muon2017_bin19_WpWpJJ_QCD
763 prop_binCRWS_electron2017_bin8_TTTo2L2Nu
764 prop_binCRWS_electron2016M_bin8_WrongSign
765 prop_binCRF_muon2017_bin13_TVX
766 prop_binCRF_muon2017_bin13_Triboson
767 prop_binSR_muon2018_bin17_TVX
768 prop_binSR_electron2018_bin16_Triboson
769 prop_binCRWS_electron2017_bin13_ZZtoLep
770 prop_binSR_muon2018_bin17_WpWpJJ_QCD
771 prop_binCRWS_muon2017_bin9_TTTo2L2Nu
772 prop_binSR_electron2018_bin11_WpWpJJ_QCD
773 prop_binCRWS_muon2016M_bin18_TTTo2L2Nu
774 prop_binCRWS_muon2016M_bin6_WrongSign
775 prop_binSR_electron2018_bin16_WpWpJJ_QCD
776 prop_binCRWS_electron2017_bin14_WrongSign
777 prop_binCRF_muon2018_bin9_Triboson
778 prop_binCRWS_muon2016M_bin18_WrongSign
779 prop_binCRF_muon2016M_bin17_ZZtoLep
780 prop_binCRWS_muon2017_bin7_VG
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



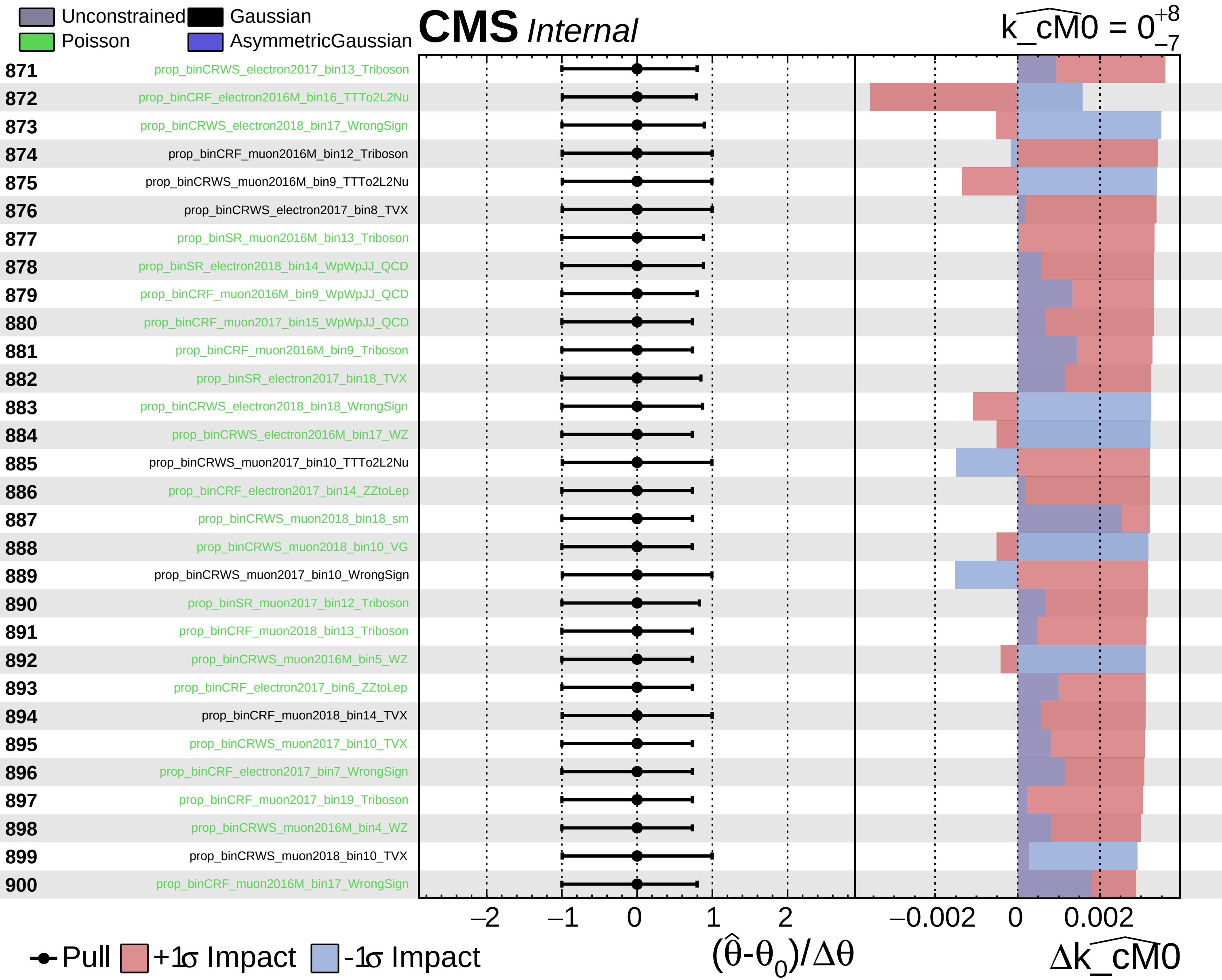
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
781 prop_binCRWS_muon2018_bin5_WrongSign
782 prop_binCRWS_electron2016M_bin7_WrongSign
783 prop_binCRWS_electron2016M_bin17_TTTo2L2Nu
784 prop_binCRF_muon2016M_bin16_WpWpJJ_QCD
785 prop_binCRTT_muon2017_bin10
786 prop_binCRWS_electron2017_bin17_Triboson
787 prop_binCRF_muon2016M_bin10_WpWpJJ_QCD
788 prop_binCRWS_muon2017_bin13_TVX
789 prop_binSR_electron2017_bin15_ZZtoLep
790 prop_binCRWS_muon2017_bin19_TTTo2L2Nu
791 prop_binCRTT_electron2017_bin12
792 prop_binCRWS_muon2017_bin9_VG
793 prop_binCRWS_muon2017_bin7_WrongSign
794 prop_binSR_electron2017_bin17_ZZtoLep
795 prop_binCRF_muon2017_bin17_WpWpJJ_QCD
796 prop_binSR_electron2016M_bin8_Triboson
797 prop_binCRF_muon2016M_bin17_Fake
798 prop_binSR_muon2016M_bin17_WpWpJJ_QCD
799 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
800 prop_binCRWS_electron2018_bin17_sm
801 prop_binSR_muon2017_bin12_WpWpJJ_QCD
802 prop_binCRWS_muon2018_bin7_TTTo2L2Nu
803 prop_binCRWS_electron2016M_bin6_TTTo2L2Nu
804 prop_binSR_electron2017_bin18_WpWpJJ_QCD
805 prop_binCRWS_electron2017_bin8_WrongSign
806 prop_binCRWS_electron2017_bin14_TTTo2L2Nu
807 prop_binCRF_muon2016M_bin9_WrongSign
808 prop_binCRWS_electron2016M_bin16_WrongSign
809 prop_binCRWS_muon2017_bin10_sm
810 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



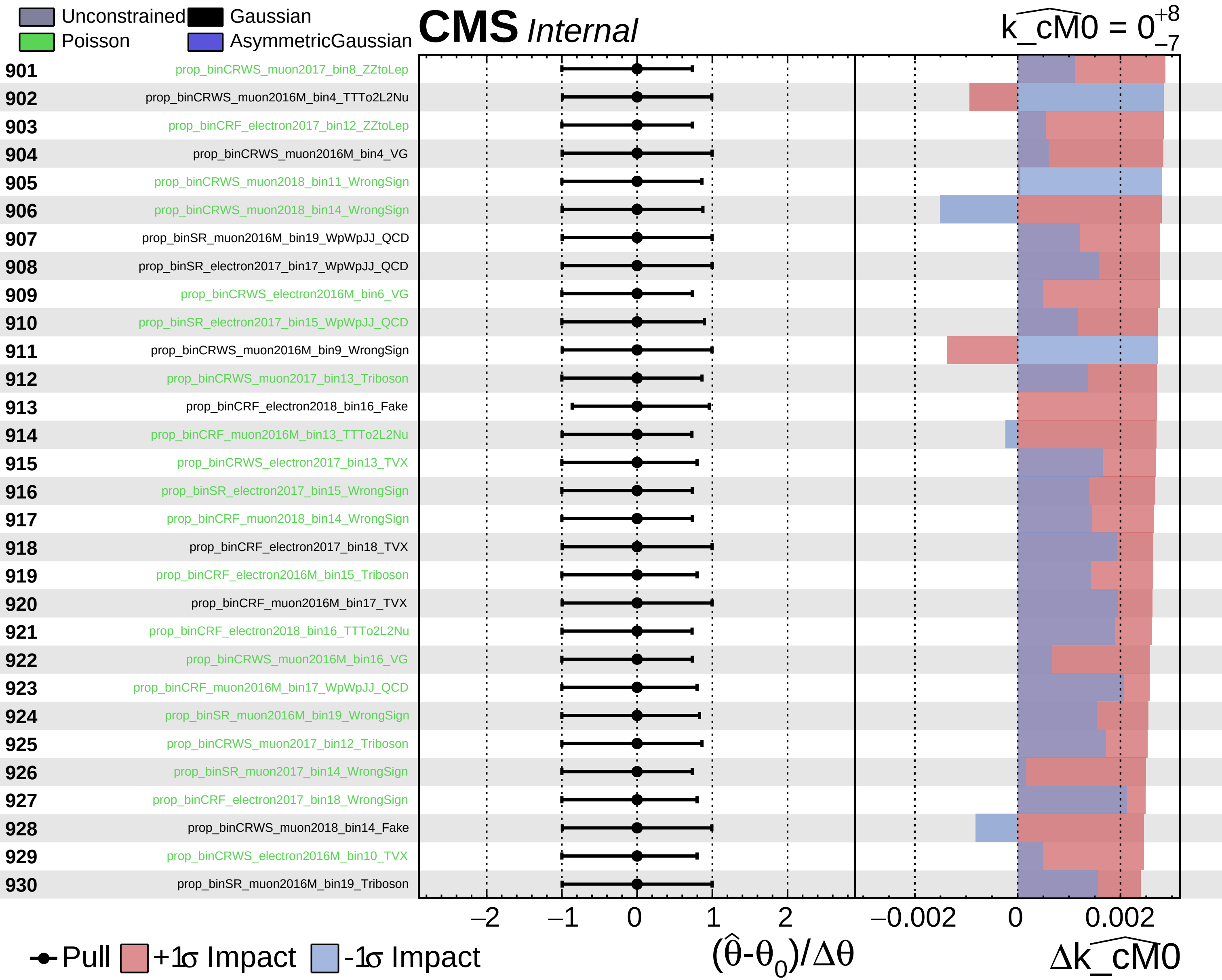
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0^{+8}_{−7}
811 prop_binSR_muon2018_bin17_Triboson
812 prop_binCRF_muon2016M_bin19_Triboson
813 prop_binCRF_electron2017_bin7_WpWpJJ_QCD
814 prop_binCRF_electron2016M_bin18_WpWpJJ_QCD
815 prop_binSR_electron2018_bin14_Triboson
816 prop_binCRWS_muon2016M_bin6_TTTo2L2Nu
817 prop_binCRWS_muon2017_bin6_VG
818 prop_binCRF_muon2017_bin12_Triboson
819 prop_binCRTT_electron2016M_bin18
820 prop_binCRF_muon2016M_bin12_WpWpJJ_QCD
821 prop_binSR_muon2017_bin12_WrongSign
822 prop_binCRWS_muon2018_bin12_sm
823 prop_binCRF_muon2018_bin19_Triboson
824 prop_binCRWS_electron2017_bin8_sm
825 prop_binCRWS_electron2017_bin6_WZ
826 prop_binCRWS_muon2018_bin7_WrongSign
827 prop_binCRWS_muon2017_bin19_sm
828 prop_binSR_muon2016M_bin19_ZZtoLep
829 prop_binCRWS_muon2017_bin8_VG
830 prop_binCRWS_muon2017_bin6_WrongSign
831 prop_binSR_electron2018_bin14_TVX
832 prop_binCRWS_muon2016M_bin9_Fake
833 prop_binSR_electron2018_bin11_Triboson
834 prop_binCRWS_electron2018_bin5_WrongSign
835 prop_binCRF_electron2017_bin18_Fake
836 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
837 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
838 prop_binCRWS_electron2018_bin18_Fake
839 prop_binSR_electron2017_bin18_Triboson
840 prop_binCRF_muon2016M_bin6_WpWpJJ_QCD
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

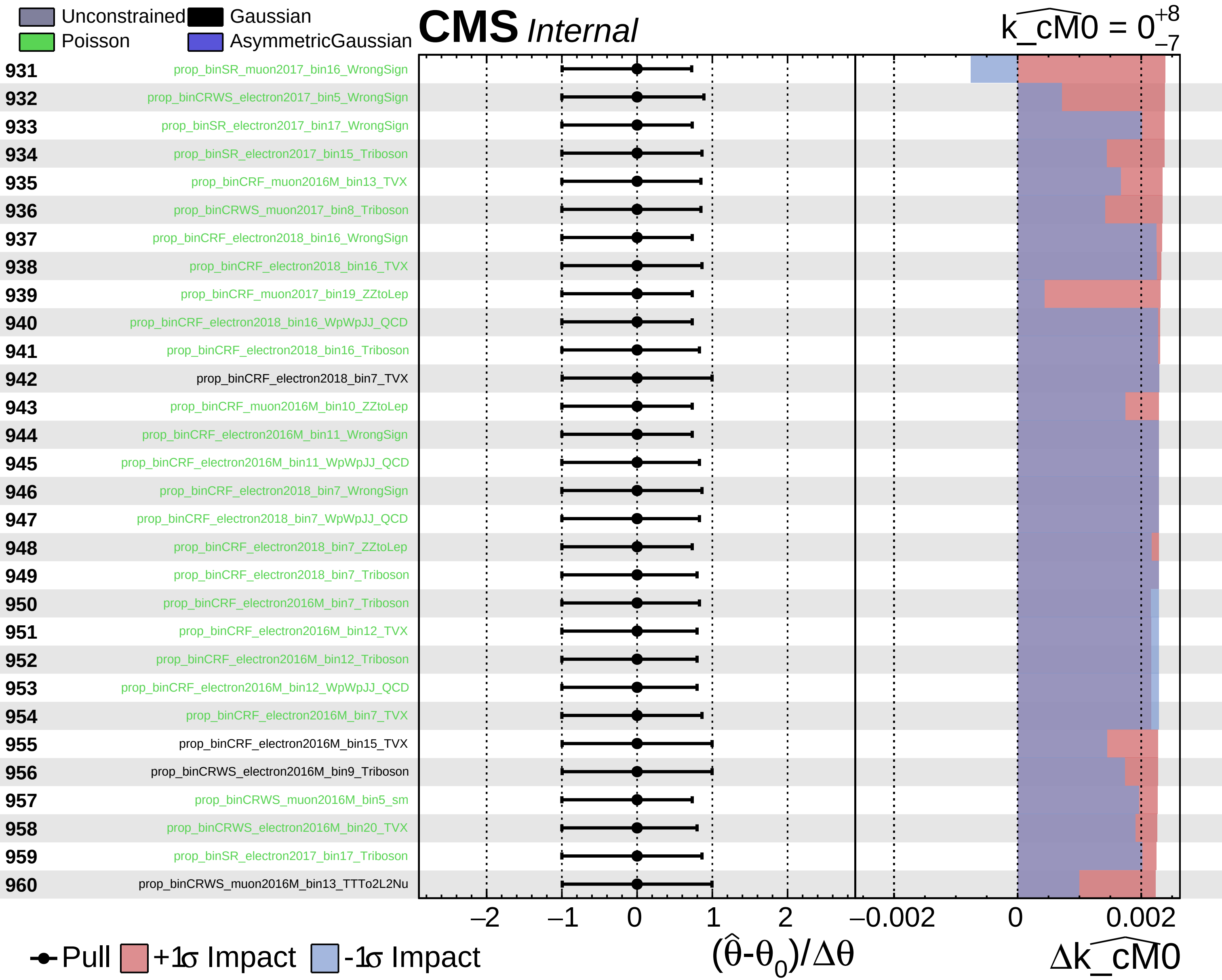


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0+8 −7
841 prop_binCRWS_muon2018_bin6_VG
842 prop_binSR_electron2016M_bin8_WrongSign
843 prop_binSR_electron2016M_bin17_Triboson
844 prop_binCRWS_electron2017_bin17_sm
845 prop_binCRTT_electron2016M_bin12
846 prop_binCRWS_muon2017_bin9_WrongSign
847 prop_binSR_electron2018_bin14_WrongSign
848 prop_binCRWS_electron2016M_bin20_Triboson
849 prop_binCRWS_electron2017_bin8_VG
850 prop_binSR_electron2016M_bin16_Triboson
851 prop_binCRF_muon2016M_bin19_WpWpJJ_QCD
852 prop_binCRF_electron2016M_bin18_Triboson
853 prop_binCRWS_muon2017_bin19_WrongSign
854 prop_binCRWS_electron2018_bin18_TTTo2L2Nu
855 prop_binCRWS_electron2016M_bin16_TTTo2L2Nu
856 prop_binCRWS_electron2018_bin17_TTTo2L2Nu
857 prop_binCRWS_electron2016M_bin9_WrongSign
858 prop_binCRWS_electron2016M_bin8_VG
859 prop_binCRF_muon2017_bin12_ZZtoLep
860 prop_binSR_muon2017_bin12_TVX
861 prop_binCRWS_muon2016M_bin13_TVX
862 prop_binCRWS_muon2017_bin5_TTTo2L2Nu
863 prop_binCRF_electron2018_bin7_Fake
864 prop_binCRWS_muon2017_bin5_WrongSign
865 prop_binSR_electron2016M_bin17_TVX
866 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
867 prop_binCRWS_muon2018_bin11_TTTo2L2Nu
868 prop_binSR_muon2016M_bin17_Triboson
869 prop_binCRF_electron2017_bin20_ZZtoLep
870 prop_binCRWS_muon2016M_bin4_WrongSign
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

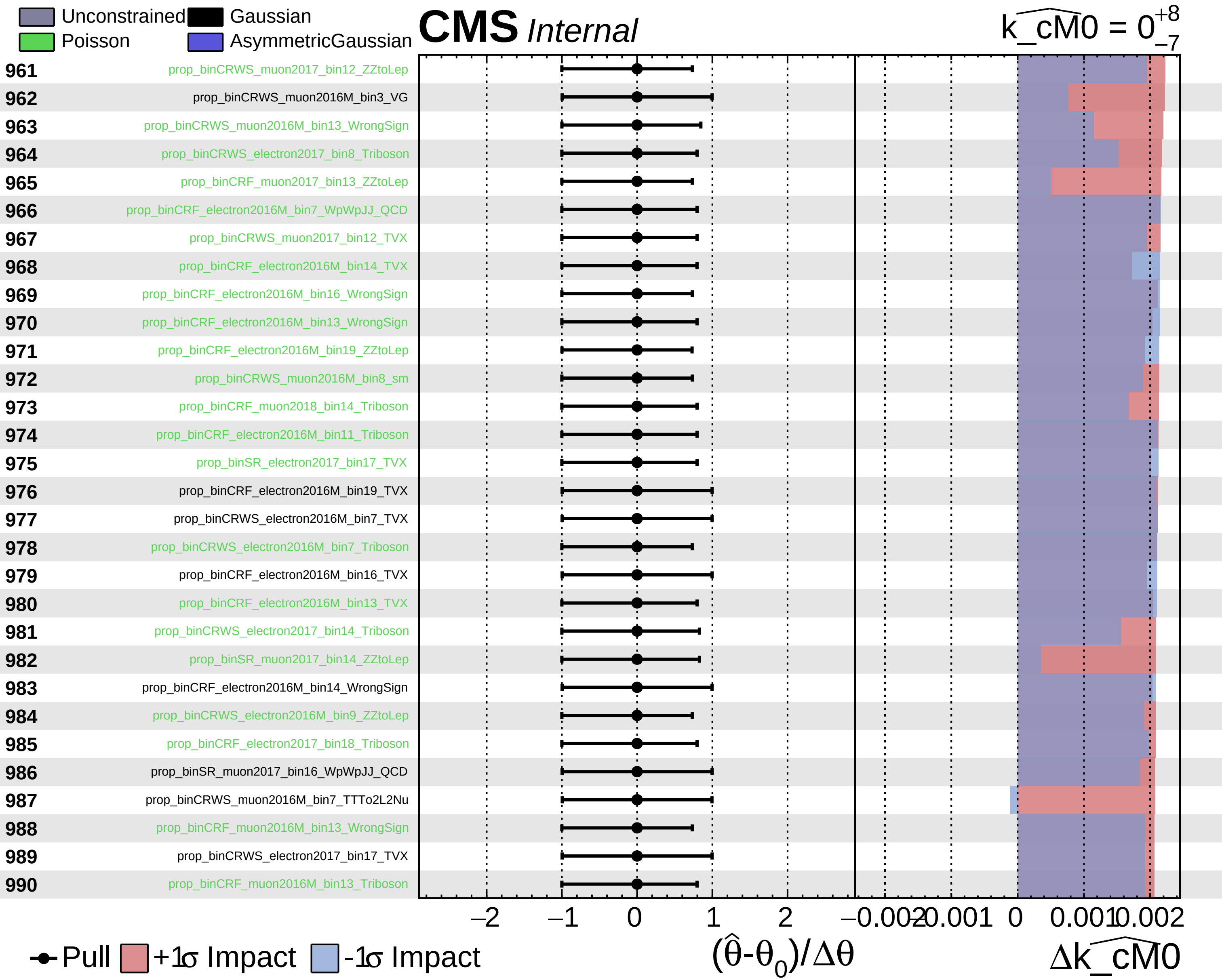


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
871 prop_binCRWS_electron2017_bin13_Triboson
872 prop_binCRF_electron2016M_bin16_TTTo2L2Nu
873 prop_binCRWS_electron2018_bin17_WrongSign
874 prop_binCRF_muon2016M_bin12_Triboson
875 prop_binCRWS_muon2016M_bin9_TTTo2L2Nu
876 prop_binCRWS_electron2017_bin8_TVX
877 prop_binSR_muon2016M_bin13_Triboson
878 prop_binSR_electron2018_bin14_WpWpJJ_QCD
879 prop_binCRF_muon2016M_bin9_WpWpJJ_QCD
880 prop_binCRF_muon2017_bin15_WpWpJJ_QCD
881 prop_binCRF_muon2016M_bin9_Triboson
882 prop_binSR_electron2017_bin18_TVX
883 prop_binCRWS_electron2018_bin18_WrongSign
884 prop_binCRWS_electron2016M_bin17_WZ
885 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
886 prop_binCRF_electron2017_bin14_ZZtoLep
887 prop_binCRWS_muon2018_bin18_sm
888 prop_binCRWS_muon2018_bin10_VG
889 prop_binCRWS_muon2017_bin10_WrongSign
890 prop_binSR_muon2017_bin12_Triboson
891 prop_binCRF_muon2018_bin13_Triboson
892 prop_binCRWS_muon2016M_bin5_WZ
893 prop_binCRF_electron2017_bin6_ZZtoLep
894 prop_binCRF_muon2018_bin14_TVX
895 prop_binCRWS_muon2017_bin10_TVX
896 prop_binCRF_electron2017_bin7_WrongSign
897 prop_binCRF_muon2017_bin19_Triboson
898 prop_binCRWS_muon2016M_bin4_WZ
899 prop_binCRWS_muon2018_bin10_TVX
900 prop_binCRF_muon2016M_bin17_WrongSign
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$

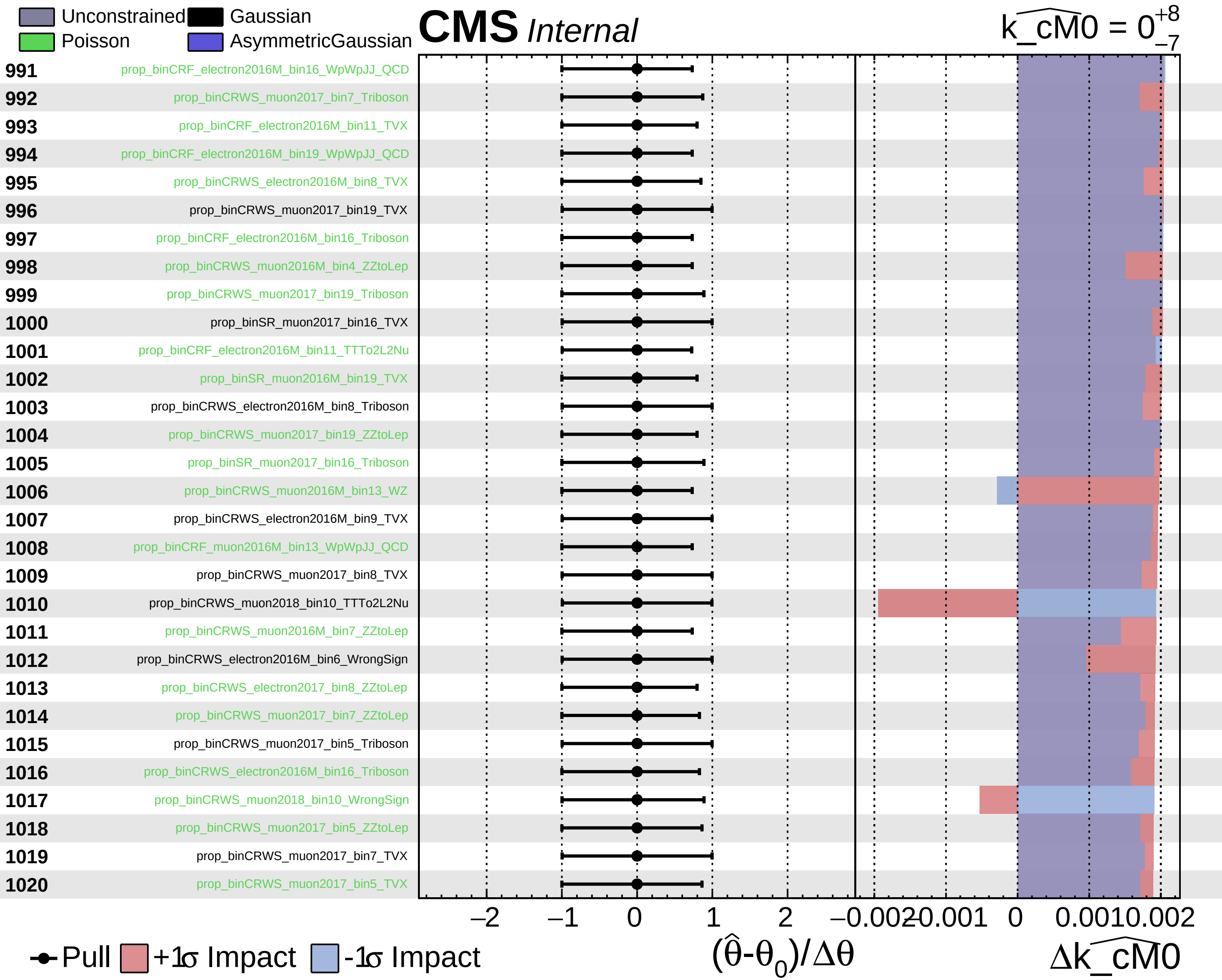

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
901 prop_binCRWS_muon2017_bin8_ZZtoLep
902 prop_binCRWS_muon2016M_bin4_TTTo2L2Nu
903 prop_binCRF_electron2017_bin12_ZZtoLep
904 prop_binCRWS_muon2016M_bin4_VG
905 prop_binCRWS_muon2018_bin11_WrongSign
906 prop_binCRWS_muon2018_bin14_WrongSign
907 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
908 prop_binSR_electron2017_bin17_WpWpJJ_QCD
909 prop_binCRWS_electron2016M_bin6_VG
910 prop_binSR_electron2017_bin15_WpWpJJ_QCD
911 prop_binCRWS_muon2016M_bin9_WrongSign
912 prop_binCRWS_muon2017_bin13_Triboson
913 prop_binCRF_electron2018_bin16_Fake
914 prop_binCRF_muon2016M_bin13_TTTo2L2Nu
915 prop_binCRWS_electron2017_bin13_TVX
916 prop_binSR_electron2017_bin15_WrongSign
917 prop_binCRF_muon2018_bin14_WrongSign
918 prop_binCRF_electron2017_bin18_TVX
919 prop_binCRF_electron2016M_bin15_Triboson
920 prop_binCRF_muon2016M_bin17_TVX
921 prop_binCRF_electron2018_bin16_TTTo2L2Nu
922 prop_binCRWS_muon2016M_bin16_VG
923 prop_binCRF_muon2016M_bin17_WpWpJJ_QCD
924 prop_binSR_muon2016M_bin19_WrongSign
925 prop_binCRWS_muon2017_bin12_Triboson
926 prop_binSR_muon2017_bin14_WrongSign
927 prop_binCRF_electron2017_bin18_WrongSign
928 prop_binCRWS_muon2018_bin14_Fake
929 prop_binCRWS_electron2016M_bin10_TVX
930 prop_binSR_muon2016M_bin19_Triboson
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



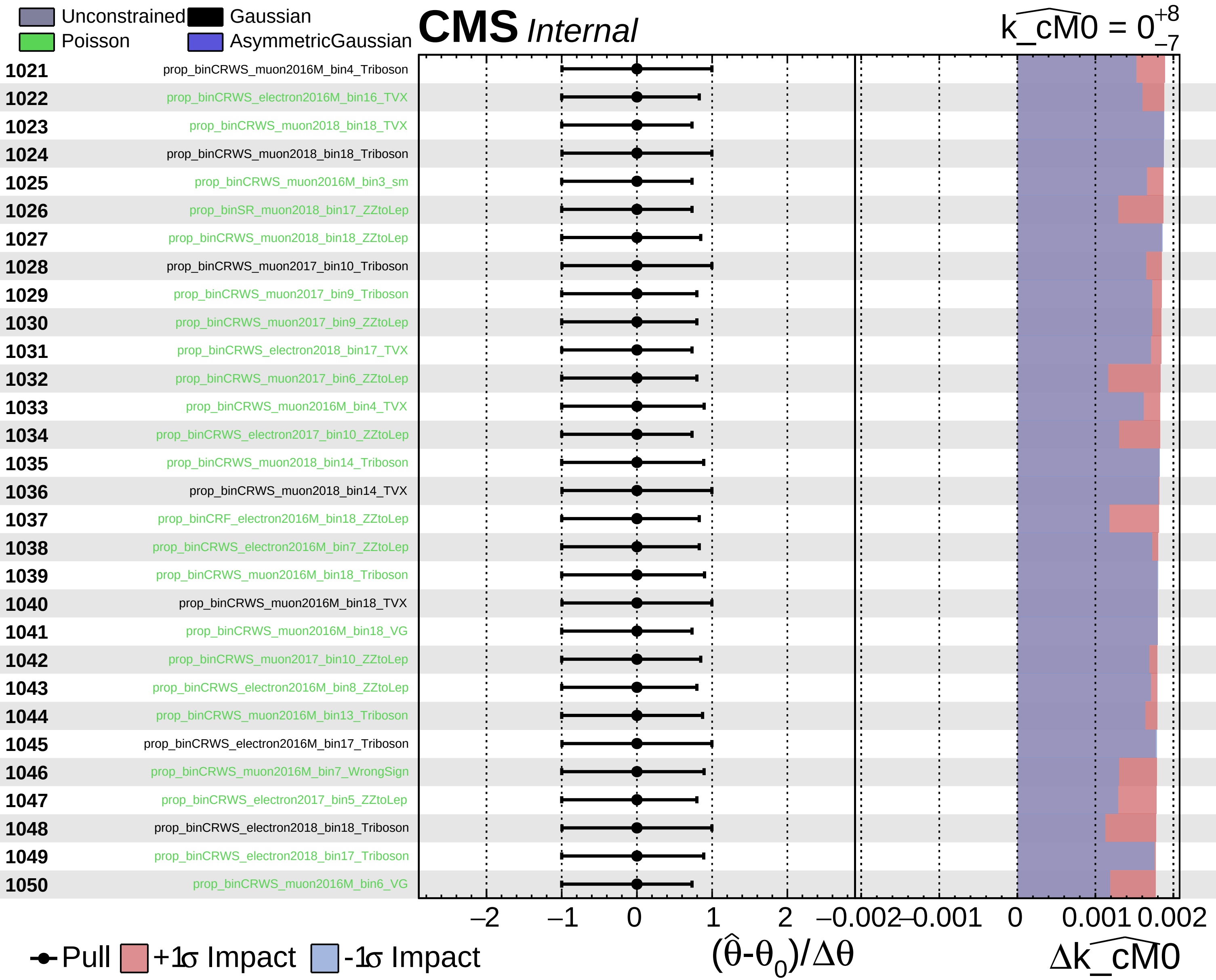
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
931 prop_binSR_muon2017_bin16_WrongSign
932 prop_binCRWS_electron2017_bin5_WrongSign
933 prop_binSR_electron2017_bin17_WrongSign
934 prop_binSR_electron2017_bin15_Triboson
935 prop_binCRF_muon2016M_bin13_TVX
936 prop_binCRWS_muon2017_bin8_Triboson
937 prop_binCRF_electron2018_bin16_WrongSign
938 prop_binCRF_electron2018_bin16_TVX
939 prop_binCRF_muon2017_bin19_ZZtoLep
940 prop_binCRF_electron2018_bin16_WpWpJJ_QCD
941 prop_binCRF_electron2018_bin16_Triboson
942 prop_binCRF_electron2018_bin7_TVX
943 prop_binCRF_muon2016M_bin10_ZZtoLep
944 prop_binCRF_electron2016M_bin11_WrongSign
945 prop_binCRF_electron2016M_bin11_WpWpJJ_QCD
946 prop_binCRF_electron2018_bin7_WrongSign
947 prop_binCRF_electron2018_bin7_WpWpJJ_QCD
948 prop_binCRF_electron2018_bin7_ZZtoLep
949 prop_binCRF_electron2018_bin7_Triboson
950 prop_binCRF_electron2016M_bin7_Triboson
951 prop_binCRF_electron2016M_bin12_TVX
952 prop_binCRF_electron2016M_bin12_Triboson
953 prop_binCRF_electron2016M_bin12_WpWpJJ_QCD
954 prop_binCRF_electron2016M_bin7_TVX
955 prop_binCRF_electron2016M_bin15_TVX
956 prop_binCRWS_electron2016M_bin9_Triboson
957 prop_binCRWS_muon2016M_bin5_sm
958 prop_binCRWS_electron2016M_bin20_TVX
959 prop_binSR_electron2017_bin17_Triboson
960 prop_binCRWS_muon2016M_bin13_TTTo2L2Nu
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$

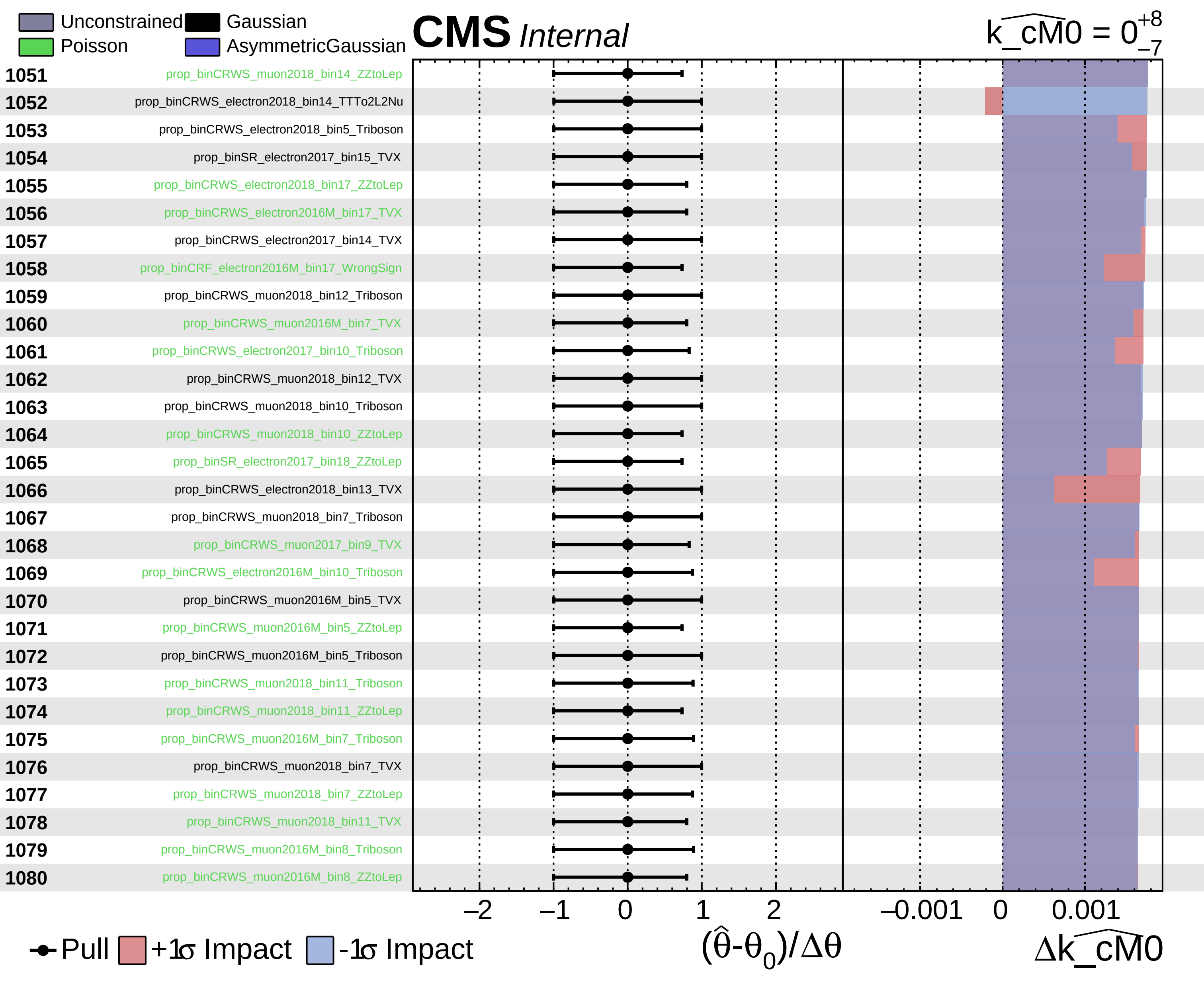


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0 +8 −7
961 prop_binCRWS_muon2017_bin12_ZZtoLep
962 prop_binCRWS_muon2016M_bin3_VG
963 prop_binCRWS_muon2016M_bin13_WrongSign
964 prop_binCRWS_electron2017_bin8_Triboson
965 prop_binCRF_muon2017_bin13_ZZtoLep
966 prop_binCRF_electron2016M_bin7_WpWpJJ_QCD
967 prop_binCRWS_muon2017_bin12_TVX
968 prop_binCRF_electron2016M_bin14_TVX
969 prop_binCRF_electron2016M_bin16_WrongSign
970 prop_binCRF_electron2016M_bin13_WrongSign
971 prop_binCRF_electron2016M_bin19_ZZtoLep
972 prop_binCRWS_muon2016M_bin8_sm
973 prop_binCRF_muon2018_bin14_Triboson
974 prop_binCRF_electron2016M_bin11_Triboson
975 prop_binSR_electron2017_bin17_TVX
976 prop_binCRF_electron2016M_bin19_TVX
977 prop_binCRWS_electron2016M_bin7_TVX
978 prop_binCRWS_electron2016M_bin7_Triboson
979 prop_binCRF_electron2016M_bin16_TVX
980 prop_binCRF_electron2016M_bin13_TVX
981 prop_binCRWS_electron2017_bin14_Triboson
982 prop_binSR_muon2017_bin14_ZZtoLep
983 prop_binCRF_electron2016M_bin14_WrongSign
984 prop_binCRWS_electron2016M_bin9_ZZtoLep
985 prop_binCRF_electron2017_bin18_Triboson
986 prop_binSR_muon2017_bin16_WpWpJJ_QCD
987 prop_binCRWS_muon2016M_bin7_TTTo2L2Nu
988 prop_binCRF_muon2016M_bin13_WrongSign
989 prop_binCRWS_electron2017_bin17_TVX
990 prop_binCRF_muon2016M_bin13_Triboson
−2
−1
0
1
2
−0.002
−0.001
0
0.001
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

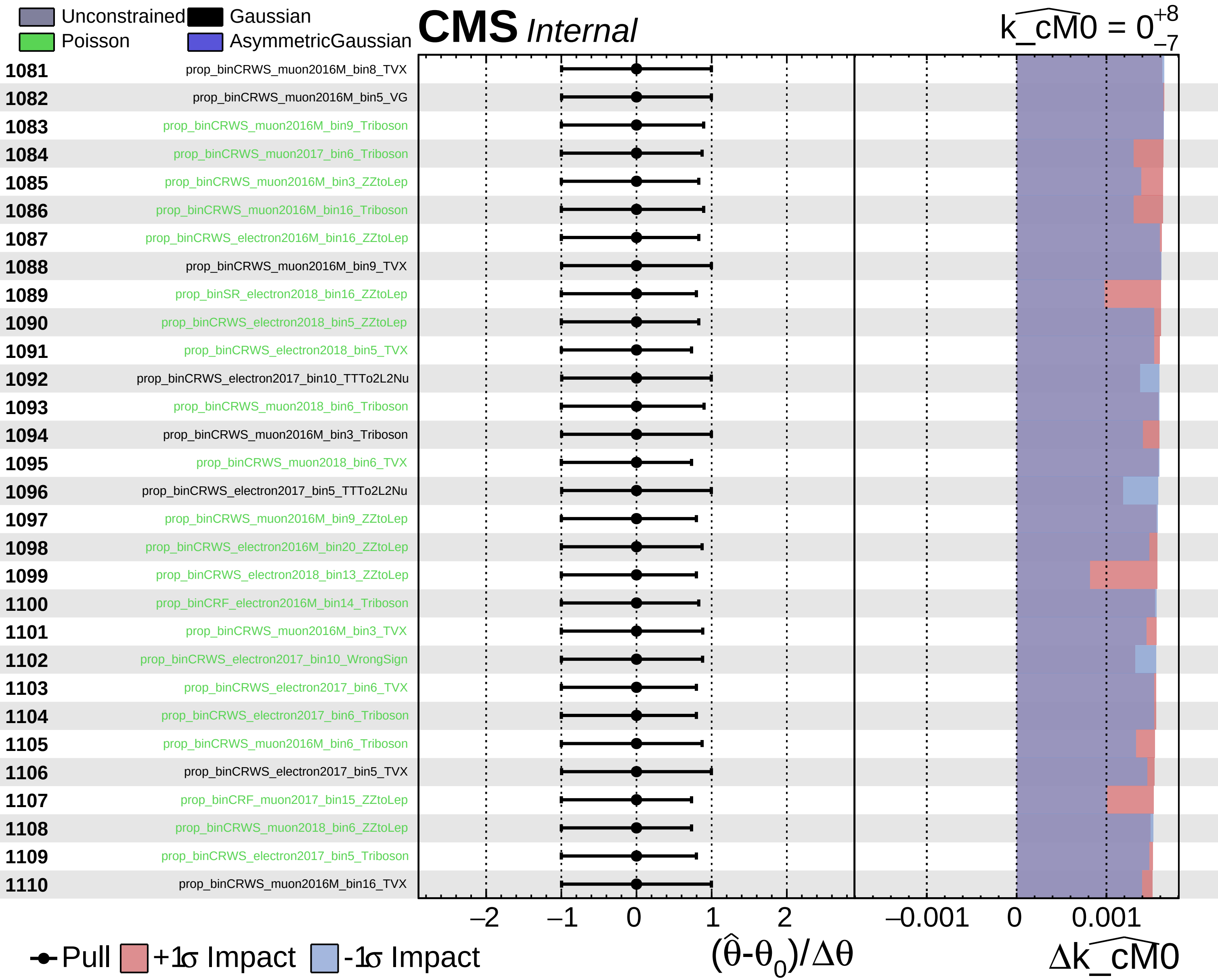


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0+8−7
991 prop_binCRF_electron2016M_bin16_WpWpJJ_QCD
992 prop_binCRWS_muon2017_bin7_Triboson
993 prop_binCRF_electron2016M_bin11_TVX
994 prop_binCRF_electron2016M_bin19_WpWpJJ_QCD
995 prop_binCRWS_electron2016M_bin8_TVX
996 prop_binCRWS_muon2017_bin19_TVX
997 prop_binCRF_electron2016M_bin16_Triboson
998 prop_binCRWS_muon2016M_bin4_ZZtoLep
999 prop_binCRWS_muon2017_bin19_Triboson
1000 prop_binSR_muon2017_bin16_TVX
1001 prop_binCRF_electron2016M_bin11_TTTo2L2Nu
1002 prop_binSR_muon2016M_bin19_TVX
1003 prop_binCRWS_electron2016M_bin8_Triboson
1004 prop_binCRWS_muon2017_bin19_ZZtoLep
1005 prop_binSR_muon2017_bin16_Triboson
1006 prop_binCRWS_muon2016M_bin13_WZ
1007 prop_binCRWS_electron2016M_bin9_TVX
1008 prop_binCRF_muon2016M_bin13_WpWpJJ_QCD
1009 prop_binCRWS_muon2017_bin8_TVX
1010 prop_binCRWS_muon2018_bin10_TTTo2L2Nu
1011 prop_binCRWS_muon2016M_bin7_ZZtoLep
1012 prop_binCRWS_electron2016M_bin6_WrongSign
1013 prop_binCRWS_electron2017_bin8_ZZtoLep
1014 prop_binCRWS_muon2017_bin7_ZZtoLep
1015 prop_binCRWS_muon2017_bin5_Triboson
1016 prop_binCRWS_electron2016M_bin16_Triboson
1017 prop_binCRWS_muon2018_bin10_WrongSign
1018 prop_binCRWS_muon2017_bin5_ZZtoLep
1019 prop_binCRWS_muon2017_bin7_TVX
1020 prop_binCRWS_muon2017_bin5_TVX
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

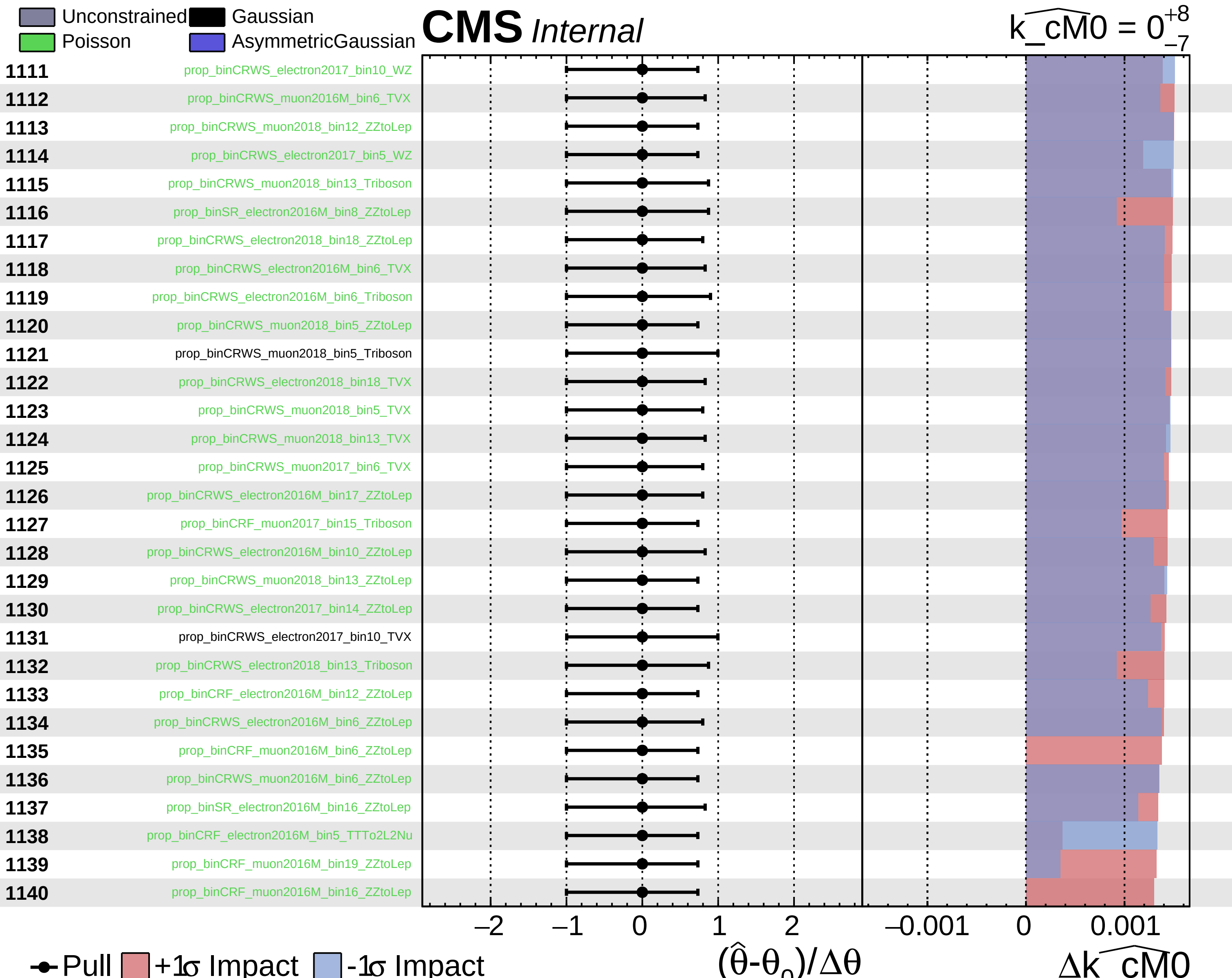

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0+8 −7
1021 prop_binCRWS_muon2016M_bin4_Triboson
1022 prop_binCRWS_electron2016M_bin16_TVX
1023 prop_binCRWS_muon2018_bin18_TVX
1024 prop_binCRWS_muon2018_bin18_Triboson
1025 prop_binCRWS_muon2016M_bin3_sm
1026 prop_binSR_muon2018_bin17_ZZtoLep
1027 prop_binCRWS_muon2018_bin18_ZZtoLep
1028 prop_binCRWS_muon2017_bin10_Triboson
1029 prop_binCRWS_muon2017_bin9_Triboson
1030 prop_binCRWS_muon2017_bin9_ZZtoLep
1031 prop_binCRWS_electron2018_bin17_TVX
1032 prop_binCRWS_muon2017_bin6_ZZtoLep
1033 prop_binCRWS_muon2016M_bin4_TVX
1034 prop_binCRWS_electron2017_bin10_ZZtoLep
1035 prop_binCRWS_muon2018_bin14_Triboson
1036 prop_binCRWS_muon2018_bin14_TVX
1037 prop_binCRF_electron2016M_bin18_ZZtoLep
1038 prop_binCRWS_electron2016M_bin7_ZZtoLep
1039 prop_binCRWS_muon2016M_bin18_Triboson
1040 prop_binCRWS_muon2016M_bin18_TVX
1041 prop_binCRWS_muon2016M_bin18_VG
1042 prop_binCRWS_muon2017_bin10_ZZtoLep
1043 prop_binCRWS_electron2016M_bin8_ZZtoLep
1044 prop_binCRWS_muon2016M_bin13_Triboson
1045 prop_binCRWS_electron2016M_bin17_Triboson
1046 prop_binCRWS_muon2016M_bin7_WrongSign
1047 prop_binCRWS_electron2017_bin5_ZZtoLep
1048 prop_binCRWS_electron2018_bin18_Triboson
1049 prop_binCRWS_electron2018_bin17_Triboson
1050 prop_binCRWS_muon2016M_bin6_VG
−2
−1
0
1
2
−0.002
−0.001
0
0.001
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0



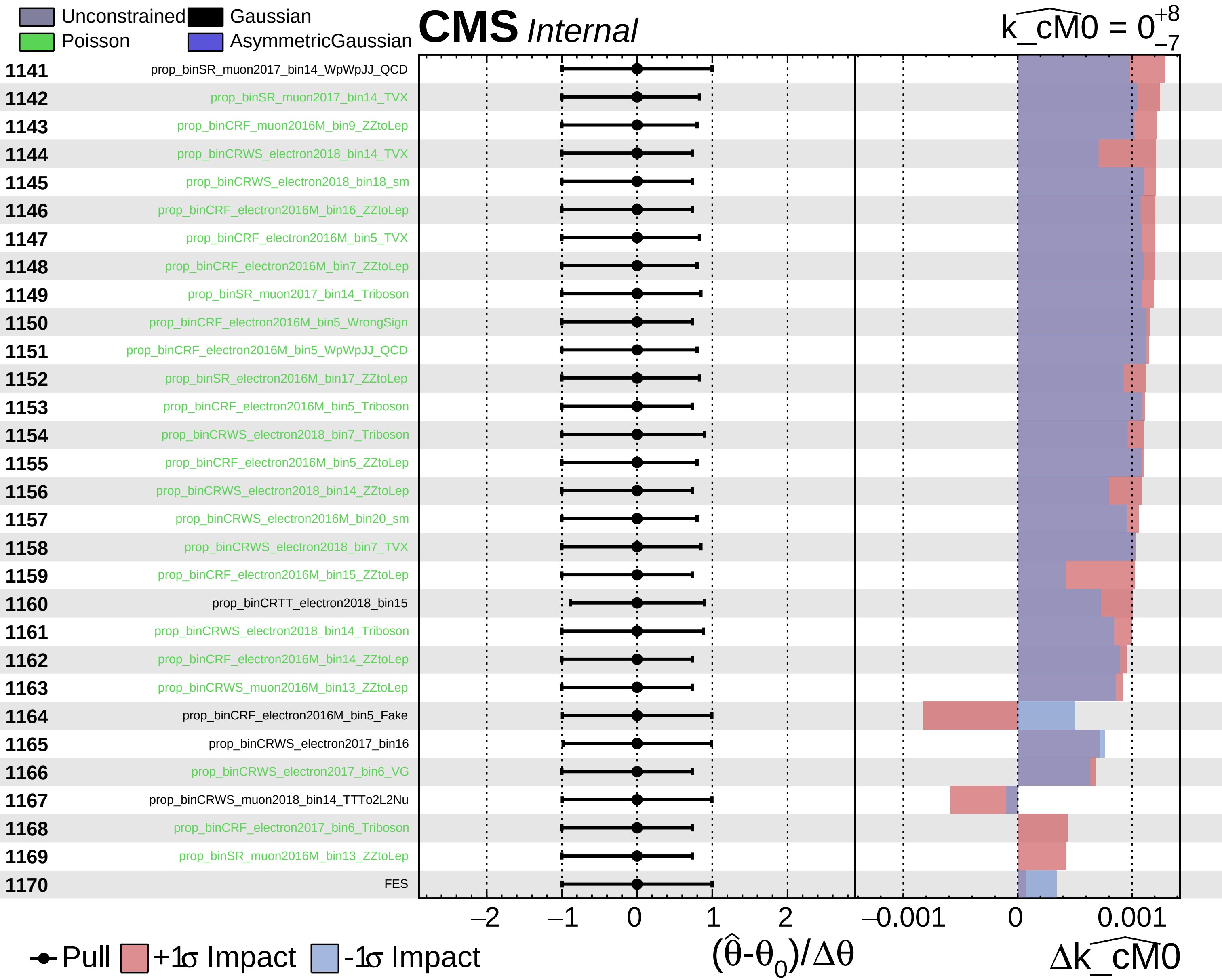
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0+8−7
1051 prop_binCRWS_muon2018_bin14_ZZtoLep
1052 prop_binCRWS_electron2018_bin14_TTTo2L2Nu
1053 prop_binCRWS_electron2018_bin5_Triboson
1054 prop_binSR_electron2017_bin15_TVX
1055 prop_binCRWS_electron2018_bin17_ZZtoLep
1056 prop_binCRWS_electron2016M_bin17_TVX
1057 prop_binCRWS_electron2017_bin14_TVX
1058 prop_binCRF_electron2016M_bin17_WrongSign
1059 prop_binCRWS_muon2018_bin12_Triboson
1060 prop_binCRWS_muon2016M_bin7_TVX
1061 prop_binCRWS_electron2017_bin10_Triboson
1062 prop_binCRWS_muon2018_bin12_TVX
1063 prop_binCRWS_muon2018_bin10_Triboson
1064 prop_binCRWS_muon2018_bin10_ZZtoLep
1065 prop_binSR_electron2017_bin18_ZZtoLep
1066 prop_binCRWS_electron2018_bin13_TVX
1067 prop_binCRWS_muon2018_bin7_Triboson
1068 prop_binCRWS_muon2017_bin9_TVX
1069 prop_binCRWS_electron2016M_bin10_Triboson
1070 prop_binCRWS_muon2016M_bin5_TVX
1071 prop_binCRWS_muon2016M_bin5_ZZtoLep
1072 prop_binCRWS_muon2016M_bin5_Triboson
1073 prop_binCRWS_muon2018_bin11_Triboson
1074 prop_binCRWS_muon2018_bin11_ZZtoLep
1075 prop_binCRWS_muon2016M_bin7_Triboson
1076 prop_binCRWS_muon2018_bin7_TVX
1077 prop_binCRWS_muon2018_bin7_ZZtoLep
1078 prop_binCRWS_muon2018_bin11_TVX
1079 prop_binCRWS_muon2016M_bin8_Triboson
1080 prop_binCRWS_muon2016M_bin8_ZZtoLep
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0



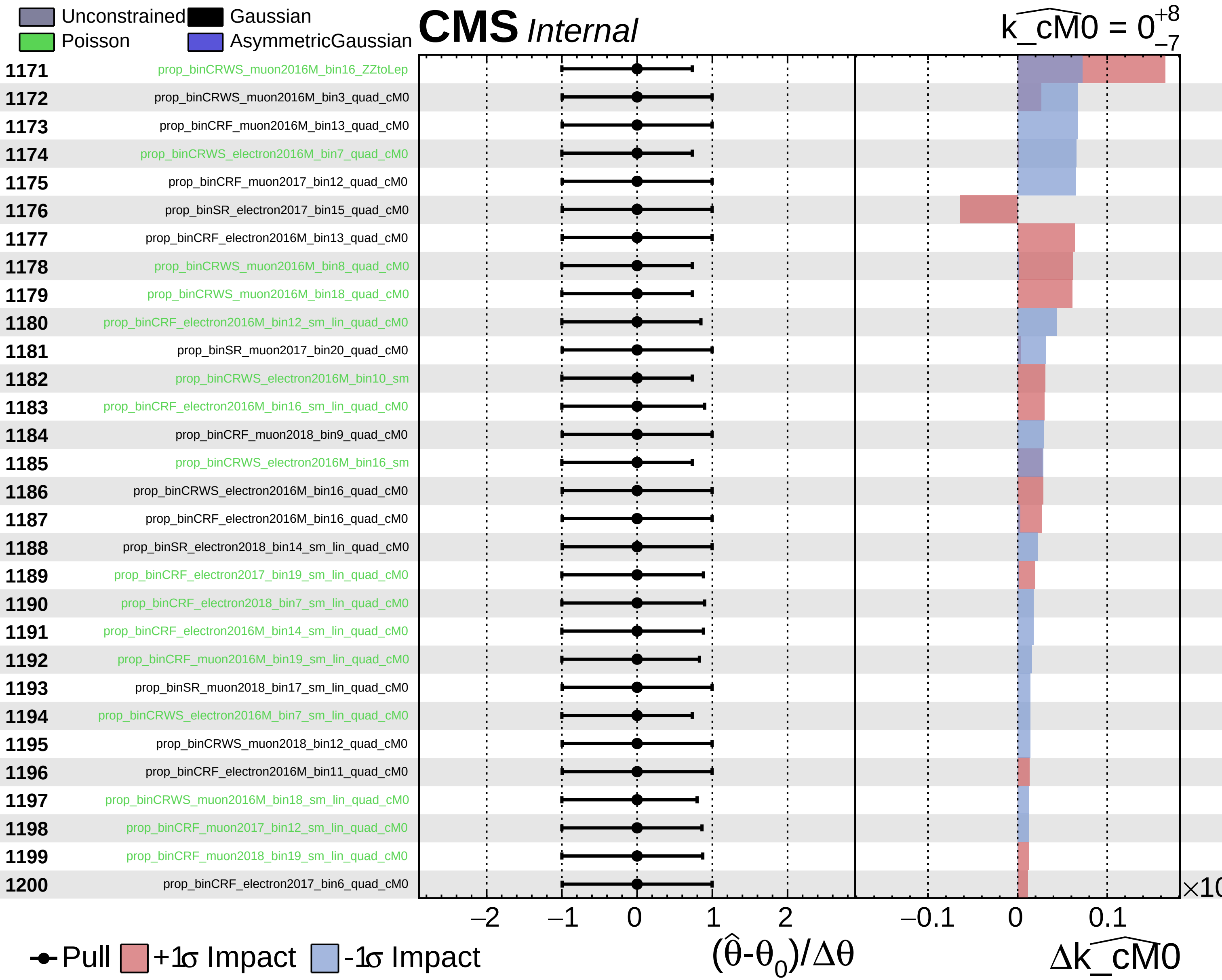
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
1081 prop_binCRWS_muon2016M_bin8_TVX
1082 prop_binCRWS_muon2016M_bin5_VG
1083 prop_binCRWS_muon2016M_bin9_Triboson
1084 prop_binCRWS_muon2017_bin6_Triboson
1085 prop_binCRWS_muon2016M_bin3_ZZtoLep
1086 prop_binCRWS_muon2016M_bin16_Triboson
1087 prop_binCRWS_electron2016M_bin16_ZZtoLep
1088 prop_binCRWS_muon2016M_bin9_TVX
1089 prop_binSR_electron2018_bin16_ZZtoLep
1090 prop_binCRWS_electron2018_bin5_ZZtoLep
1091 prop_binCRWS_electron2018_bin5_TVX
1092 prop_binCRWS_electron2017_bin10_TTTo2L2Nu
1093 prop_binCRWS_muon2018_bin6_Triboson
1094 prop_binCRWS_muon2016M_bin3_Triboson
1095 prop_binCRWS_muon2018_bin6_TVX
1096 prop_binCRWS_electron2017_bin5_TTTo2L2Nu
1097 prop_binCRWS_muon2016M_bin9_ZZtoLep
1098 prop_binCRWS_electron2016M_bin20_ZZtoLep
1099 prop_binCRWS_electron2018_bin13_ZZtoLep
1100 prop_binCRF_electron2016M_bin14_Triboson
1101 prop_binCRWS_muon2016M_bin3_TVX
1102 prop_binCRWS_electron2017_bin10_WrongSign
1103 prop_binCRWS_electron2017_bin6_TVX
1104 prop_binCRWS_electron2017_bin6_Triboson
1105 prop_binCRWS_muon2016M_bin6_Triboson
1106 prop_binCRWS_electron2017_bin5_TVX
1107 prop_binCRF_muon2017_bin15_ZZtoLep
1108 prop_binCRWS_muon2018_bin6_ZZtoLep
1109 prop_binCRWS_electron2017_bin5_Triboson
1110 prop_binCRWS_muon2016M_bin16_TVX
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



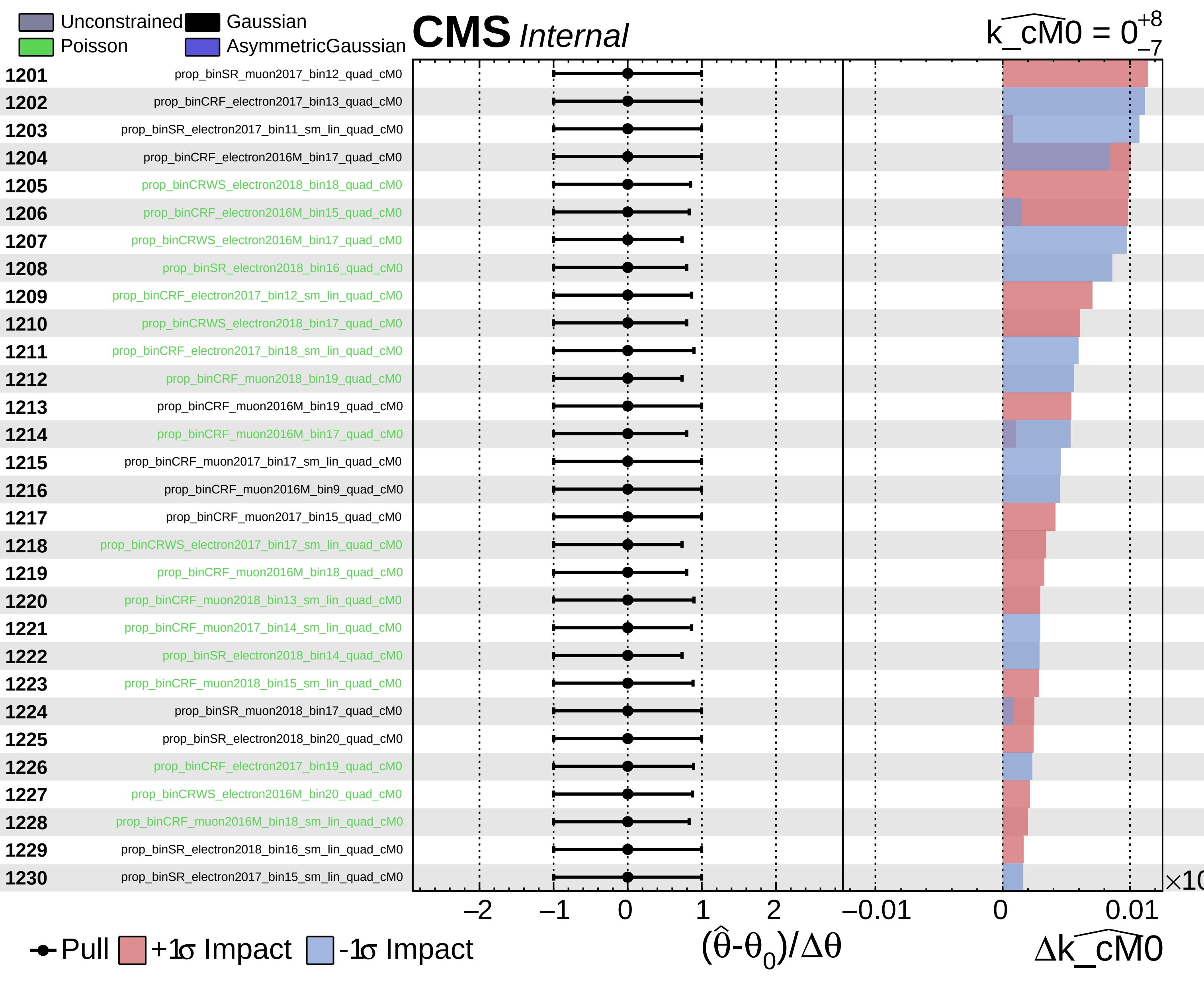
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0 +8 −7
1111 prop_binCRWS_electron2017_bin10_WZ
1112 prop_binCRWS_muon2016M_bin6_TVX
1113 prop_binCRWS_muon2018_bin12_ZZtoLep
1114 prop_binCRWS_electron2017_bin5_WZ
1115 prop_binCRWS_muon2018_bin13_Triboson
1116 prop_binSR_electron2016M_bin8_ZZtoLep
1117 prop_binCRWS_electron2018_bin18_ZZtoLep
1118 prop_binCRWS_electron2016M_bin6_TVX
1119 prop_binCRWS_electron2016M_bin6_Triboson
1120 prop_binCRWS_muon2018_bin5_ZZtoLep
1121 prop_binCRWS_muon2018_bin5_Triboson
1122 prop_binCRWS_electron2018_bin18_TVX
1123 prop_binCRWS_muon2018_bin5_TVX
1124 prop_binCRWS_muon2018_bin13_TVX
1125 prop_binCRWS_muon2017_bin6_TVX
1126 prop_binCRWS_electron2016M_bin17_ZZtoLep
1127 prop_binCRF_muon2017_bin15_Triboson
1128 prop_binCRWS_electron2016M_bin10_ZZtoLep
1129 prop_binCRWS_muon2018_bin13_ZZtoLep
1130 prop_binCRWS_electron2017_bin14_ZZtoLep
1131 prop_binCRWS_electron2017_bin10_TVX
1132 prop_binCRWS_electron2018_bin13_Triboson
1133 prop_binCRF_electron2016M_bin12_ZZtoLep
1134 prop_binCRWS_electron2016M_bin6_ZZtoLep
1135 prop_binCRF_muon2016M_bin6_ZZtoLep
1136 prop_binCRWS_muon2016M_bin6_ZZtoLep
1137 prop_binSR_electron2016M_bin16_ZZtoLep
1138 prop_binCRF_electron2016M_bin5_TTTo2L2Nu
1139 prop_binCRF_muon2016M_bin19_ZZtoLep
1140 prop_binCRF_muon2016M_bin16_ZZtoLep
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0



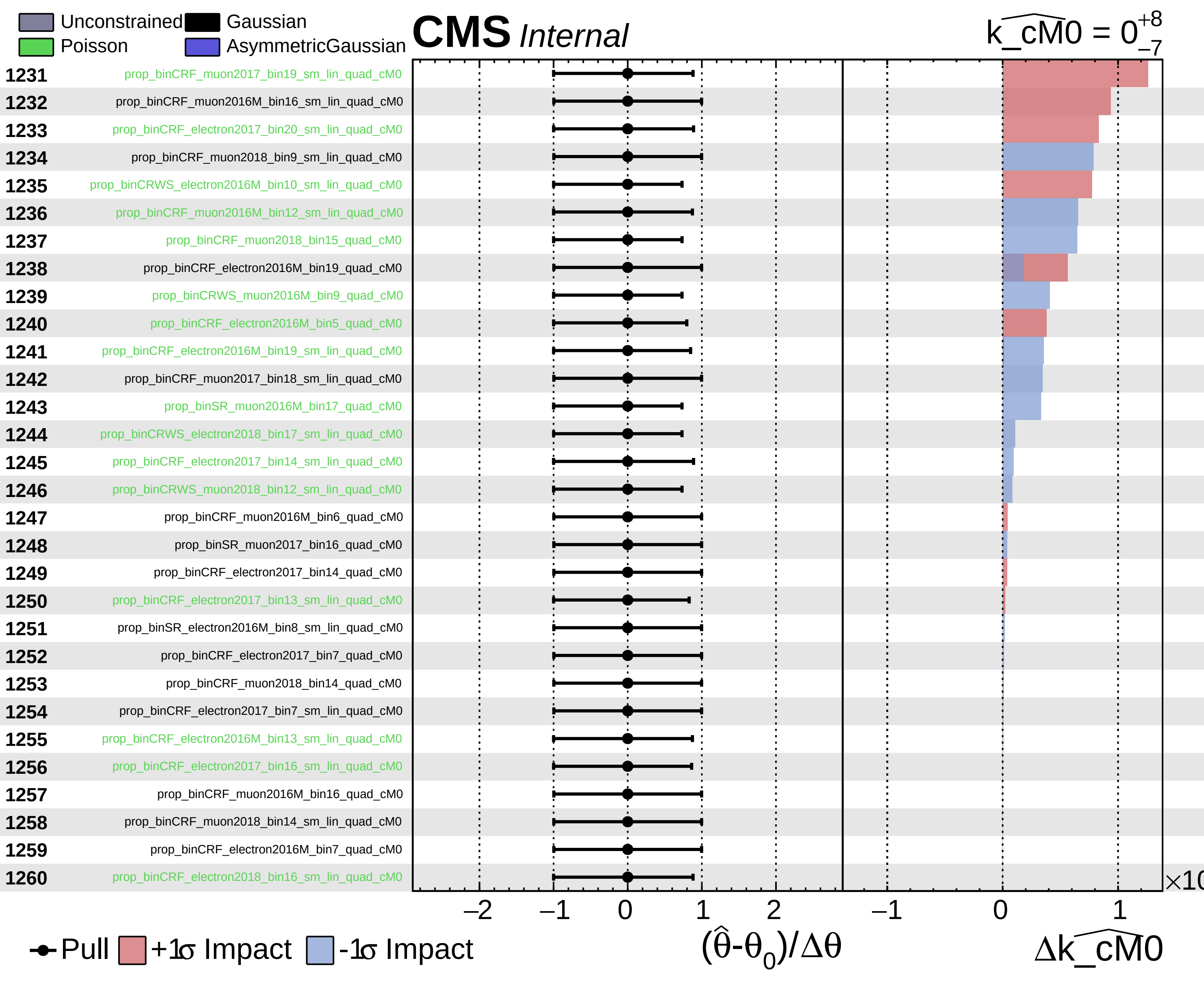
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0+8−7
1141 prop_binSR_muon2017_bin14_WpWpJJ_QCD
1142 prop_binSR_muon2017_bin14_TVX
1143 prop_binCRF_muon2016M_bin9_ZZtoLep
1144 prop_binCRWS_electron2018_bin14_TVX
1145 prop_binCRWS_electron2018_bin18_sm
1146 prop_binCRF_electron2016M_bin16_ZZtoLep
1147 prop_binCRF_electron2016M_bin5_TVX
1148 prop_binCRF_electron2016M_bin7_ZZtoLep
1149 prop_binSR_muon2017_bin14_Triboson
1150 prop_binCRF_electron2016M_bin5_WrongSign
1151 prop_binCRF_electron2016M_bin5_WpWpJJ_QCD
1152 prop_binSR_electron2016M_bin17_ZZtoLep
1153 prop_binCRF_electron2016M_bin5_Triboson
1154 prop_binCRWS_electron2018_bin7_Triboson
1155 prop_binCRF_electron2016M_bin5_ZZtoLep
1156 prop_binCRWS_electron2018_bin14_ZZtoLep
1157 prop_binCRWS_electron2016M_bin20_sm
1158 prop_binCRWS_electron2018_bin7_TVX
1159 prop_binCRF_electron2016M_bin15_ZZtoLep
1160 prop_binCRTT_electron2018_bin15
1161 prop_binCRWS_electron2018_bin14_Triboson
1162 prop_binCRF_electron2016M_bin14_ZZtoLep
1163 prop_binCRWS_muon2016M_bin13_ZZtoLep
1164 prop_binCRF_electron2016M_bin5_Fake
1165 prop_binCRWS_electron2017_bin16
1166 prop_binCRWS_electron2017_bin6_VG
1167 prop_binCRWS_muon2018_bin14_TTTo2L2Nu
1168 prop_binCRF_electron2017_bin6_Triboson
1169 prop_binSR_muon2016M_bin13_ZZtoLep
1170 FES
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM0

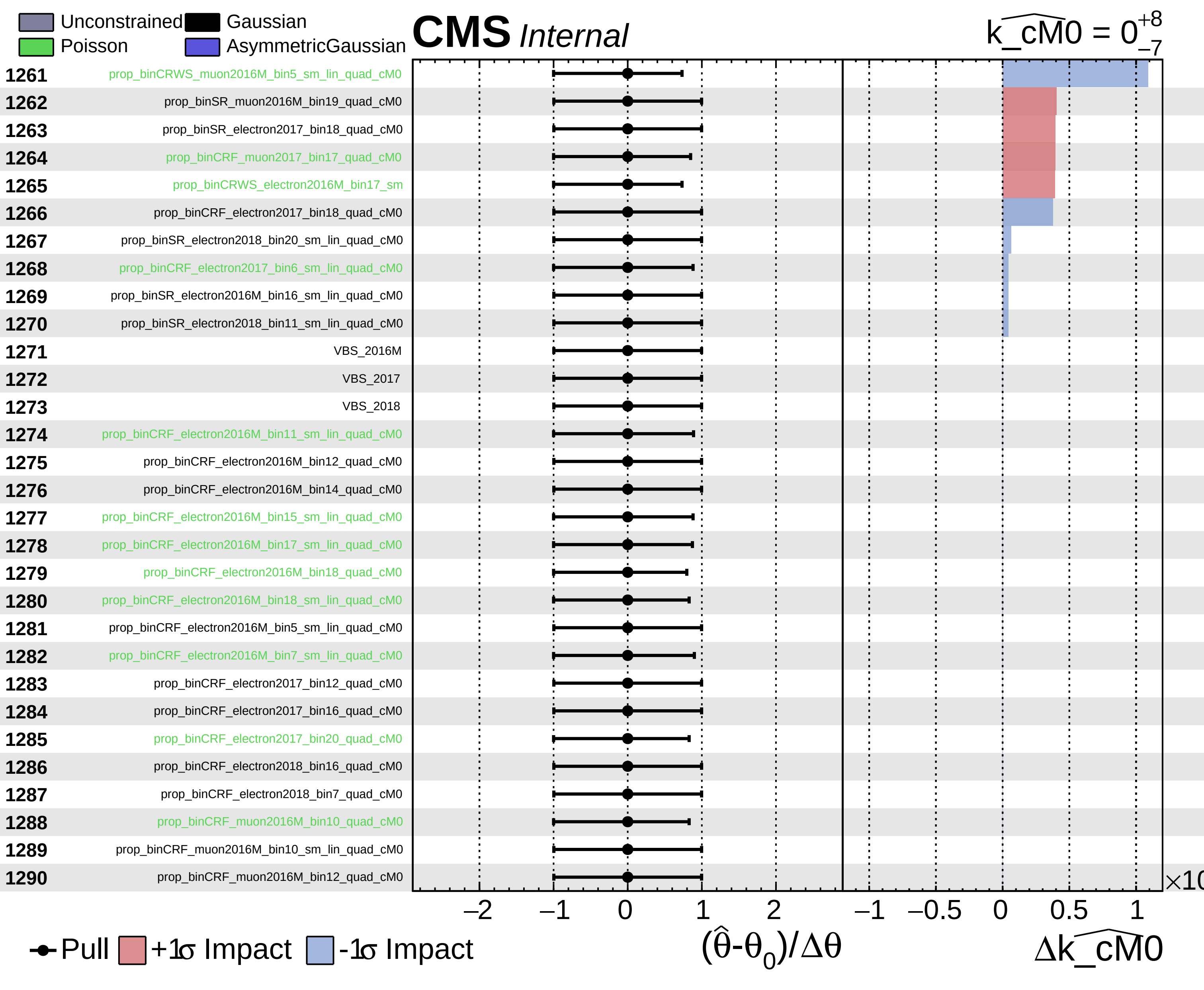


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0 +8 −7
1171 prop_binCRWS_muon2016M_bin16_ZZtoLep
1172 prop_binCRWS_muon2016M_bin3_quad_cM0
1173 prop_binCRF_muon2016M_bin13_quad_cM0
1174 prop_binCRWS_electron2016M_bin7_quad_cM0
1175 prop_binCRF_muon2017_bin12_quad_cM0
1176 prop_binSR_electron2017_bin15_quad_cM0
1177 prop_binCRF_electron2016M_bin13_quad_cM0
1178 prop_binCRWS_muon2016M_bin8_quad_cM0
1179 prop_binCRWS_muon2016M_bin18_quad_cM0
1180 prop_binCRF_electron2016M_bin12_sm_lin_quad_cM0
1181 prop_binSR_muon2017_bin20_quad_cM0
1182 prop_binCRWS_electron2016M_bin10_sm
1183 prop_binCRF_electron2016M_bin16_sm_lin_quad_cM0
1184 prop_binCRF_muon2018_bin9_quad_cM0
1185 prop_binCRWS_electron2016M_bin16_sm
1186 prop_binCRWS_electron2016M_bin16_quad_cM0
1187 prop_binCRF_electron2016M_bin16_quad_cM0
1188 prop_binSR_electron2018_bin14_sm_lin_quad_cM0
1189 prop_binCRF_electron2017_bin19_sm_lin_quad_cM0
1190 prop_binCRF_electron2018_bin7_sm_lin_quad_cM0
1191 prop_binCRF_electron2016M_bin14_sm_lin_quad_cM0
1192 prop_binCRF_muon2016M_bin19_sm_lin_quad_cM0
1193 prop_binSR_muon2018_bin17_sm_lin_quad_cM0
1194 prop_binCRWS_electron2016M_bin7_sm_lin_quad_cM0
1195 prop_binCRWS_muon2018_bin12_quad_cM0
1196 prop_binCRF_electron2016M_bin11_quad_cM0
1197 prop_binCRWS_muon2016M_bin18_sm_lin_quad_cM0
1198 prop_binCRF_muon2017_bin12_sm_lin_quad_cM0
1199 prop_binCRF_muon2018_bin19_sm_lin_quad_cM0
1200 prop_binCRF_electron2017_bin6_quad_cM0
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

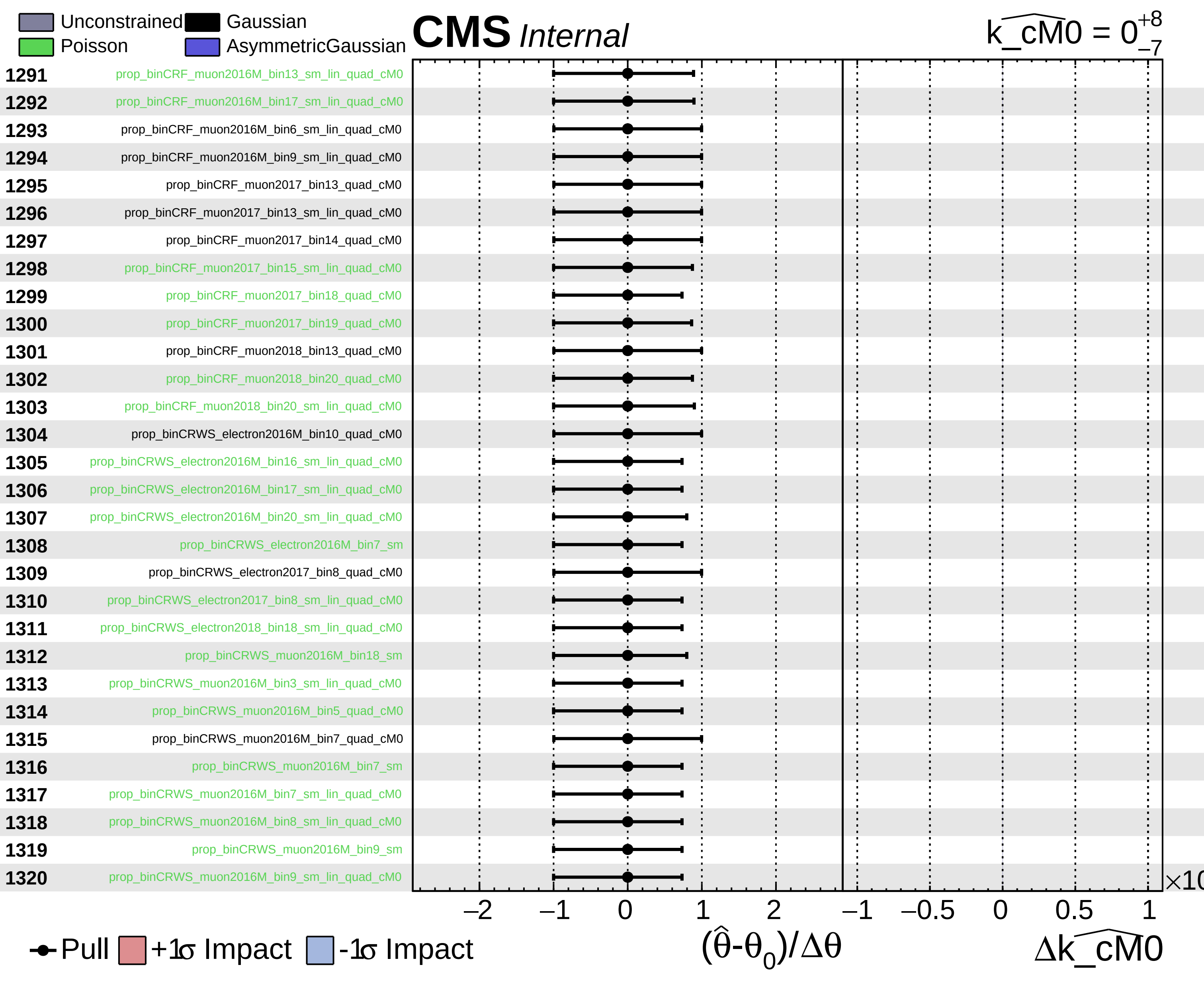


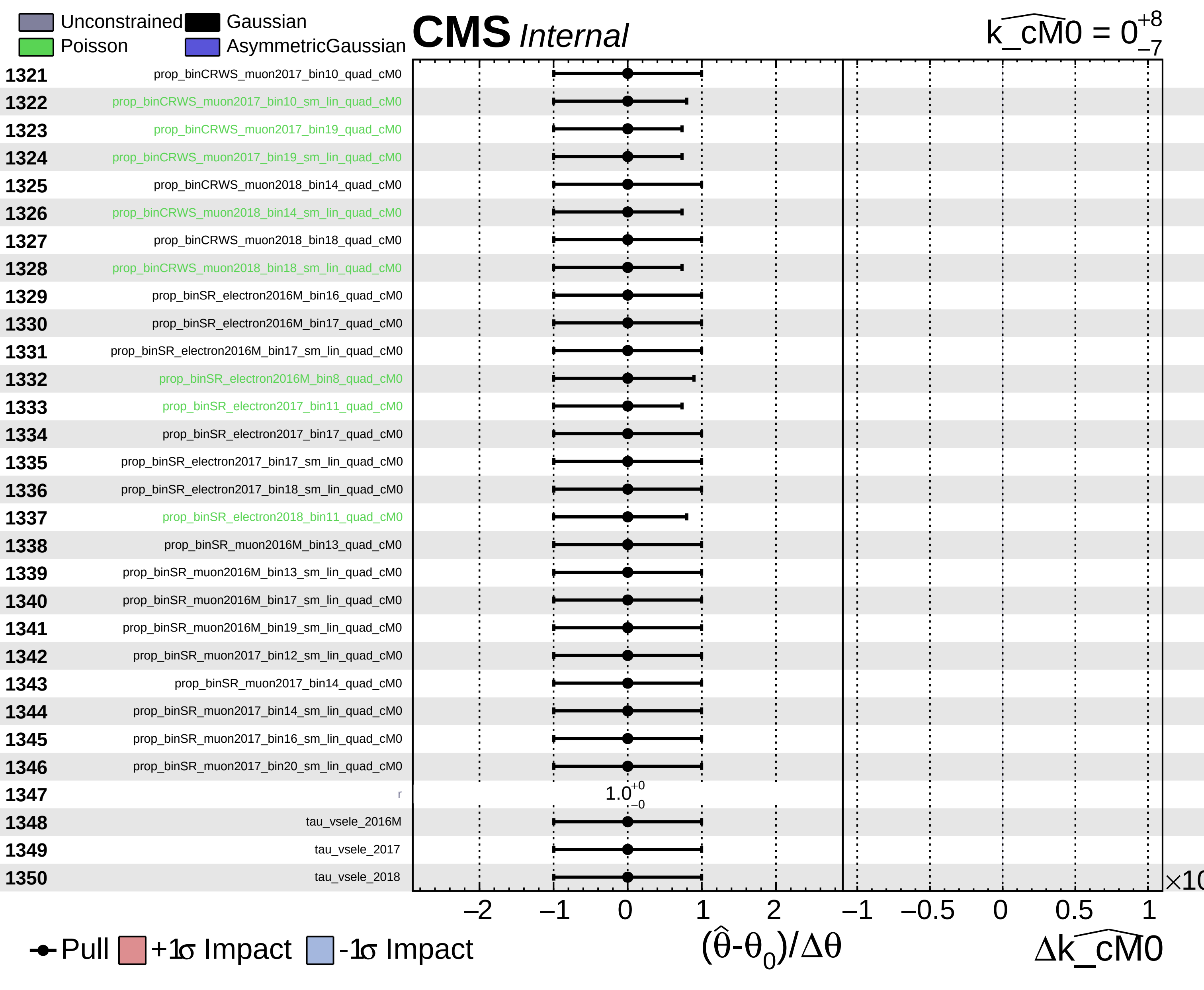
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0+8−7
1201 prop_binSR_muon2017_bin12_quad_cM0
1202 prop_binCRF_electron2017_bin13_quad_cM0
1203 prop_binSR_electron2017_bin11_sm_lin_quad_cM0
1204 prop_binCRF_electron2016M_bin17_quad_cM0
1205 prop_binCRWS_electron2018_bin18_quad_cM0
1206 prop_binCRF_electron2016M_bin15_quad_cM0
1207 prop_binCRWS_electron2016M_bin17_quad_cM0
1208 prop_binSR_electron2018_bin16_quad_cM0
1209 prop_binCRF_electron2017_bin12_sm_lin_quad_cM0
1210 prop_binCRWS_electron2018_bin17_quad_cM0
1211 prop_binCRF_electron2017_bin18_sm_lin_quad_cM0
1212 prop_binCRF_muon2018_bin19_quad_cM0
1213 prop_binCRF_muon2016M_bin19_quad_cM0
1214 prop_binCRF_muon2016M_bin17_quad_cM0
1215 prop_binCRF_muon2017_bin17_sm_lin_quad_cM0
1216 prop_binCRF_muon2016M_bin9_quad_cM0
1217 prop_binCRF_muon2017_bin15_quad_cM0
1218 prop_binCRWS_electron2017_bin17_sm_lin_quad_cM0
1219 prop_binCRF_muon2016M_bin18_quad_cM0
1220 prop_binCRF_muon2018_bin13_sm_lin_quad_cM0
1221 prop_binCRF_muon2017_bin14_sm_lin_quad_cM0
1222 prop_binSR_electron2018_bin14_quad_cM0
1223 prop_binCRF_muon2018_bin15_sm_lin_quad_cM0
1224 prop_binSR_muon2018_bin17_quad_cM0
1225 prop_binSR_electron2018_bin20_quad_cM0
1226 prop_binCRF_electron2017_bin19_quad_cM0
1227 prop_binCRWS_electron2016M_bin20_quad_cM0
1228 prop_binCRF_muon2016M_bin18_sm_lin_quad_cM0
1229 prop_binSR_electron2018_bin16_sm_lin_quad_cM0
1230 prop_binSR_electron2017_bin15_sm_lin_quad_cM0
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0^{+8}_{−7}
1231 prop_binCRF_muon2017_bin19_sm_lin_quad_cM0
1232 prop_binCRF_muon2016M_bin16_sm_lin_quad_cM0
1233 prop_binCRF_electron2017_bin20_sm_lin_quad_cM0
1234 prop_binCRF_muon2018_bin9_sm_lin_quad_cM0
1235 prop_binCRWS_electron2016M_bin10_sm_lin_quad_cM0
1236 prop_binCRF_muon2016M_bin12_sm_lin_quad_cM0
1237 prop_binCRF_muon2018_bin15_quad_cM0
1238 prop_binCRF_electron2016M_bin19_quad_cM0
1239 prop_binCRWS_muon2016M_bin9_quad_cM0
1240 prop_binCRF_electron2016M_bin5_quad_cM0
1241 prop_binCRF_electron2016M_bin19_sm_lin_quad_cM0
1242 prop_binCRF_muon2017_bin18_sm_lin_quad_cM0
1243 prop_binSR_muon2016M_bin17_quad_cM0
1244 prop_binCRWS_electron2018_bin17_sm_lin_quad_cM0
1245 prop_binCRF_electron2017_bin14_sm_lin_quad_cM0
1246 prop_binCRWS_muon2018_bin12_sm_lin_quad_cM0
1247 prop_binCRF_muon2016M_bin6_quad_cM0
1248 prop_binSR_muon2017_bin16_quad_cM0
1249 prop_binCRF_electron2017_bin14_quad_cM0
1250 prop_binCRF_electron2017_bin13_sm_lin_quad_cM0
1251 prop_binSR_electron2016M_bin8_sm_lin_quad_cM0
1252 prop_binCRF_electron2017_bin7_quad_cM0
1253 prop_binCRF_muon2018_bin14_quad_cM0
1254 prop_binCRF_electron2017_bin7_sm_lin_quad_cM0
1255 prop_binCRF_electron2016M_bin13_sm_lin_quad_cM0
1256 prop_binCRF_electron2017_bin16_sm_lin_quad_cM0
1257 prop_binCRF_muon2016M_bin16_quad_cM0
1258 prop_binCRF_muon2018_bin14_sm_lin_quad_cM0
1259 prop_binCRF_electron2016M_bin7_quad_cM0
1260 prop_binCRF_electron2018_bin16_sm_lin_quad_cM0
−2 −1 0 1 2
−1 0 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

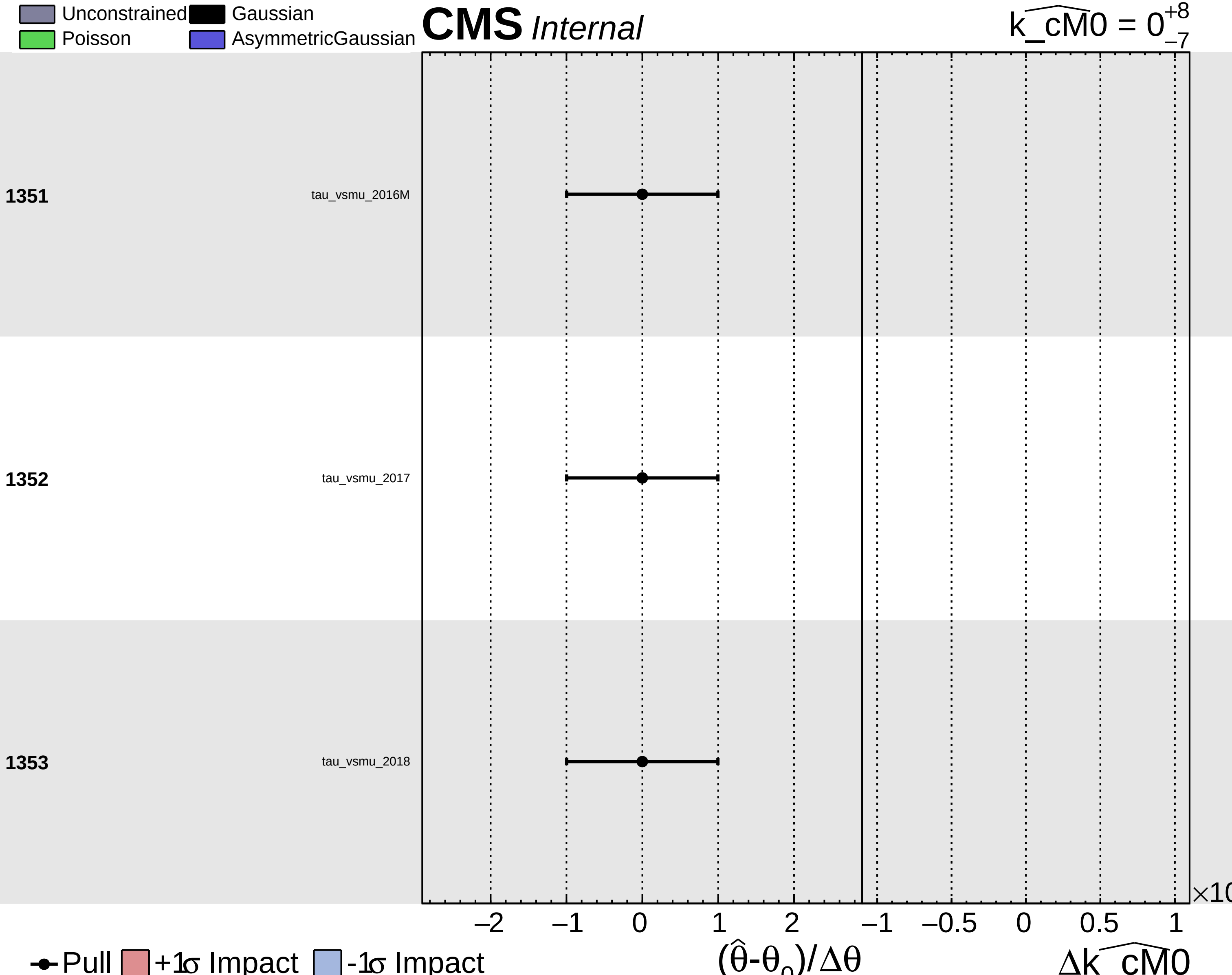


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0 +8 −7
1261 prop_binCRWS_muon2016M_bin5_sm_lin_quad_cM0
1262 prop_binSR_muon2016M_bin19_quad_cM0
1263 prop_binSR_electron2017_bin18_quad_cM0
1264 prop_binCRF_muon2017_bin17_quad_cM0
1265 prop_binCRWS_electron2016M_bin17_sm
1266 prop_binCRF_electron2017_bin18_quad_cM0
1267 prop_binSR_electron2018_bin20_sm_lin_quad_cM0
1268 prop_binCRF_electron2017_bin6_sm_lin_quad_cM0
1269 prop_binSR_electron2016M_bin16_sm_lin_quad_cM0
1270 prop_binSR_electron2018_bin11_sm_lin_quad_cM0
1271 VBS_2016M
1272 VBS_2017
1273 VBS_2018
1274 prop_binCRF_electron2016M_bin11_sm_lin_quad_cM0
1275 prop_binCRF_electron2016M_bin12_quad_cM0
1276 prop_binCRF_electron2016M_bin14_quad_cM0
1277 prop_binCRF_electron2016M_bin15_sm_lin_quad_cM0
1278 prop_binCRF_electron2016M_bin17_sm_lin_quad_cM0
1279 prop_binCRF_electron2016M_bin18_quad_cM0
1280 prop_binCRF_electron2016M_bin18_sm_lin_quad_cM0
1281 prop_binCRF_electron2016M_bin5_sm_lin_quad_cM0
1282 prop_binCRF_electron2016M_bin7_sm_lin_quad_cM0
1283 prop_binCRF_electron2017_bin12_quad_cM0
1284 prop_binCRF_electron2017_bin16_quad_cM0
1285 prop_binCRF_electron2017_bin20_quad_cM0
1286 prop_binCRF_electron2018_bin16_quad_cM0
1287 prop_binCRF_electron2018_bin7_quad_cM0
1288 prop_binCRF_muon2016M_bin10_quad_cM0
1289 prop_binCRF_muon2016M_bin10_sm_lin_quad_cM0
1290 prop_binCRF_muon2016M_bin12_quad_cM0
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0 +8 −7
1291 prop_binCRF_muon2016M_bin13_sm_lin_quad_cM0
1292 prop_binCRF_muon2016M_bin17_sm_lin_quad_cM0
1293 prop_binCRF_muon2016M_bin6_sm_lin_quad_cM0
1294 prop_binCRF_muon2016M_bin9_sm_lin_quad_cM0
1295 prop_binCRF_muon2017_bin13_quad_cM0
1296 prop_binCRF_muon2017_bin13_sm_lin_quad_cM0
1297 prop_binCRF_muon2017_bin14_quad_cM0
1298 prop_binCRF_muon2017_bin15_sm_lin_quad_cM0
1299 prop_binCRF_muon2017_bin18_quad_cM0
1300 prop_binCRF_muon2017_bin19_quad_cM0
1301 prop_binCRF_muon2018_bin13_quad_cM0
1302 prop_binCRF_muon2018_bin20_quad_cM0
1303 prop_binCRF_muon2018_bin20_sm_lin_quad_cM0
1304 prop_binCRWS_electron2016M_bin10_quad_cM0
1305 prop_binCRWS_electron2016M_bin16_sm_lin_quad_cM0
1306 prop_binCRWS_electron2016M_bin17_sm_lin_quad_cM0
1307 prop_binCRWS_electron2016M_bin20_sm_lin_quad_cM0
1308 prop_binCRWS_electron2016M_bin7_sm
1309 prop_binCRWS_electron2017_bin8_quad_cM0
1310 prop_binCRWS_electron2017_bin8_sm_lin_quad_cM0
1311 prop_binCRWS_electron2018_bin18_sm_lin_quad_cM0
1312 prop_binCRWS_muon2016M_bin18_sm
1313 prop_binCRWS_muon2016M_bin3_sm_lin_quad_cM0
1314 prop_binCRWS_muon2016M_bin5_quad_cM0
1315 prop_binCRWS_muon2016M_bin7_quad_cM0
1316 prop_binCRWS_muon2016M_bin7_sm
1317 prop_binCRWS_muon2016M_bin7_sm_lin_quad_cM0
1318 prop_binCRWS_muon2016M_bin8_sm_lin_quad_cM0
1319 prop_binCRWS_muon2016M_bin9_sm
1320 prop_binCRWS_muon2016M_bin9_sm_lin_quad_cM0
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = $0^{+8}_{-7}$
1321 prop_binCRWS_muon2017_bin10_quad_cM0
1322 prop_binCRWS_muon2017_bin10_sm_lin_quad_cM0
1323 prop_binCRWS_muon2017_bin19_quad_cM0
1324 prop_binCRWS_muon2017_bin19_sm_lin_quad_cM0
1325 prop_binCRWS_muon2018_bin14_quad_cM0
1326 prop_binCRWS_muon2018_bin14_sm_lin_quad_cM0
1327 prop_binCRWS_muon2018_bin18_quad_cM0
1328 prop_binCRWS_muon2018_bin18_sm_lin_quad_cM0
1329 prop_binSR_electron2016M_bin16_quad_cM0
1330 prop_binSR_electron2016M_bin17_quad_cM0
1331 prop_binSR_electron2016M_bin17_sm_lin_quad_cM0
1332 prop_binSR_electron2016M_bin8_quad_cM0
1333 prop_binSR_electron2017_bin11_quad_cM0
1334 prop_binSR_electron2017_bin17_quad_cM0
1335 prop_binSR_electron2017_bin17_sm_lin_quad_cM0
1336 prop_binSR_electron2017_bin18_sm_lin_quad_cM0
1337 prop_binSR_electron2018_bin11_quad_cM0
1338 prop_binSR_muon2016M_bin13_quad_cM0
1339 prop_binSR_muon2016M_bin13_sm_lin_quad_cM0
1340 prop_binSR_muon2016M_bin17_sm_lin_quad_cM0
1341 prop_binSR_muon2016M_bin19_sm_lin_quad_cM0
1342 prop_binSR_muon2017_bin12_sm_lin_quad_cM0
1343 prop_binSR_muon2017_bin14_quad_cM0
1344 prop_binSR_muon2017_bin14_sm_lin_quad_cM0
1345 prop_binSR_muon2017_bin16_sm_lin_quad_cM0
1346 prop_binSR_muon2017_bin20_sm_lin_quad_cM0
1347 r
$1.0^{+0}_{-0}$
1348 tau_vsele_2016M
1349 tau_vsele_2017
1350 tau_vsele_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0^{+8}_{-7}$
1351
tau_vsmu_2016M
1352
tau_vsmu_2017
1353
tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$